Forecasting on the Accuracy–Timeliness Frontier: Decoupling From Present and Max-Tau Predictors

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September 1, 2026

Abstract

We re-examine the Mean-Squared Error (MSE) forecasting paradigm by formally integrating an accuracy–timeliness trade-off, where accuracy is defined by MSE (or target correlation) and timeliness by advancement (or phase excess). While MSE-optimal predictors track levels accurately, they can lag behind changing targets, sacrificing dynamic lead. We introduce a Decoupling-from-Present (DFP) predictor with closed-form solutions, of which the classical MSE predictor is a special case. The DFP achieves maximum advancement for any given accuracy level in the plane spanned by fore- and nowcast, tracing a conditional efficient frontier of the accuracy–timeliness trade-off, under the geometric constraint. In this framework, we derive an upper bound on lead over MSE under a consistency constraint, and show that the DFP approaches this bound arbitrarily closely. The framework extends to singular cases and non-stationary integrated processes, with applications in forecasting and real-time signal extraction. We then derive a Max-Tau predictor, which maximizes lead across all linear predictors, yielding an unconditional efficient frontier between accuracy and timeliness. Although Max-Tau outperforms any linear predictor on the frontier, including DFP, the latter can generate additional look-ahead behaviour, beyond Max-Tau's upper limit of an infinite lead, at the detriment of sign inversion tendency.

1 Introduction

Petropoulos et al. (2022) contend that "the theory of forecasting appears mature today ... the fact that forecasting is mature does not mean that all has been done." Exemplifying this, we formalize a novel forecast accuracy–timeliness dilemma.

Forecasting involves partly competing objectives: accuracy (predicting future levels), timeliness (lead-lag behavior relative to a benchmark), and smoothness (suppressing high-frequency noise) — equally relevant to real-time signal extraction (nowcasting). Jointly optimizing all three yields a formal prediction-trilemma: improving accuracy, smoothness, and timeliness (AST) simultaneously is impossible (Wildi and McElroy, 2019), though that frequency-domain approach lacks the time domain's more direct appeal and an exact closed-form solution is missing. Wildi (2024, 2026a, 2026b) addresses the accuracy-smoothness dilemma; here we instead focus on the accuracy-timeliness trade-off, providing interpretable closed-form time-domain solutions.

We introduce the Decoupling From Present (DFP) approach, generalizing the classical MSE predictor, recovered as a special case. Achieving an effective lead by a predictor requires a dynamic pattern that differs from the contemporaneous behavior of the process (the last data point): a degree of decoupling is intrinsic to constructing a genuinely forward-looking predictor. The problem is non-trivial since the last data point is often the single most important observation, which somewhat conflicts with decoupling from it.

We prove a duality result: for any target accuracy, our predictor achieves the maximum possible phase excess or lead relative to the MSE benchmark, among all predictors in the plane spanned by the MSE fore- and nowcast. This yields a conditional efficient frontier, valid under this geometric constraint, formalizing the AT tradeoff. While MSE is a single point on this frontier, DFP traces the full curve. We also derive an upper bound on the maximum admissible lead under a consistency constraint; our predictor comes arbitrarily close to it.

Next, we relax the geometric restriction of DFP, proposing the Max-Tau predictor, which maximizes lead within *all* linear predictors of a given length, formalizing a new unconditional efficient frontier between target correlation (TC) and timeliness (advancement/retardation) through optimal decoupling. This frontier intersects the DFP frontier at the common MSE point. Unlike DFP, Max-Tau relies on decoupling vectors largely independent of the data generating process (DGP), with simple linear or purely sinusoidal profiles—i.e., filter coefficients that do not decay to zero with increasing lag. In both DFP and Max-Tau, decoupling can be interpreted as freeing MSE-optimal predictors from lag-inducing or lead-inhibiting components.

Addressing predictor lead can involve counterintuitive and technically challenging aspects. While the MSE predictor ordinarily leads the nowcast—the *standard case*—the opposite, non-standard case is not uncommon. There, the DFP geometry for a lead inverts, and the classic minimum MSE objective becomes an error maximization under a finite length constraint. Counterintuitively, in the DFP, decoupling the predictor from the most recent observation—commonly regarded as most influential—can increase the lead, and overweighting older data can extend it further. Strong forward-looking behavior can also generate sign inversion, beginning with trend/mean inversion, whereby a linear trend or non-vanishing mean is mirrored at the zero line. We propose a variant, the *time-shift DFP*, which guards against such undesirable inversion. Notably, while the latter's lead can grow unboundedly, our approach can extend beyond its natural infinity bound, providing additional forward-looking behavior; to prevent excessive sign inversion in these extreme scenarios, we propose a consistency rule requiring a positive TC. The Max-Tau predictor, in contrast, preserves trend orientation and signs while maximizing lead on the efficient frontier, where it consistently outperforms DFP by design. However, DFP can generally exceed Max-Tau's natural upper bound on lead, assuming substantially reduced TC and concomitant sign inversion are permitted.

While our main results target stationary, regular, full-rank problems, we further extend to singular, rank-deficient cases and non-stationary integrated processes in the univariate setting, omitting extensions to multivariate frameworks.

Section 2 outlines limitations of traditional MSE forecasting. Section 3 introduces the DFP, derives closed-form solutions, and specifies a conditional accuracy–timeliness efficient frontier, extending results to singular cases (DFP II) and integrated processes (I-DFP). Section 4 defines lead time, relates it to key hyperparameters, derives an upper bound on admissible look-ahead behaviour in the DFP, and introduces the Max-Tau predictor, generating an unconditional frontier over all linear predictors. Section 5 presents forecasting and signal extraction business-cycle applications, with examples from the open-source R package accompanying the DFP and Max-Tau.[1] Section 6 summarizes the main conclusions.

2 Limitations of Classic Forecast Approach: a Case Study

For illustration we consider forecasting an MA(q) process

$$x_t = \mu + \gamma_0\epsilon_t + \gamma_1\epsilon_{t-1} + ... + \gamma_q\epsilon_{t-q},$$

where $\gamma_0 = 1$. To simplify exposition, we assume that $\mu = 0$, that $\gamma_1, ..., \gamma_q$ are known and that the white noise innovations ϵ_t are observed.

The MSE h-step ahead forecast $\hat{x}_{hT}^{MSE}$ of x_{T+h} at the sample end $t = T$ is

$$\hat{x}_{hT}^{MSE} = \begin{cases} \sum_{k=0}^{q-h} \gamma_{h+k}\epsilon_{T-k} & 1 \leq h \leq q \\ 0 & \text{otherwise.} \end{cases}$$

[1] See `https://github.com/wiaidp/R-Package-Look-Ahead` and `https://marcwildi.com` for a synopsis.

For $h \leq q$ and $t \in \{q-h+1, \ldots, T\}$, consider the *forecast filter* $\hat{x}_{ht}^{MSE} = \sum_{k=0}^{q-h} \gamma_{k+h}\epsilon_{t-k}$ with weights $\gamma_h, \ldots, \gamma_q$. The cross correlation function (CCF) between $x_{t+\delta}$ and $\hat{x}_{ht}^{MSE}$ at lead δ ($q \geq \delta > 0$) or at lag δ ($h - q \leq \delta \leq 0$) is

$$\rho(x_{t+\delta}, \hat{x}_{ht}^{MSE}) \quad = \quad \frac{\sum_{k=\max(0,-\delta)}^{\min(q-h,q-\delta)} \gamma_{k+\delta}\gamma_{k+h}}{\sqrt{\sum_{k=0}^{q} \gamma_k^2 \sum_{k=0}^{q-h} \gamma_{k+h}^2}}.$$

Figure 1 presents the CCF for an exponentially weighted MA(9) process, $x_t = \sum_{k=0}^{9} 0.9^k \epsilon_{t-k}$, with optimal $h = 5$ step-ahead forecast $\hat{x}_{5t}^{MSE} = \sum_{k=0}^{4} 0.9^{k+5}\epsilon_{t-k}$, at leads and lags $\delta \in \{-4, -3, ..., 9\}$ (top panel). By design, $\hat{x}_{5t}^{MSE}$ maximizes the CCF at $h = 5$ (green vertical line), with correlation peaking at 0.86 at $\delta = 0$ (black vertical line). The lower panel shows that rather than clearly leading x_t, the predictor is largely time-aligned with it — at odds with the intent of a forward-looking predictor. This tight coupling is implied by MSE optimality and is not resolved by changing h. Although deliberately simple, the example reflects common challenges in economic forecasting, including real-time signal extraction and trend nowcasting (see Section 5.2). Our new approach addresses this problem, incorporating a controllable left shift in forecasts for general stationary and integrated processes.

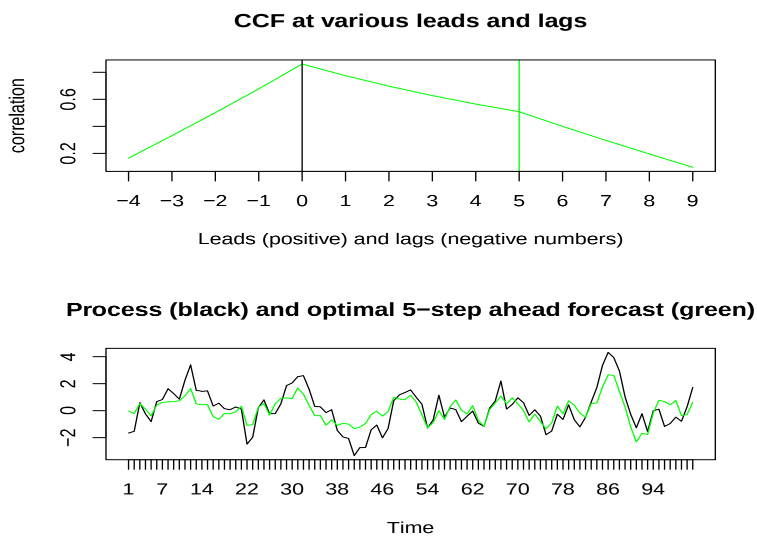

Figure 1: Top graph: CCF of MSE 5-step ahead forecast filter and MA(9) process at leads $9 \geq \delta > 0$ and lags $-4 \leq \delta \leq 0$. Bottom graph: a comparison of realizations of the process (black line) and of the optimal 5-step ahead forecast filter (green line)

We consider a stationary purely non-deterministic linear process

$$x_t = \sum_{k=-\infty}^{\infty} \gamma_k \epsilon_{t-k}, \tag{1}$$

where γ_k is square-summable, $\sum_{k=-\infty}^{\infty} \gamma_k^2 < \infty$. We assume iid innovations ϵ_t, though results extend to uncorrelated white noise when focusing on best *linear* prediction. For simplicity, ϵ_t is taken as observed; otherwise it can be recovered via standard inversion.[2] Allowing $k < 0$

[2]Compact AR or ARMA formulations are not considered since the MA impulse-response representation is generally more informative about predictor dynamics and, in any case, the particular representation is irrelevant for the problem at hand.

in (1) accommodates signal-extraction with two-sided filters (cf. Section 5.2). Specifically, let $z_t = \sum_{h=0}^{\infty} \xi_k \epsilon_{t-k}$ be a stationary process with (purely non deterministic) Wold decomposition $\xi_k, k \geq 0$, and let $x_t = \sum_{k=-\infty}^{\infty} \eta_k z_{t-k}$ be a signal obtained by applying the two-sided filter η_k to z_t. Then Equation (1) is obtained with $\gamma_k = (\eta * \xi)_k$, the convolution of the Wold decomposition and the filter.

Consider a generic (not necessarily MSE-optimal) predictor for x_{t+h} generated by a causal finite-length filter of length $L > 1$ with coefficient vector $\mathbf{b} = (b_0, ..., b_{L-1})'$:

$$\hat{x}_{ht} = \mathbf{b}'\boldsymbol{\epsilon}_t = \sum_{k=0}^{L-1} b_k \epsilon_{t-k},$$

where $\mathbf{b} = \mathbf{b}(h)$ generally depends on the forecast horizon h (suppressed in notation for brevity). Define the length-L coefficient vector for horizon h as $\boldsymbol{\gamma}_h := (\gamma_h, \gamma_{h+1}, ..., \gamma_{h+L-1})'$, which yields the (length-L) MSE-optimal linear predictor

$$\hat{x}_{ht}^{MSE} = \sum_{k=0}^{L-1} \gamma_{h+k} \epsilon_{t-k}.$$

For simplicity we assume L to be 'large', so that the MA(L)-representation is close to the (infinite length) MSE predictor, i.e., $\sum_{k=0}^{L-1} \gamma_k^2 / \sum_{k=0}^{\infty} \gamma_k^2 \approx 1$; we also assume zero means throughout, as static level adjustments do not affect dynamic lead. For brevity, we refer to the 'MSE predictor at horizon h' interchangeably as the weight vector $\boldsymbol{\gamma}_h$ or its implied predictor $\hat{x}_{ht}^{MSE}$. When $h = 0$, this reduces to the nowcast, and $\boldsymbol{\gamma}_0$ denotes the contemporaneous coefficient vector. In a stationary ARMA(p, q) setting, the nowcast coincides with the truncated MA inversion (Wold decomposition). Later, we extend this framework to non-stationary integrated processes.

3 Look-Ahead Forecast Optimization Criteria

We develop a look-ahead variant by 'Decoupling From Present' (DFP) to advance the predictor relative to the MSE benchmark, inducing a controllable lead via a leftward shift. We provide closed-form solutions for regular stationary problems, extending to singular formulations and non-stationary integrated processes, and derive the distribution of the predictor. We further establish equivalence with the dual optimization problem and derive an efficient accuracy–timeliness (AT) frontier for the DFP, conditional on the geometric decoupling.

3.1 Decoupling from Present

As established in Section 2, the classical MSE predictor in this examples achieves its CCF maximum at the contemporaneous lag $\delta = 0$, yielding a coincident rather than a leading forecast, essentially invariant to increases in $h \leq q$. To induce a systematic lead, we seek to decouple the predictor from the present and formulate the following decoupling-from-present (DFP) problem:

$$\begin{aligned} &\max_{\mathbf{b}} \boldsymbol{\gamma}_h'\mathbf{b} \\ \text{s.t.} \quad &\boldsymbol{\gamma}_0'\mathbf{b}/\|\boldsymbol{\gamma}_0\| = \alpha_0 \\ &||\mathbf{b}|| = 1, \end{aligned} \tag{2}$$

where $\boldsymbol{\gamma}_0 \neq \mathbf{0}$ and where $|| \cdot ||$ means the L^2 norm, i.e., $\|\boldsymbol{\gamma}_0\| = \sqrt{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0}$. The unit-norm constraint implies that the linear objective $\boldsymbol{\gamma}_h'\mathbf{b}$ is proportional to the correlation $\rho(x_{t+h}, \hat{x}_{ht})$ at $\delta = h$, yielding a tractable optimization while preserving the correlation-based interpretation at the target horizon (up to scaling $||\boldsymbol{\gamma}_h||$). Likewise, α_0 specifies the contemporaneous correlation $\rho(x_t, \hat{x}_{ht})$ at $\delta = 0$. Setting $\alpha_0 = 0$ achieves complete decoupling of the predictor from the present, or the nowcast; see Section 5.1 for an example. Observe that $\alpha_0 = \cos(\theta_{0b})$, where θ_{0b} denotes the angle between the nowcast $\boldsymbol{\gamma}_0$ and $\mathbf{b}$, so α_0 parametrizes the phase between the predictor $\mathbf{b}$ and the nowcast $\boldsymbol{\gamma}_0$; see Fig.2. Typically, only one of the two solutions ($\pm\theta_{0b}$) is genuinely forward-looking. This connects conceptually to Wildi and McElroy (2019), who frame a trilemma in terms of frequency-domain time-shift and amplitude functions. We assume $|\alpha_0| \leq 1$ (feasibility) and a strictly positive objective value (strict positivity).[3]

Remark: Decoupling the predictor from the present may seem counterintuitive, as x_T is typically regarded as highly informative for forecasting. However, decoupling does not imply that x_T is irrelevant. Rather, it acknowledges that achieving an effective lead requires a dynamic pattern that differs from the contemporaneous behavior of x_T (necessary condition): a degree of decoupling is intrinsic to constructing a genuinely forward-looking predictor.

The following Proposition provides the closed-form solution to the DFP criterion.

Proposition 1. *Assume $\boldsymbol{\gamma}_0 \not\propto \boldsymbol{\gamma}_h$, meaning that $\boldsymbol{\gamma}_0$ and $\boldsymbol{\gamma}_h$ ($h > 0$) are linearly independent, and $|\alpha_0| < 1$. Then the solution to criterion* (2) *lies in span$\{\boldsymbol{\gamma}_0, \boldsymbol{\gamma}_h\}$:*

$$\mathbf{b} = \lambda_1 \boldsymbol{\gamma}_h + \lambda_2 \boldsymbol{\gamma}_0 \tag{3}$$

for some scalars λ_1, λ_2. If $\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h \neq 0$, then

$$\lambda_1 = \frac{\alpha_0 \|\boldsymbol{\gamma}_0\| - \lambda_2 \boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0}{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h} \quad ,\textit{and} \quad \lambda_2 = \frac{-b - sign(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h)\sqrt{b^2 - 4ac}}{2a}, \tag{4}$$

with coefficients $a = \boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0 \left(\frac{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0 \boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}{(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h)^2} - 1 \right)$, $b = 2\alpha_0 \|\boldsymbol{\gamma}_0\| \left(1 - \frac{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0 \boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}{(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h)^2} \right)$, and $c = \frac{\alpha_0^2 \boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0 \boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}{(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h)^2} - 1$.

If $\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h = 0$, then

$$\lambda_2 = \frac{\alpha_0}{\|\boldsymbol{\gamma}_0\|} \quad , \quad \lambda_1 = \sqrt{\frac{1 - \alpha_0^2}{\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}},$$

Finally, if $|\alpha_0| = 1$, then $\mathbf{b} = \mathrm{sign}(\alpha_0)\boldsymbol{\gamma}_0 / \|\boldsymbol{\gamma}_0\|$.

The proof is given in the Appendix. The singular case $\boldsymbol{\gamma}_0 \propto \boldsymbol{\gamma}_h$, violating full rank, is addressed in Section 3.2. The two solutions for (λ_1, λ_2) correspond to $\pm\theta_{0b}$ in Fig.2, so for the objective-maximizing $\mathbf{b}$, $\mathrm{sign}(\theta_{0b}) = \mathrm{sign}(\theta_{0h})$; the case $\mathrm{sign}(\theta_{0h}) = 1$, i.e., $\boldsymbol{\gamma}_h$ 'above' $\boldsymbol{\gamma}_0$, defines the standard case, where the MSE predictor leads the nowcast (see below). Geometrically, Fig. 2 shows that for root $\lambda_2 < 0$ and $\theta_{0h} > 0$ (standard case), θ_{0b}—the angle between $\mathbf{b}$ and $\boldsymbol{\gamma}_0$—exceeds θ_{0h}, the angle between $\boldsymbol{\gamma}_h$ and $\boldsymbol{\gamma}_0$; the excess $\theta_{hb} = \theta_{0b} - \theta_{0h} > 0$ is the additional phase induced by decoupling, associated with the predictor's lead (Section 4). Root $\lambda_2 > 0$ then gives the phase-reversed, lagging solution. In the non-standard case, where $\theta_{0h} < 0$, $\boldsymbol{\gamma}_h$ is 'below' $\boldsymbol{\gamma}_0$ and the predictor actually *lags* the nowcast—not uncommon, and observable even for simple ARMA(1,1) processes (Appendix E). To obtain a lead $\theta_{hb} > 0$ here, either $\lambda_1 > 0$ and $\lambda_2 > 0$, with $\mathbf{b}$ between $\boldsymbol{\gamma}_0, \boldsymbol{\gamma}_h$ (so DFP still lags the nowcast); or $\lambda_1 < 0$, $\lambda_2 > 0$ (inversion of the standard case), with $\mathbf{b}$ 'above' $\boldsymbol{\gamma}_0$ (so DFP leads *both* the MSE predictor and the nowcast). Interestingly, this sign combination may require sign inversion, minimizing the TC objective instead of maximizing it. This largely counterintuitive case is addressed in Appendix E. Complete decoupling occurs at $\theta_{0b} = \pi/2$, though this is generally too extreme for applications (see below). Note that the geometric time-domain depiction in this figure is merely an analogy for the relevant frequency-domain geometry elaborated in Section 4, which connects angles θ_{ij} to physical leads/lags (Corollary 2); still, the analogy here usefully renders the various cases and corresponding geometric inversions, expressed via the sign combinations on λ_1, λ_2. For simplicity, we henceforth focus on the standard case, where the MSE forecast leads the nowcast and a lead of DFP is associated with maximization of the TC; we refer to the appendix for the necessary sign inversions in the non-standard case.

[3] If $\boldsymbol{\gamma}_0$ and $\boldsymbol{\gamma}_h$ are closely aligned, $\alpha_0 \approx -1$ may breach this assumption. Strict positivity guards against sign-inverting designs at the forecast horizon h.

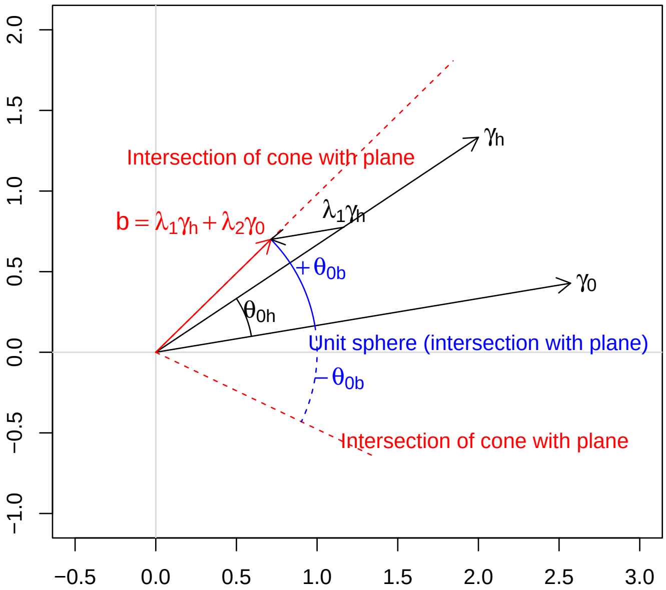

Figure 2: Geometry of the DFP predictor. The solution $\mathbf{b} = \lambda_1\boldsymbol{\gamma}_h + \lambda_2\boldsymbol{\gamma}_0$ lies at the intersection of the plan spanned by $(\boldsymbol{\gamma}_0, \boldsymbol{\gamma}_h)$, the circular cone with semi-angle θ_{0b} (red), and the unit-sphere (blue). The cone has axis $\boldsymbol{\gamma}_0$ and half-angle $\theta_{0b} = \arccos(\alpha_0)$. The figure depicts one of the two solutions of the quadratic in λ_2 associated with $+\theta_{0b}$ (with $\lambda_1 > 0, \lambda_2 < 0$); the mirrored solution at $-\theta_{0b}$ (with $\lambda_1 < 0, \lambda_2 > 0$) does not maximize the objective $\boldsymbol{\gamma}_h'\mathbf{b}$. If the MSE predictor leads the nowcast (so-called standard case, see below for details), then the mirrored solution corresponds to a lag, so it is omitted in the figure. In the standard case, the angle $\theta_{hb} = \theta_{0b} - \theta_{0h} > 0$ represents the phase excess of the DFP relative to the MSE predictor $\boldsymbol{\gamma}_h$ and is associated with its advancement (lead), see Section 4.

We can derive the distribution of the DFP predictor conditional on the distribution of an estimate $\hat{\boldsymbol{\gamma}}_0$ of $\boldsymbol{\gamma}_0$. To this end, we approximate Equation (3) with

$$\mathbf{b} \approx (\lambda_1\mathbf{F}^h + \lambda_2\mathbf{I})\boldsymbol{\gamma}_0, \tag{5}$$

where $\mathbf{F}$ is the $L \times L$ forward operator, with ones on its first superdiagonal and zeroes elsewhere, and $\mathbf{I}$ is the identity matrix. Note that the last h entries of $\mathbf{F}^h\boldsymbol{\gamma}_0$ are zero, so the approximation in (5) is valid provided $\gamma_k \approx 0$ for $L-1 \geq k \geq L-1-h$, which holds for sufficiently large L under stationarity.

Corollary 1. *Let $\hat{\boldsymbol{\gamma}}_0$ be an estimator of $\boldsymbol{\gamma}_0$ with mean $\boldsymbol{\mu}_{\gamma_0}$ and covariance matrix $\boldsymbol{\Sigma}_{\gamma_0}$ and assume λ_1, λ_2 are fixed. Then the DFP predictor in Equation (3) has mean and covariance*

$$\boldsymbol{\mu}_b \approx (\lambda_1\mathbf{F}^h + \lambda_2\mathbf{I})\boldsymbol{\mu}_{\gamma_0} \text{ and } \boldsymbol{\Sigma}_b \approx (\lambda_1\mathbf{F}^h + \lambda_2\mathbf{I})\boldsymbol{\Sigma}_{\gamma_0}(\lambda_1\mathbf{F}^h + \lambda_2\mathbf{I})', \tag{6}$$

where the approximations are valid for L sufficiently large. Moreover, if $\hat{\boldsymbol{\gamma}}_0$ is (asymptotically) Gaussian, then $\hat{\mathbf{b}}$ is (asymptotically) Gaussian as well.

The result follows immediately from (5) and standard properties of linear transformations. For the (asymptotic) distribution of the classic MSE estimate $\hat{\boldsymbol{\gamma}}_0$ of $\boldsymbol{\gamma}_0$, see Brockwell and Davis (1993) and Cox (1990)[4].

A limitation of the corollary is that it treats λ_1 and λ_2 as deterministic, even though both depend on $\hat{\boldsymbol{\gamma}}_0$ through Proposition 1. We address this below.

As an alternative to the above unit-length DFP, we propose a DFP-MSE decoupling criterion:

$$\begin{aligned} &\min_{\mathbf{b}} (\mathbf{b}-\boldsymbol{\gamma}_h)'(\mathbf{b}-\boldsymbol{\gamma}_h) \\ \text{s.t.} \quad &\boldsymbol{\gamma}_0'\mathbf{b} = \alpha_0. \end{aligned} \tag{7}$$

This minimizes the mean-squared prediction error for x_{t+h} under DFP; if $\alpha_0 = \boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h$, the solution recovers $\mathbf{b} = \boldsymbol{\gamma}_h$. The unit-norm constraint is omitted, simplifying the geometry and optimization, though α_0 loses its interpretation as a correlation, complicating its selection. The objective captures accuracy while the constraint enforces lead, formalizing the accuracy-timeliness trade-off in (7) (see Section 4.3). While minimizing MSE is the standard goal, the non-standard case (Appendix E) calls for maximizing it, under a length constraint.

Proposition 2. *Assume that $\boldsymbol{\gamma}_0$ and $\boldsymbol{\gamma}_h$ ($h > 0$) are linearly independent. Then the solution to criterion* (7) *is*

$$\mathbf{b} = \boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0, \tag{8}$$

where

$$\lambda = \frac{\alpha_0 - \boldsymbol{\gamma}_h'\boldsymbol{\gamma}_0}{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0}. \tag{9}$$

Proof: The solution lies on the affine hyperplane $\boldsymbol{\gamma}_0'\mathbf{b} = \alpha_0$ and is the orthogonal projection of $\boldsymbol{\gamma}_h$ onto it. Since $\boldsymbol{\gamma}_0$ is normal to the hyperplane, the optimizer has the form $\mathbf{b} = \boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0$, with λ determined by enforcing the constraint. □

The distribution of the MSE-DFP predictor follows from Corollary 1 with $\lambda_1 = 1$ and $\lambda_2 = \lambda$ fixed. We can extend this to incorporate the randomness of λ, as a function of the empirical $\hat{\boldsymbol{\gamma}}_0, \hat{\boldsymbol{\gamma}}_h$ in Eq.(9), see Appendix D. Both DFP criteria, (2) and (7), yield predictors as linear combinations of $\boldsymbol{\gamma}_h$ and $\boldsymbol{\gamma}_0$, but entail distinct trade-offs. Criterion (7) delivers a geometrically simple construction with a unique solution, convenient for extensions to integrated processes (Section 3.3). Criterion (2) induces a richer geometry with two candidate solutions and greater interpretability: under the unit-norm constraint, $\alpha_0 = \rho(x_t, \hat{x}_{ht})$ has the meaning of a correlation, facilitating hyperparameter selection. Finally, the unit-length formulation admits a dual result underlying the efficient Accuracy–Timeliness frontier (Section (4.3)) which cannot be obtained as easily in the absence of the additional length constraint.

To conclude, we introduce the MSE-DFP triangle in Fig. 3, with $\theta_{0b}, \theta_{0h}, \theta_{hb}$, the angles between $\boldsymbol{\gamma}_0$, $\boldsymbol{\gamma}_h$ and $\mathbf{b}$. In the standard case, $\theta_{0h} > 0$ and under a lead, $\theta_{hb} > 0$, with $\lambda < 0$. Here, the phase excess $\theta_{hb} > 0$ further increases the lead, and doing so in this precise way can be optimal, as captured by the efficient frontier (see below). Strict positivity $\boldsymbol{\gamma}_h'\mathbf{b} > 0$ implies $\beta = \theta_{hb} < \pi/2$. As before, this time-domain rendition of the DFP triangle is merely an analogy of the relevant frequency-domain representation—with matching signs and orientation—linking angles to leads/lags (Section 4).

[4]For an ARMA-process, the distribution of the MA-inverted weights γ_k, $k = 1, ..., L-1$ (noting that $\gamma_0 = 1$) can be derived from the distribution of the AR- and MA-estimates via the delta method as elaborated in Cox (1990).

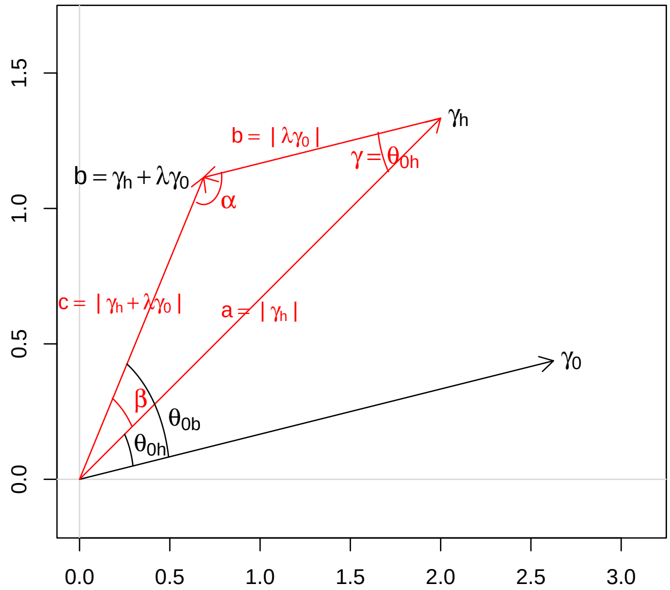

Figure 3: Side lengths a, b and c (red) as well as angles α, β and γ of the MSE-DFP triangle spanned by the vectors $\boldsymbol{\gamma}_h$ and $\mathbf{b} = \boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0$ (black), with $\lambda < 0$.

3.2 Generalized DFP II: Addressing the Singular Rank One Case

The DFP predictor lies in the plane spanned by $\boldsymbol{\gamma}_0$ and $\boldsymbol{\gamma}_h$: if $\boldsymbol{\gamma}_0 \propto \boldsymbol{\gamma}_h$, this plane collapses to a line (rank-one case), so the conflict between decoupling and target correlation maximization cannot be resolved. We propose a generalization, DFP II, which resolves this by specifying an alternative decoupling predictor $\tilde{\boldsymbol{\gamma}}_0$ that cannot lie along $\boldsymbol{\gamma}_h$ when $\boldsymbol{\gamma}_0 \propto \boldsymbol{\gamma}_h$. Specifically, DFP II decouples from both present *and* past:

$$\begin{aligned}
&\boldsymbol{\gamma}_h'\mathbf{b} \to \max \\
&\sum_{k=-h_1}^{0} w_{-k}\mathrm{CCF}(k) = \alpha_0 \\
&\|\mathbf{b}\| = 1,
\end{aligned}$$

where $h_1 \in \mathbb{N}$ is a non-negative integer and $w_i \geq 0,\ i = 0, ..., h_1$ are weights. For simplicity, we assume $w_i \equiv 1$. DFP II reduces to the original DFP for $h_1 = 0$. The generalized decoupling constraint can be expressed as

$$\begin{aligned}
\sum_{k=-h_1}^{0} \mathrm{CCF}(k) &= \sum_{k=-h_1}^{0} (b_{-k}, \ldots, b_{L-1})(\gamma_0, \ldots, \gamma_{L-1+k})'/\|\boldsymbol{\gamma}_0\| \\
&= \left(\boldsymbol{\Sigma}_{h_1}\boldsymbol{\gamma}_0\right)'\mathbf{b}/\|\boldsymbol{\gamma}_0\|,
\end{aligned}$$

where $\boldsymbol{\Sigma}_{h_1}$ is an integrator matrix whose i-th row is

$$\left(\mathbf{0}_{\max(0,\,i-(h_1+1))},\ \mathbf{1}_{\min(i,\,h_1+1)},\ \mathbf{0}_{L-\max(0,\,i-(h_1+1))-\min(i,\,h_1)}\right),$$

with $\mathbf{m}_n$ denoting a length-n vector with entries m. For $h_1 > 0$, the normalized eigenvectors of $\mathbf{\Sigma}_{h_1}$ are $\pm\mathbf{e}_L$, where $\mathbf{e}_L$ is the last Cartesian unit vector of length L (with entries δ_{iL}, the Kronecker symbol).Since $\gamma_0 \neq 0$ in general (with $\gamma_0 = 1$ for the one-sided Wold decomposition), $\boldsymbol{\gamma}_0 \not\propto \mathbf{e}_L$; however, $\gamma_0 = 0$ is not excluded in signal extraction. But if $\boldsymbol{\gamma}_0 \propto \boldsymbol{\gamma}_h$ and $h < L$, then $\boldsymbol{\gamma}_0 \not\propto \mathbf{e}_L$, since otherwise the last non-zero entry γ_{L-1} of $\boldsymbol{\gamma}_0$ would shift h steps forward in $\boldsymbol{\gamma}_h$, conflicting with proportionality. Hence we may assume that $\boldsymbol{\gamma}_0$ is not an eigenvector of the integrator, giving $\mathbf{\Sigma}_{h_1}\boldsymbol{\gamma}_0 \not\propto \boldsymbol{\gamma}_0$, and thus $\mathbf{\Sigma}_{h_1}\boldsymbol{\gamma}_0 \not\propto \boldsymbol{\gamma}_h$ when $\boldsymbol{\gamma}_0 \propto \boldsymbol{\gamma}_h$. Consequently, in the singular rank one case the generalized DFP II

$$\begin{aligned} &\boldsymbol{\gamma}_h'\mathbf{b} \to \max \\ &\left(\mathbf{\Sigma}_{h_1}\boldsymbol{\gamma}_0\right)'\mathbf{b}/\|\boldsymbol{\gamma}_0\| = \alpha_0 \\ &\|\mathbf{b}\| = 1 \end{aligned} \tag{10}$$

is full-rank and feasible, assuming $h_1 > 0$.

For illustration we consider the challenging AR(1) forecast problem, where $\boldsymbol{\gamma}_h \propto \boldsymbol{\gamma}_0$ for all h – a property we refer to as self-similarity. The CCF at positive lags $k > 0$ satisfies

$$\mathrm{CCF}(k) = \boldsymbol{\gamma}_k'\mathbf{b}/\|\boldsymbol{\gamma}_0\| = a_1\,\mathrm{CCF}(k-1),$$

so that CCF(k) decreases exponentially (in absolute value) at positive lags, irrespective of $\mathbf{b}$. In contrast, the CCF at negative lags $k < 0$ is

$$\mathrm{CCF}(k) = (b_{-k}, \ldots, b_{L-1})(\gamma_0, ..., \gamma_{L-1+k})'/\|\boldsymbol{\gamma}_0\|,$$

whose pattern can be controlled explicitly by $\mathbf{b}$. Instead of re-shaping the future, that is the positive lags of the CCF, which is vain in the case of self-similarity, the DFP II addresses the left tail of the CCF (negative lags) by re-shaping the link between the predictor and the past. Look-ahead behaviour can be obtained by right-skewing the CCF through a leveraging effect, removing mass from the left tail or even assigning it negative mass, while maximizing CCF(h) at $h > 0$ on the right tail (see Section 5.4 for illustration).

Note that in the AR(1) case, Criterion (10) is invariant to the forecast horizon for any $h > 0$, due to self-similarity. To incorporate the forecast horizon and to calibrate leveraging and right-skewing of the CCF, we may set $h_1 := h$ in the constraint, so that the impact of the forecast horizon materializes in $\left(\mathbf{\Sigma}_h\boldsymbol{\gamma}_0\right)'\mathbf{b}/\|\boldsymbol{\gamma}_0\| = \alpha_0$, which encodes an anti-symmetric right-skew of the CCF, anchored in $-h$, effectively compensating for the invariance of leverage at $h > 0$ in the right tail. An alternative approach would be to link α_0 with a measure for the left-shift τ of the predictor and to select $\alpha_0 = \alpha_0(\tau)$ with $\tau = h$; see Sections 4 (theory) and 5.4 (application).

3.3 Extension to Integrated Processes: I-DFP

For stationary processes we relied on the MA form (1). For integrated processes the MA inversion no longer converges, so the final predictor is obtained in AR form, though optimization relies on a convergent MA form of the (stationary) filter error and resulting pseudo-covariances of the process (to be defined).

Let $\tilde{x}_t = \sum_{k=-\infty}^{\infty}\gamma_k\tilde{z}_{t-k}$ where $\tilde{z}_t$ is a non-stationary I(d) process such that $\Delta(B)\tilde{z}_t = (1-B)^d\tilde{z}_t =: z_t$ is stationary and invertible with purely non-deterministic Wold decomposition $z_t = \sum_{k=0}^{\infty}\xi_k\epsilon_{t-k}$, with $\xi_0 = 1$. Assume $\sum_{k=-\infty}^{\infty}|\gamma_k k^{d-1}| < \infty$ and let $\mathbf{\Xi}$ denote the $L \times L$ matrix with i-th row $\mathbf{\Xi}_{i\cdot} := (\xi_{i-1}, \xi_{i-2}, ..., \xi_0, \mathbf{0}_{L-i})$, $i = 1, ..., L$, where $\mathbf{0}_{L-i}$ is a zero vector of length $L-i$, and $L > d+1$. We define $\mathbf{z}_t := (z_t, ..., z_{t-(L-1)})'$, $\boldsymbol{\epsilon}_t := (\epsilon_t, ..., \epsilon_{t-(L-1)})'$, and for a length-$L$ filter $\mathbf{b} = (b_0, ..., b_{L-1})$ we set $\mathbf{b}_\epsilon := \mathbf{\Xi}\mathbf{b}$. Consequently:

$$\mathbf{b}'\mathbf{z}_t \approx \left(\mathbf{\Xi}\mathbf{b}_z\right)'\boldsymbol{\epsilon}_t = \mathbf{b}_\epsilon'\boldsymbol{\epsilon}_t, \tag{11}$$

where the approximation holds if filter coefficients decay sufficiently rapidly or, equivalently, if L is sufficiently large. The MSE predictor of the target $\tilde{x}_{t+h}$ is $y_{t,h,MSE} = \hat{\boldsymbol{\gamma}}'_{h,MSE}\tilde{\mathbf{z}}_t$, with weights $(\hat{\gamma}_{\tilde{x},h,k})_{k=0,...,L-1}$ and $\tilde{\mathbf{z}}_t := (\tilde{z}_t, ..., \tilde{z}_{t-(L-1)})'$, derivable under general assumptions (McElroy and Wildi, 2016; McElroy and Livsey, 2022; McElroy, 2022). For $L \geq d$ (and a fortiori for $L > d+1$), the MSE principle guarantees stationarity of the filter error $e_t := \tilde{x}_{t+h} - y_{t,h,MSE}$, so $\tilde{x}_{t+h}$ and $y_{t,h,MSE}$ are cointegrated with vector $(1, -1)$. To cancel the unit root(s) of $\tilde{z}_t$, the error filter $\gamma_{k+h} - \hat{\gamma}_{\tilde{x},h,k}$ must satisfy the so-called cointegration constraints:

$$\sum_{k=-\infty}^{\infty} (\gamma_{k+h} - \hat{\gamma}_{\tilde{x},h,k})k^j = 0, \; j = 0, ..., d-1, \tag{12}$$

where $\hat{\gamma}_{\tilde{x},h,k} = 0$ for $k \notin \{0, ..., L-1\}$, see McElroy and Wildi (2016).[5]

Conceptually, a generalization of the DFP criterion to integrated processes can rely on

$$\begin{aligned} \min_{\mathbf{b}_{\tilde{z}}} \quad & E[(\tilde{x}_{t+h} - y_t)^2] \\ & E[(\tilde{x}_t - y_t)^2] = \beta_0 \end{aligned} \tag{13}$$

where $y_t := \mathbf{b}_{\tilde{z}}\tilde{\mathbf{z}}_t$ is the I-DFP predictor; cointegration ensures finiteness of the objective mean-square error. However, finiteness of the mean-square error in the constraint may additionally require suitable initialization of the process, compatible with the data (see the Appendix proof). Setting $\beta_0 > E[(\tilde{x}_t - y_{t,h,MSE})^2]$ in the constraint ensures decoupling from the present. Since covariances of the original DFP are not well-defined in the non-stationary case, we used mean-square errors in 13; as shown below, these can be replaced by 'pseudo' covariances, so the decoupling framework and its interpretation extend straightforwardly to I(d) processes.

Before presenting the I-DFP optimization problem and its solution let us define the $L \times L$ summation and differentiation matrices:

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0 & 0 & ... & 0 \\ 1 & 1 & 0 & ... & 0 \\ ... & & & & \\ 1 & 1 & 1 & ... & 1 \end{pmatrix} \text{ and } \boldsymbol{\Delta} := \boldsymbol{\Sigma}^{-1} = \begin{pmatrix} 1 & 0 & 0 & ... & 0 & 0 \\ -1 & 1 & 0 & ... & 0 & 0 \\ & & & & & \\ 0 & 0 & 0 & ... & -1 & 1 \end{pmatrix}.$$

The following Theorem generalizes the stationary DFP to I-DFP in the case of integrated processes.

Theorem 1. *Let $\tilde{z}_t$ be an I(d) process and $\tilde{x}_t = \sum_{k=-\infty}^{\infty} \gamma_k \tilde{z}_{t-k}$ with $\sum_{k=-\infty}^{\infty} |k^{d-1}\gamma_k| < \infty$. Assume the length-L filter $\mathbf{b}_{\tilde{z}} = (b_{\tilde{z},0}, ..., b_{\tilde{z},L-1})'$ satisfies the cointegration constraints $\sum_{k=-\infty}^{\infty} \gamma_{k+h}k^j - \sum_{k=0}^{L-1} b_{\tilde{z},k}k^j = 0$ for $j = 0, ..., d-1$, where $L > d+1$, and assume $\hat{\boldsymbol{\gamma}}_{0,MSE} \not\propto \hat{\boldsymbol{\gamma}}_{h,MSE}$ (full rank), where $\hat{\boldsymbol{\gamma}}_{0,MSE}, \hat{\boldsymbol{\gamma}}_{h,MSE}$ denote the MSE nowcast and forecast.*

1. *The stationary DFP criterion generalizes to any of the following four I-DFP criteria*

$$\left.\begin{array}{cc} \begin{array}{c} \min_{\mathbf{b}_{\tilde{x}}} ||\boldsymbol{\Sigma}^d\boldsymbol{\Xi}(\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})||^2 \\ \mathbf{b}'_{\tilde{z}}\boldsymbol{\Xi}'\boldsymbol{\Sigma}^{d\,\prime}\boldsymbol{\Sigma}^d\boldsymbol{\Xi}\hat{\boldsymbol{\gamma}}_{0,MSE} = \alpha_0 \end{array}, & \begin{array}{c} \min_{\mathbf{b}_{\tilde{x}}} ||\boldsymbol{\Sigma}^d\boldsymbol{\Xi}(\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})||^2 \\ ||\boldsymbol{\Sigma}^d\boldsymbol{\Xi}(\hat{\boldsymbol{\gamma}}_{0,MSE} - \mathbf{b}_{\tilde{z}})||^2 = \beta_0 \end{array} \\ & \\ \begin{array}{c} \max_{\mathbf{b}_{\tilde{z}}} \mathbf{b}'_{\tilde{z}}\boldsymbol{\Xi}'\boldsymbol{\Sigma}^{d\,\prime}\boldsymbol{\Sigma}^d\boldsymbol{\Xi}\hat{\boldsymbol{\gamma}}_{h,MSE} \\ \mathbf{b}'_{\tilde{z}}\boldsymbol{\Xi}'\boldsymbol{\Sigma}^{d\,\prime}\boldsymbol{\Sigma}^d\boldsymbol{\Xi}\hat{\boldsymbol{\gamma}}_{0,MSE} = \nu_0 \\ ||\boldsymbol{\Sigma}^d\boldsymbol{\Xi}\mathbf{b}_{\tilde{z}}|| = 1 \end{array}, & \begin{array}{c} \max_{\mathbf{b}_{\tilde{z}}} \mathbf{b}'_{\tilde{z}}\boldsymbol{\Xi}'\boldsymbol{\Sigma}^{d\,\prime}\boldsymbol{\Sigma}^d\boldsymbol{\Xi}\hat{\boldsymbol{\gamma}}_{h,MSE} \\ ||\boldsymbol{\Sigma}^d\boldsymbol{\Xi}(\hat{\boldsymbol{\gamma}}_{0,MSE} - \mathbf{b}_{\tilde{z}})||^2 = \mu_0 \\ ||\boldsymbol{\Sigma}^d\boldsymbol{\Xi}\mathbf{b}_{\tilde{z}}|| = 1 \end{array} \end{array}\right\} \tag{14}$$

where $\alpha_0, \beta_0, \nu_0, \mu_0$ in the constraints are decoupling hyperparameters controlling decoupling strength. All four criteria are equivalent, up to eventual re-scaling.

[5] For an I(2) process, both $\tilde{z}_t$ and $\tilde{z}_t - \tilde{z}_{t-1}$ are asymptotically unbounded, so $y_{t,h,MSE}$ must track both level and slope closely to ensure a stationary error. The I(1) constraint $\sum_{k=-\infty}^{\infty} \gamma_k = \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x},h,k}$ ensures preservation of levels; the additional I(2) constraint $\sum_{k=-\infty}^{\infty} k\gamma_k = \sum_{k=0}^{L-1} k\hat{\gamma}_{\tilde{x},h,k}$ ensures preservation of slopes (and similarly for $d > 2$).

2. *For $d = 1$, the I-DFP solution is*

$$\begin{aligned}\mathbf{b}_{\tilde{z}} &= \Gamma(0)\mathbf{e}_1^L + \mathbf{B}\tilde{\mathbf{b}}, \text{ where } \tilde{\mathbf{b}} = \tilde{\mathbf{B}}\tilde{\tilde{\mathbf{b}}} + d_0(\alpha_0)\mathbf{e}_1^{L-1}, \text{ and}\\ \tilde{\tilde{\mathbf{b}}} &= \left(\tilde{\mathbf{B}}'\mathbf{B}'\boldsymbol{\Xi}'\boldsymbol{\Sigma}'\boldsymbol{\Sigma}\boldsymbol{\Xi}\mathbf{B}\tilde{\mathbf{B}}\right)^{-1}\tilde{\mathbf{B}}'\mathbf{B}'\boldsymbol{\Xi}'\boldsymbol{\Sigma}'\boldsymbol{\Sigma}\boldsymbol{\Xi}\left(\hat{\gamma}_{h,MSE} - \Gamma(0)\mathbf{e}_1^L - d_0(\alpha_0)\mathbf{B}\mathbf{e}_1^{L-1}\right).\end{aligned}$$

where $\mathbf{e}_1^L, \mathbf{e}_1^{L-1}$ are the first canonical vectors of lengths L and $L-1$, and $\Gamma(0) = \sum_{k=-\infty}^{\infty} \gamma_k$. The vectors $\tilde{\mathbf{b}}$ and $\tilde{\tilde{\mathbf{b}}}$ have lengths $L-1$ and $L-2$, respectively. The cointegration and decoupling matrices $\mathbf{B}$ ($L \times (L-1)$) and $\tilde{\mathbf{B}}$ ($(L-1) \times (L-2)$), and $d_0(\alpha_0)$, depending on the decoupling constraint, are derived in the proof.

3. *The above solution extends to $d > 1$ with suitable modifications (see the proof).*

The proof is provided in Appendix A. The four I-DFP criteria represent equivalent extensions of the DFP to non-stationary processes. The top two criteria emphasize mean-square objective functions with a (generalized) unit-length constraint omitted. The bottom two rely on a generalized pseudo target covariance in their objective function, necessitating an additional generalized length constraint to render the maximization meaningful. This pseudo covariance concept can also be applied to the decoupling constraint, yielding the 2×2 combination of criteria. The cointegration constraints ensure (asymptotic) finiteness of the target mean-square filter error, imposing additional structure reflected in the more complex solution in the I(1) case (second assertion). To ensure finitiness, the I-DFP predictor $\mathbf{b}_{\tilde{z}}$ must be applied to $\tilde{z}_t$ (AR form).

4 Connecting Hyperparameters and Phase-Excess with Lead

The phase excess $\theta_{hb} > 0$ of the DFP predictor does not admit an immediate interpretation as a temporal lead. To establish a relationship between θ_{hb} (or α_0) and the relative lead of the DFP predictor over the MSE benchmark, we must first define the concept of a lead.

4.1 Time-Shift at Frequency Zero

Define the (complex) transfer function of the length-L filter $\boldsymbol{\gamma} = (\gamma_0, ..., \gamma_{L-1})'$ by

$$\Gamma(\omega) = \sum_{k=0}^{L-1} \gamma_k \exp(-ik\omega), \ \omega \in [0, \pi][6].$$

Write its polar form as

$$\Gamma(\omega) = A(\omega)\exp(i\Phi(\omega)),$$

where $A(\omega) = |\Gamma(\omega)| \geq 0$ is the amplitude response and $\Phi(\omega) = \arg(\Gamma(\omega))$ is the phase response ($\mathrm{mod}(2\pi)$). For $\omega \neq 0$, define the time-shift

$$\tau(\omega) := \Phi(\omega)/\omega. \tag{15}$$

For a sinusoid $x_t = \exp(it\omega)$, the filtered output is

$$y_t = \sum_{k=0}^{L-1} \gamma_k x_{t-k} = \sum_{k=0}^{L-1} \gamma_k \exp(i(t-k)\omega) = \Gamma(\omega)x_t = A(\omega)\exp(i(t+\tau(\omega))\omega) = A(\omega)x_{t+\tau(\omega)},$$

so the output is scaled by $A(\omega)$ and shifted by $\tau(\omega)$: $\tau(\omega) > 0$ indicates a lead and $\tau(\omega) < 0$ a lag. If $A(\omega) = 0$, the sinusoid is suppressed and the time-shift is not meaningful.

[6]Negative frequencies $\omega \in [-\pi, 0[$ can be ignored due to symmetry.

To extend the definition to $\omega = 0$, differentiate $\Gamma(\omega)$ with respect to ω. Enforcing $\Gamma(0) > 0$ yields $A(0) = \Gamma(0) > 0$ and $\Phi(0) = 0$. Since A is even, $\dot{A}(0) = 0$, hence

$$\dot{\Gamma}(0) = i\Gamma(0)\dot{\Phi}(0), \quad \text{so } \dot{\Phi}(0) = -i\frac{\dot{\Gamma}(0)}{\Gamma(0)}.$$

Consequently,

$$\tau(0) := \lim_{\omega\to 0} \tau(\omega) = \dot{\Phi}(0) = -i\frac{\dot{\Gamma}(0)}{\Gamma(0)} = -\frac{\sum_{k=1}^{L-1} k\gamma_k}{\sum_{k=0}^{L-1} \gamma_k}. \tag{16}$$

This is confirmed by applying the filter to a linear trend $x_t = t$:

$$y_t = \sum_{k=0}^{L-1} \gamma_k(t-k) = (t+\tau(0))\sum_{k=0}^{L-1} \gamma_k = A(0)x_{t+\tau(0)}.$$

Furthermore, $\partial y_t/\partial t = \Gamma(0) > 0$ confirms that the filter preserves trend direction. Since $\tau(\omega)$ varies with ω, we fix the reference frequency $\omega_0 = 0$ to emphasize low-frequency (trend) components relevant for forecasting/nowcasting[7] (our proceeding can be extended to $\omega_0 > 0$, see Max-Tau). We assume a standard scenario in which the MSE predictor leads the nowcast, $\tau_h(0) > \tau_0(0)$ (the non-standard case $\tau_h(0) \leq \tau_0(0)$ is addressed in Appendix (E)). If the DFP predictor additionally leads the MSE predictor, then $\tau_b(0) > \tau_h(0) > \tau_0(0)$; we show below how to tune DFP hyperparameters to achieve a desired $\tau_b(0)$.

Let $\Gamma_0(\omega), \Gamma_h(\omega), \Gamma_b(\omega)$ denote the transfer functions of the nowcast and the MSE and DFP predictors, with amplitude and phase functions A_0, A_h, A_b and Φ_0, Φ_h, Φ_b. We assume $\Gamma_0(0), \Gamma_h(0), \Gamma_b(0) > 0$, ruling out trend suppression or reversal; if this fails, an alternative reference frequency $\omega_0 > 0$ may be chosen. The lead-chain $\tau_b(0) > \tau_h(0) > \tau_0(0)$ implies

$$\Phi_b(\omega) > \Phi_h(\omega) > \Phi_0(\omega) \tag{17}$$

for sufficiently small $\omega > 0$, by continuity. Analogously to the time-domain DFP triangle in Fig. 3, we define a frequency-domain DFP triangle with vertices $\mathbf{0}$, $\Gamma_h(\omega)$ and $\Gamma_b(\omega) = \Gamma_h(\omega) + \lambda\Gamma_0(\omega)$ in the complex plane. The phase ordering (17) implies $\lambda < 0$, which in turn implies a phase excess $\theta_{0b} > \theta_{0h}$ in Fig. 3. Consequently, $\theta_{0b} > \theta_{0h}$ in the time-domain corresponds to $\Phi_b(\omega) - \Phi_0(\omega) > \Phi_h(\omega) - \Phi_0(\omega)$, i.e., $\Phi_b(\omega) > \Phi_h(\omega)$ in the frequency-domain, justifying the terminology (phase excess) and providing an informal link between $\theta_{hb} > 0$ (in the time-domain) and lead (in the frequency domain). We now derive exact expressions linking the lead to the angles and the DFP hyperparameters, denoting $\tau := \tau(0)$ and omitting the reference to zero-frequency for brevity.

4.2 Linking Hyperparameters to the Time-Shift

We analyze the DFP-MSE predictor (8), $\mathbf{b} = \boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0$, linking λ, α_0, and θ_{hb} to the DFP's lead $\tau = \tau_b - \tau_h > 0$. Unless stated otherwise, we maintain the following baseline assumptions: (i) $\tau_h > \tau_0$ (standard case); (ii) $\tau > 0$ (lead) and $|\theta_{hb}| < \pi/2$ (strict positivity); and (iii) $\Gamma_0(0), \Gamma_h(0), \Gamma_b(0) > 0$ (no trend reversal or suppression). Note that (i) implies $\boldsymbol{\gamma}_0 \not\propto \boldsymbol{\gamma}_h$ (full rank): in the rank-deficient case, DFP II applies; if $\tau_h < \tau_0$, see Appendix E. Assumption (ii) presupposes a 'meaningful' (not excessive) lead, which is addressed specifically in time-shift DFP and Max-Tau below. If (iii) fails, an alternative reference frequency $\omega_0 \neq 0$ can be selected. These assumptions mainly affect interpretability, allowing an explicit link between hyperparameters and lead; violating them may reduce interpretability but not necessarily impair look-ahead behavior.

The following proposition connects λ to the relative shift $\tau := \tau_b - \tau_h$.

[7] Under continuity, a lead at frequency zero extends to adjacent frequencies — such as the relevant business-cycle frequencies, see Section (5.2) — though possibly in attenuated form (see the DFP R-package for further evidence).

Proposition 3. *Let the baseline assumptions hold, and let $\tau = \tau_b - \tau_h > 0$ be a desired lead of DFP relative to MSE. Then, λ is determined by τ as*

$$\lambda(\tau) = -\frac{\tau\Gamma_h(0)}{(\tau + \tau_h - \tau_0)\Gamma_0(0)}. \tag{18}$$

A proof is provided in Appendix A; we henceforth refer to the DFP predictor based on $\lambda(\tau)$ in (18) as the *time-shift DFP*, a special case of the ordinary MSE DFP. For $\tau = 0$, (18) gives $\lambda = 0$, so $\mathbf{b} = \boldsymbol{\gamma}_h$, as expected. As $\tau \to \infty$, $\lambda \to -\Gamma_h(0)/\Gamma_0(0)$, giving $\Gamma_b(0) \to 0$: the time-shift DFP asymptotically removes linear trends and mean levels. Choosing $\lambda < -\Gamma_h(0)/\Gamma_0(0)$ in the ordinary MSE-DFP exceeds the time-shift DFP's reach, causing increasing sign inversion starting with trend and mean reversal; typically the left-shift (advancement) increases further, as demonstrated in our examples, though Eq. (16) for τ_b becomes meaningless (it would represent the time-shift of a sign-inverted predictor). Because of the positive offset $\tau_h - \tau_0 > 0$ in Eq.(18), $\lambda(\tau) < 0$ is strictly decreasing in τ and since $\Gamma_0(0), \Gamma_h(0) > 0$, it follows that $\Gamma_b(0) = \Gamma_0(0) + \lambda(\tau)\Gamma_h(0) > 0$, so the time-shift DFP preserves trend orientation and mean sign.

For $\lambda < \lambda_\infty := \lim_{\tau\to\infty} \lambda(\tau)$, $\Gamma_b(0) < 0$ (see Proposition 5), implying mean and trend are reverted; increasingly negative $\lambda < \lambda_\infty$ causes this inversion to spread to adjacent frequencies $\omega > 0$; to limit contamination and avoid excessive sign inversion, we propose a consistency rule in Section 4.4. While controlled sign inversion consistent with this rule may enhance look-ahead behaviour further, beyond the natural infinite bound on the lead of time-shift DFP, an inverted mean is nevertheless problematic for stationary non-zero mean processes (trend inversion is irrelevant for stationary series); in these extreme cases, where trend and mean inversion are tolerated, we therefore recommend centering the data first, before applying the DFP, and then adding back the mean to the centered forecast.

Next, we express α_0 and the phase excess θ_{hb} in terms of $\tau = \tau_b - \tau_h$.

Corollary 2. *Let the baseline assumptions hold with $\tau = \tau_b - \tau_h > 0$ given. Then*

1.

$$\alpha_0 = \alpha_0(\tau) = \boldsymbol{\gamma}_0' \left(\boldsymbol{\gamma}_h + \lambda(\tau)\boldsymbol{\gamma}_0\right) \tag{19}$$

$$\theta_{hb} = \theta_{hb}(\tau) = \arccos\left(\frac{\boldsymbol{\gamma}_h'(\boldsymbol{\gamma}_h+\lambda(\tau)\boldsymbol{\gamma}_0)}{\|\boldsymbol{\gamma}_h\|\|\boldsymbol{\gamma}_h+\lambda(\tau)\boldsymbol{\gamma}_0\|}\right) \approx \arccos\left(\frac{(\mathbf{F}^h\boldsymbol{\gamma}_0)'\left(\mathbf{F}^h+\lambda(\tau)\mathbf{I}\right)\boldsymbol{\gamma}_0}{\|\mathbf{F}^h\boldsymbol{\gamma}_0\|\|(\mathbf{F}^h+\lambda(\tau)\mathbf{I})\boldsymbol{\gamma}_0\|}\right), \tag{20}$$

where $\lambda(\tau)$ is defined in (18).

2. *$\theta_{hb}(\tau)$ is strictly monontonically increasing in τ.*

Eq.(19) follows from (9) and Eq.(20) from $\cos(\theta_{hb}) = \boldsymbol{\gamma}_h'\mathbf{b}/(\|\boldsymbol{\gamma}_h\|\|\mathbf{b}\|)$, using $\boldsymbol{\gamma}_h \approx \mathbf{F}^h\boldsymbol{\gamma}_0$ for L sufficiently large. A proof of the second statement is provided in Appendix A. Eq. (19) enables principled hyperparameter selection based on τ, improving interpretability, as illustrated in our examples.

4.3 Accuracy-Timeliness Dilemma, Dual Problem and Efficient AT Frontier

We rely on the geometric picture of the DFP triangle in Fig. 3, introducing the DFP frontier in terms of the angles θ_{ij}. Since $\theta_{hb}(\tau)$ is strictly increasing in τ, maximizing θ_{hb} or τ is equivalent. Therefore, the discussion applies indifferently to leads or angles; we select the latter here for ease of exposition.

The decoupling parameter and phase θ_{0b} are linked by $\alpha_0 = \cos(\theta_{0b})$, making the accuracy–timeliness trade-off explicit: accuracy is encoded in the objective, timeliness in the phase-based constraint. Geometrically, as the phase excess $\theta_{hb} = \theta_{0b} - \theta_{0h}$—or equivalently the lead—increases, the projection of $\mathbf{b}$ onto $\boldsymbol{\gamma}_h$ declines. Under $\|\mathbf{b}\| = 1$,

$$\mathbf{b}'\boldsymbol{\gamma}_h = \cos(\theta_{0b} - \theta_{0h})\|\boldsymbol{\gamma}_h\|,$$

so a larger phase excess reduces the objective $\mathbf{b}'\boldsymbol{\gamma}_h$, and conversely.

Consider the 'dual' DFP criterion

$$\begin{aligned} &\min_{\mathbf{b}} \boldsymbol{\gamma}_0'\mathbf{b} \\ \text{s.t.}\quad &\boldsymbol{\gamma}_h'\mathbf{b}/\|\boldsymbol{\gamma}_h\| = \alpha_h \\ &\|\mathbf{b}\| = 1, \end{aligned} \tag{21}$$

obtained from the 'primal' by swapping objective and correlation constraint and switching to minimization (our terminology differs from the classic Lagrangian primal/dual). While the dual's algebra parallels the primal, its geometry differs: the feasible intersection of cone and unit sphere in Fig. (2) is now based on a cone with axis $\boldsymbol{\gamma}_h$ rather than $\boldsymbol{\gamma}_0$, with boundary rays symmetric about $\boldsymbol{\gamma}_h$. Let $\theta_{primal} = \theta_{0b}$ (the primal constraint) and $\theta_{dual} = \theta_{hb}$ (the dual constraint). Since $\theta_{0b} = \theta_{0h} + \theta_{hb}$,

$$\theta_{primal} = \theta_{dual} + \theta_{0h} = \arccos(\alpha_h) + \theta_{0h}\ ,\ \alpha_0 = \cos(\theta_{primal}) = \cos(\arccos(\alpha_h) + \theta_{0h}).$$

The dual's constraints yield two candidate solutions, $\pm\theta_{dual}$, with objective values $\boldsymbol{\gamma}_0'\mathbf{b} = \cos(\pm\theta_{dual}+\theta_{0h})\|\boldsymbol{\gamma}_0\|$. Selecting the minimizing branch in (21) gives $+\theta_{dual}$, matching the primal DFP solution. In the standard case we assume $0 < \theta_{bh} < \theta_{0b} < \pi$—the first inequality expressing a leading DFP, the last avoiding sign inversion (not strictly necessary but simplifying)—ensuring uniqueness and strict monotonicity. Hence, when $\alpha_0 := \cos(\arccos(\alpha_h) + \theta_{0h})$, primal and dual coincide: DFP is the fastest predictor —maximal phase excess θ_{dual} or lead via Corollary 2— subject to prescribed TC $\alpha_h = \boldsymbol{\gamma}_h'\mathbf{b}/\|\boldsymbol{\gamma}_h\|$.
This equivalence shows that DFP traces an efficient frontier between accuracy and timeliness. Unlike the MSE predictor, which is confined to a single point on this frontier, DFP spans the full curve. Efficiency claims, however, rely on geometric decoupling from the present via $\boldsymbol{\gamma}_0$ and this limitation is resolved in Section 4.5.

4.4 Limits on the Attainable Lead (or Phase Excess) under Strict Positivity

To illustrate, assume $\boldsymbol{\gamma}_0$ and $\boldsymbol{\gamma}_h$ are nearly (but not exactly) aligned. As $\lambda < 0$ becomes more negative, $\mathbf{b} = \boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0$ rotates to point almost opposite $\boldsymbol{\gamma}_h$, so pushing λ too far to increase lead can (almost) flip the predictor's sign. Componentwise, such sign flips can sometimes increase lead, but the limit is extreme, so a criterion is needed to rule it out.

A basic requirement is the strict positivity condition: $\boldsymbol{\gamma}_h'\mathbf{b}(\lambda) > 0$, or equivalently, $\boldsymbol{\gamma}_h'(\boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0) > 0$, where $\lambda < 0$. If $\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_0 > 0$, then

$$\lambda > -\frac{\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}{\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_0} =: \lambda_{\text{lim}},$$

establishing a lower bound $\lambda_{\text{lim}} < 0$ on λ. If $\Gamma_b(0) = \lambda_{\text{lim}}\Gamma_0(0) + \Gamma_h(0) > 0$ (no trend inversion), substituting into (18) yields

$$\tau \leq -\frac{(\tau_h - \tau_0)\lambda_{\text{lim}}\Gamma_0(0)}{\lambda_{\text{lim}}\Gamma_0(0) + \Gamma_h(0)} =: \tau_{\text{lim}},$$

establishing an upper bound $\tau_{\text{lim}} > 0$ on the admissible lead at frequency zero: for $0 < \tau < \tau_{\text{lim}}$ the target correlation remains strictly positive. In addition, no trend inversion $\Gamma_{b(\tau)}(0) < 0$ occurs for the time-shift DFP, see the previous section. If instead $\lambda_{\text{lim}}\Gamma_0(0) + \Gamma_h(0) = 0$, the predictor $\mathbf{b}(\lambda_{\text{lim}})$ eliminates the trend and $|\tau_{\text{lim}}| = \infty$ (see Proposition 6). Since the predictor is effectively looking ahead, we conclude $\tau_{\text{lim}} = \infty$, so any τ is compatible with strict positivity. If $\lambda_{\text{lim}}\Gamma_0(0)+\Gamma_h(0) < 0$, there exists $0 > \lambda_\infty > \lambda_{\text{lim}}$ with $\lambda_\infty\Gamma_0(0)+\Gamma_h(0) = 0$ giving again $\tau_{\lambda_\infty} = \infty$;

all τ are thus compatible with strict positivity. However, DFP designs with $\lambda \in]\lambda_{\lim}, \lambda_\infty]$ satisfy strict positivity —though they invert trend and mean— yet lie beyond the reach of the time-shift DFP; see the business-cycle application in Section 5.2 for illustration. Completing our treatment, if $\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_0 \leq 0$, any $\lambda < 0$ ensures strict positivity since $\theta_{0h} \in [\pi/2, \pi[$ and $\theta_{0b} \in]\theta_{0h}, \pi[\subset]\pi/2, \pi[$, i.e., $\theta_{hb} \in]0, \pi/2[$, irrespective of λ or $\tau > 0$.

Strict positivity ensures the predictor's dynamics remain aligned with those of the target at horizon h, preventing sign flips and ill-conditioned leads. Thus $\theta_{hb} < \pi/2$—equivalently $\lambda_{\lim}$ or $\tau_{\lim}$—places a hard cap on how far DFP can 'meaningfully' look ahead of MSE, possibly beyond the natural infinity bound on the time-shift-DFP.

By primal–dual equivalence, DFP maximizes the phase excess and thus the lead, through Corollary 2, in the DFP geometry specified by predictors with nowcast correlation α_0. Thus $\tau_{\lim}$ is an upper bound on the frequency-zero lead among such predictors, subject to strict positivity, which DFP approaches arbitrarily closely. These efficiency claims, however, are tied to decoupling from the nowcast, a limitation that is resolved next.

4.5 Max-Tau Predictor

The Max-Tau predictor solves the optimization problem

$$\begin{aligned} &\max \tau_b && (22)\\ &\mathbf{b}'\boldsymbol{\gamma}_h = \alpha_h\\ &\mathbf{b}'\mathbf{b} = 1, \end{aligned}$$

where the unit-length constraint ensures proportionality $\mathbf{b}'\boldsymbol{\gamma}_h \propto \rho(x_{t+h}, y_t)$ (henceforth denoted ρ_b or ρ) of linearized objective and TC, i.e., $\rho_b = \alpha_h/\|\boldsymbol{\gamma}_h\|$. Unfortunately, the objective $\tau_b = -\mathbf{b}'\mathbf{k}/\mathbf{b}'\mathbf{1}$ is non-linear in $\mathbf{b}$. We propose two alternative routes that match on the efficient frontier (but differ elsewhere). The first relies on the 'primal' problem

$$\begin{aligned} &\max_{\mathbf{b}} \mathbf{b}'\boldsymbol{\gamma}_h && (23)\\ &\mathbf{b}'(\mathbf{k} + \tau_b\mathbf{1}) = 0 && (24)\\ &\mathbf{b}'\mathbf{b} = 1, \end{aligned}$$

where $\mathbf{k} = (0, 1, ..., L-1)'$ and $\mathbf{b} = (1, ..., 1)'$ are length-L vectors, and $\tau_b = -\mathbf{b}'\mathbf{k}/\mathbf{b}'\mathbf{1}$ is the time-shift at frequency zero. This problem shares the structure of the original DFP (2), with the decoupling vector $\mathbf{v}(\tau) := \mathbf{k} + \tau\mathbf{1}$ replacing $\boldsymbol{\gamma}_0$. However, whereas in the original DFP the decoupling parameter α_0 shifts the plane, here $\alpha_0 = 0$ throughout, and τ_b instead rotates the plane (centered at the origin). This geometric specificity affects uniqueness, invertibility, and the efficient frontier. The primal yields $\rho_b^*(\tau) = \max_{\mathbf{b}|\tau_b=\tau} \mathbf{b}'\boldsymbol{\gamma}_h/\|\boldsymbol{\gamma}_h\|$ conditional on $\tau_b = \tau$; on the efficient frontier the function $\rho_b^*(\tau)$ is strictly monotonically decreasing with inverse $\tau^*(\rho)$. The Max-Tau predictor follows by inverting the primal in this sense: given ρ_0 on the frontier, insert $\tau^*(\rho_0)$ into (24) and solve for $\mathbf{b}$. Theorem 2 formalizes this inverted primal approach. Alternatively, the dual (22) can be solved directly via a stacked optimization that linearizes the problem, as discussed in Appendix B. The two approaches agree on the efficient frontier, by uniqueness, but generally differ elsewhere.

Theorem 2. *Assume $L \geq 2$, $\omega_0 = 0$ (trend reference frequency), $\Gamma_h(\omega_0), \Gamma_b(\omega_0) > 0$ (no trend inversion), $\tau_h > -(L-1)/2$ (full rank and/or positive time orientation), and $\rho \in]U, 1]$ (frontier), where $U = \sqrt{1 - \frac{\Gamma_h(0)^2}{\|\boldsymbol{\gamma}_h\|^2 L}} < 1$ (lead).*

1. *The Max-Tau predictor for $\rho \in]U, 1]$ is then, up to arbitrary scaling,*

$$\mathbf{b}^{Max-Tau}(\rho) = \boldsymbol{\gamma}_h - \lambda(\rho)\mathbf{v}(\rho) \quad , where \quad \lambda(\rho) = \frac{\boldsymbol{\gamma}_h'\mathbf{v}(\rho)}{\mathbf{v}(\rho)'\mathbf{v}(\rho)} \tag{25}$$

with $\mathbf{v}(\rho) = \mathbf{k} + \tau(\rho)\mathbf{1}$ *and*

$$\tau(\rho) = \frac{-B(\rho) + \sqrt{B(\rho)^2 - 4A(\rho)C(\rho)}}{2A(\rho)}, \tag{26}$$

with

$$A(\rho) = (\boldsymbol{\gamma}_h'\mathbf{1})^2 - \|\boldsymbol{\gamma}_h\|^2(1-\rho^2)\mathbf{1}'\mathbf{1},\ B(\rho) = 2\boldsymbol{\gamma}_h'\mathbf{k}\cdot\boldsymbol{\gamma}_h'\mathbf{1} - 2\|\boldsymbol{\gamma}_h\|^2(1-\rho^2)\mathbf{k}'\mathbf{1},\ C(\rho) = (\boldsymbol{\gamma}_h'\mathbf{k})^2 - \|\boldsymbol{\gamma}_h\|^2(1-\rho^2)\mathbf{k}'\mathbf{k}.$$

Here $\tau(\rho)$ *decreases strictly monotonically, tracing the efficient frontier between* ρ *and* τ*.*

2. *When* $\rho = 1$, $\mathbf{b}^{Max-Tau}(\rho) \propto \boldsymbol{\gamma}_h$, *generalizing the MSE predictor (up to unit-length). The Max-Tau predictor is uniquely defined, and no other linear predictor of length* L *can simultaneously improve both* τ *and* ρ *(efficient frontier). Strict positivity* $\mathbf{b}^{Max-Tau}(\rho)'\boldsymbol{\gamma}_h > 0$ *holds automatically.*

3. *An MSE-optimal Max-Tau predictor is obtained by rescaling:*

$$\mathbf{b}^{Max-Tau}_{MSE} = \frac{\boldsymbol{\gamma}_h'\mathbf{b}^{Max-Tau}}{\|\mathbf{b}^{Max-Tau}\|^2}\mathbf{b}^{Max-Tau} \tag{27}$$

The proof is in the Appendix; we now briefly provide some intuition. The solution $\boldsymbol{\gamma}_h - \lambda(\rho)\mathbf{v}(\rho)$ reflects the MSE-DFP, where $\mathbf{v}(\rho)$ now replaces the nowcast as the decoupling vector, i.e., Max-Tau lives in span$\{\mathbf{v}, \boldsymbol{\gamma}_h\}$. The vectors $\mathbf{1}$ and $\mathbf{k}$ constituting $\mathbf{v}$ emphasize large lags, assigning equal or dominant weight to the remote past; consequently, $\mathbf{v}$ also exhibits a large lag (see Proposition 7 in the appendix). The assumption $\tau_h > -(L-1)/2$ ensures $\boldsymbol{\gamma}_h$ has a smaller lag than $\mathbf{v}$ for any τ, i.e., $\tau_h > \tau_v$, so the problem has full rank. This also ensures validity of the standard case, i.e., Max-Tau avoids the counterintuitive sign inversion of the non-standard case in Appendix E. Decoupling from $\mathbf{v}$ in the Max-Tau design removes this linear high-lag component from $\boldsymbol{\gamma}_h$: projecting $\boldsymbol{\gamma}_h$ onto the subspace orthogonal to $\mathbf{v}$, via the optimal weight $\lambda(\rho)$ in (25), makes this decoupling efficient, yielding the Max-Tau predictor on the efficient frontier between ρ and τ, determined by $\rho \in]U, 1]$, signifying a lead $\tau_b > \tau_h$.

Remarkably, the optimal decoupling vector $\mathbf{v}$ has a simple linear weight profile with fixed unit slope in $\mathbf{k}$; only the intercept $\tau\mathbf{1}$ depends on the data-generating process, via $\|\boldsymbol{\gamma}_h\|$ and the projections $\boldsymbol{\gamma}_h'\mathbf{k}$ and $\boldsymbol{\gamma}_h'\mathbf{1}$ onto $\mathbf{k}$ and $\mathbf{1}$, as in (26). The condition $\tau_h > -(L-1)/2$ has a further interpretation: it rules out cases where $\boldsymbol{\gamma}_h$ weights the remote past more than the present (time inversion), typically satisfied in a prediction setting. This condition is key for defining the 'correct' frontier (see the proof).

For $\rho = U$, $\tau = \infty$ and $\Gamma_b(0) = 0$, violating positivity (no trend inversion); for $\rho_0 \in]0, U[$, no predictor can lie on the efficient frontier, since for any $\tau_0 > 0$ there exists a linear length-L predictor $\mathbf{b}$ with $\tau_b > \tau_0$ and $\rho_b > \rho_0$.[8] While typically uninteresting in prediction, the case $\rho < U$ is nevertheless addressed in the appendix. There, inversion of the primal and dual Max-Tau based on (22) generally differ. The condition $\rho \in]U, 1]$ expresses leads of Max-Tau over MSE on the frontier.

For $L \to \infty$, $U \to 1$ and $\rho \to 1$ on the efficient frontier, so the frontier shrinks along ρ (but not along τ). This is because, as L increases, more degrees of freedom are available to Max-Tau to optimize performance at the single ω_0, focusing arbitrarily tightly on ω_0 (overfitting). This is undesirable, since we generally want the lead to extend to adjacent frequencies over a sizeable interval around ω_0. [9] To address this Max-Tau overfitting, we propose two alternative and complementary solutions: i) extend Max-Tau to a set $\omega_i \in \Omega$ of reference frequencies ω_i, with $1 < |\Omega| < \infty$ (Appendix A), and ii) control the curvature of $\Gamma_b(\omega_0)$ at ω_0 through an additional

[8] For given τ_0 select $\epsilon > 0$ such that $\tau(U+\epsilon) > \tau_0$ (always possible since $\tau(U) = \infty$). Then $\rho(\tau(U+\epsilon)) = U+\epsilon > \rho_0$ (the functions invert on the frontier), as claimed.

[9] In contrast, DFP resides in span$\{\boldsymbol{\gamma}_0, \boldsymbol{\gamma}_h\}$, so overfitting is not an issue, at least if $\boldsymbol{\gamma}_0, \boldsymbol{\gamma}_h$ are known (or estimated but not overfitted).

curvature constraint, thereby controlling the spread of the lead over adjacent frequencies, especially when L is large (Appendix C).

The following corollary extends Max-Tau to reference frequency $\omega_0 > 0$, addressing, e.g., a maximal lead at a specific business-cycle frequency (see our applications). Recall that if $\omega_0 > 0$, then $\tau_b(\omega_0) = \Phi_b(\omega_0)/\omega_0 = \arctan\left(\frac{\Im(\Gamma_b(\omega_0))}{\Re(\Gamma_b(\omega_0))}\right)/\omega_0$. Denote $\mathbf{s} = \mathbf{s}(\omega_0) = (\sin(k\omega_0))_{k=0,...,L-1}$ and $\mathbf{c} = \mathbf{c}(\omega_0) = (\cos(k\omega_0))_{k=0,...,L-1}$, so that $\Im(\Gamma_b(\omega_0)) = -\mathbf{b}'\mathbf{s}$ and $\Re(\Gamma_b(\omega_0)) = \mathbf{b}'\mathbf{c}$. The primal thus becomes

$$\begin{aligned}
&\max_{\mathbf{b}} \mathbf{b}'\boldsymbol{\gamma}_h \\
&\mathbf{b}'(\mathbf{s} + d_b\mathbf{c}) = 0 \\
&\mathbf{b}'\mathbf{b} = 1,
\end{aligned} \tag{28}$$

where $d_b = \tan(\omega_0\tau_b(\omega_0))$. Relative to $\omega_0 = 0$, the key changes are: i) the decoupling vector $\tilde{\mathbf{v}} = \mathbf{s} + d_b\mathbf{c}$, where $\mathbf{1}$ is replaced by $\mathbf{s}$ and $\mathbf{k}$ by $\mathbf{c}$; ii) the hyperparameter $d_b = \tan(\omega_0\tau_b(\omega_0))$, which depends nonlinearly on τ_b: if $|\omega_0\tau_b(\omega_0)| < \pi/2$, then $d_b(\tau_b)$ is strictly increasing, so Max-Tau (maximizing τ_b) and Max-d (maximizing d_b) are equivalent.

Unlike at $\omega_0 = 0$, the wavelength (periodicity) $P = 2\pi/\omega_0$ at $\omega_0 > 0$ is finite. Since we typically target the component at ω_0 (e.g., a business-cycle component), predictors should not fall out of sync, i.e., $-P/4 < \tau_h(\omega_0) < P/4$, and we seek a leading predictor $\mathbf{b}$ with $\tau_h < \tau_b(\omega_0) < P/4$, see Corollary 3. A modification allowing $\tau_h < \tau_b(\omega_0) < \tau_h + P/4$, ensuring that Max-Tau does not fall out-of-sync of MSE, is proposed in Corollary 4. In applications, typically, the reference frequency is bounded by $\omega_0 < \pi/2$, ensuring $P/4 > 1$. Otherwise the lead $\tau_b \leq 1$ would not be properly observable, though the proofs allow $\omega_0 \geq \pi/2$.

Corollary 3. *Assume $L \geq 2$, $\omega_0 > 0$, $\Re(\Gamma_h(\omega_0)) > 0, \Re(\Gamma_b(\omega_0)) > 0$ (no signal inversion), $\boldsymbol{\gamma}_h$ linearly independent of $\mathbf{c}$ and $\mathbf{s}$ (full rank), $-P/4 < \tau_h < P/4$ (synced) and $\rho \in]U, 1]$ with $U = \sqrt{1 - \frac{\Re\left(\Gamma_h(0)\right)^2}{\|\boldsymbol{\gamma}_h\|^2\mathbf{c}'\mathbf{c}}}$ (lead).*

1. *The Max-Tau predictor $\mathbf{b}^{Max-tau}$ follows from (25), replacing $\mathbf{k}$ by $\mathbf{s}$ and $\mathbf{1}$ by $\mathbf{c}$ everywhere.*

2. *At $\rho = 1$, $\mathbf{b}^{Max-tau} \propto \boldsymbol{\gamma}_h$. The shift $\tau(\rho)$ decreases strictly monotonically in ρ, tracing the efficient frontier between ρ and τ. The Max-Tau predictor is uniquely defined, and no other linear predictor of length L can simultaneously improve both τ and ρ. Strict positivity $\mathbf{b}^{Max-Tau}(\rho)'\boldsymbol{\gamma}_h > 0$ holds automatically.*

3. *Unlike the case $\omega_0 = 0$, $\tau(\omega_0)$ is now bounded on the frontier, $\tau_h \leq \tau < P/4$, the quarter-wavelength at ω_0. Consequently, the predictor correlates positively with the identity filter $(1, 0, ..., 0)'$ at frequency ω_0 under a corresponding sinusoidal input.*

4. *The case $\omega_0 = 0$ of Theorem 2 arises as the limit $\omega_0 \to 0$.*

The proof is in the appendix. The above corollary references the time-shift against the identity, implying an upper bound $\tau_b < P/4$ approached as $d_b \to \infty$, assuming feasibility. In some applications, however, one may instead wish to measure the lead against $\boldsymbol{\gamma}_h$, via the time-shift excess $\tau = \tau_b - \tau_h$, and maximize τ rather than τ_b—the so-called Max-Tau-Excess predictor. This predictor correlates positively with $\boldsymbol{\gamma}_h$, rather than the identity, at frequency ω_0 under a corresponding sinusoidal input, i.e., $\tau_h \leq \tau < \tau_h + P/4$ instead of $\tau_h \leq \tau < P/4$ as before. The latter bound could put Max-Tau-Excess out of sync with MSE when $\tau_h < 0$ (a lag), since then $\tau_b - \tau_h \approx P/4 - \tau_h > P/4$ (assuming very large d_b, giving $\tau_b \approx P/4$, are admissible). Conversely, if $\tau_h > 0$ (a lead), the bound $\tau_b < P/4$ given in the previous corollary is unnecessarily restrictive. The Max-Tau-Excess design in the following corollary addresses these limitations.

Corollary 4. *Let the assumptions of the above corollary hold. Maximizing the time-shift excess* $\tau = \tau_b - \tau_h$ *is obtained by replacing* $\mathbf{s}$ *with*

$$\mathbf{s}^{excess} = \Re(1/\Gamma_h(\omega_0))\mathbf{s} + \Im(1/\Gamma_h(\omega_0))\mathbf{c}$$

and $\mathbf{c}$ *with*

$$\mathbf{c}^{excess} = \Re(1/\Gamma_h(\omega_0))\mathbf{c} - \Im(1/\Gamma_h(\omega_0))\mathbf{s}.$$

The bound on τ_b *on the frontier becomes* $\tau_h < \tau_b < \tau_h + P/4$, *and the predictor correlates positively with* $\boldsymbol{\gamma}_h$ *at frequency* ω_0 *under a corresponding sinusoidal input.*

The proof is in the appendix. Unlike the classic DFP, the above Max-Tau variants—whether at $\omega_0 = 0$ or $\omega_0 > 0$, and, in the latter case, referenced against the identity (Corollary 3) or against MSE (Corollary 4)—avoid the counter-intuitive non-standard case with its peculiar objective inversion. Signal inversion ($\Re(\Gamma_b(\omega_0)) < 0$) and out-of-sync behavior ($\tau \notin]-P/4, P/4[$ or $\tau \notin]-P/4+\tau_h, P/4+\tau_h[$) are likewise avoided, and the problem is virtually always feasible (full rank), as $\boldsymbol{\gamma}_h$ typically does not lie in the plane spanned by $\mathbf{1}$ and $\mathbf{k}$ when $\omega_0 = 0$, or by the strictly periodic $\mathbf{c}, \mathbf{s}$ when $\omega_0 > 0$, in typical prediction problems.

For a given TC on the frontier, Max-Tau generates a larger lead at ω_0 than DFP. However, extreme DFP designs can generate look-ahead behavior out of reach of Max-Tau, whose lead is bounded by $\tau < \infty$ for $\omega_0 = 0$ or by $\tau < P/4$ for $\omega_0 > 0$—at the cost of substantially reduced TC and increasing sign inversion (e.g., trend/mean reversion); see Section 4.4 and our applications.

5 Empirical Examples

DFP and Max-Tau sit on efficient (conditional or unconditional) frontiers and cannot be outperformed by univariate linear benchmarks in target correlation and time-shift simultaneously. We thus focus on examples illustrating key features of both approaches, omitting comparisons against linear benchmarks other than the MSE predictor, which actually sits on the frontier: an off-frontier benchmark is outperformed in both AT-dimensions, while an on-frontier one is replicated—both uninteresting outcomes.

First, we apply the unit-length DFP criterion (2) to the MA process of Section 2, comparing partial and complete decoupling. Second, we apply the MSE DFP (7), with its interpretable time-shift decoupling constraint, to business-cycle analysis using quarterly GDP and the Hodrick–Prescott filter. Third, we compare this MSE DFP with time-shift DFP and Max-Tau predictors on the same example. Finally, we apply the generalized DFP II to the AR(1) forecast problem, which is especially challenging due to self-similarity (rank-defficiency).

5.1 Illustration of the DFP-Criterion in the MA(9)-Case

We apply the unit-length DFP criterion (2) to the MA(9) example with forecast horizon $h = 5$, treating ϵ_t as observed (see Section 2). Figure 1 shows that the CCF $\rho(\hat{x}_{5,t}^{\text{MSE}}, x_{t+\delta})$ of the MSE predictor peaks at 0.86 at $\delta = 0$, indicating a predominantly coincident predictor. To mitigate this, we consider two DFP designs: (i) partial decoupling, setting CCF(0) $= \alpha_0 = 0.43$, noting that for the unit-length DFP α_0 admits a straightforward correlation interpretation, and (ii) complete decoupling, imposing CCF(0) $= \alpha_0 = 0$. Solving the quadratic for λ_2 in (4) and selecting the root maximizing the objective yields

$$\begin{aligned} \mathbf{b}^{\text{partial}} &= 1.62\boldsymbol{\gamma}_h - 0.51\boldsymbol{\gamma}_0 \\ \mathbf{b}^{\text{complete}} &= 1.8\boldsymbol{\gamma}_h - 0.79\boldsymbol{\gamma}_0. \end{aligned}$$

Figure 4 compares predictors (top-left), CCF (top-right), and forecasts (bottom). Unlike the MSE design, the DFP predictor places nonzero weights up to lag $k = L$ (a similar result is reported in Wildi (2024) and (2026a), when prioritizing smoothness instead of lead). The criterion enforces decoupling at $\delta = 0$ while maximizing performance at horizon $h = 5$. Ideally, any degradation at $\delta = 0$, due to decoupling, does not propagate to $\delta = h$ and the criterion minimizes such spillover losses at h (efficient frontier). As decoupling strengthens, forecasts shift progressively left, indicating anticipation achieved—counterintuitively—by placing more (negative) weight on higher-lag observations.

This example highlights the core trade-off of the DFP criterion: timeliness (advancement via decoupling) versus accuracy (MSE performance at $\delta = h$). Table 1 quantifies this trade-off via the CCF and the relative lead of the DFP design, defined as the shift maximizing the *sample* correlation between the nowcast and the DFP predictors (De Jong and Nijman, 1997), averaged over 10 000 realizations of length 200. This average shift corresponds, up to sample error, to the right-shift of the population CCF peak in Fig. 4 (top-right panel). Stronger decoupling (smaller α_0) shifts this peak rightward, signaling endogenous look-ahead behavior, whereas the MSE predictor's CCF peak remains fixed at $\delta = 0$ regardless of h. This suggests an alternative look-ahead mechanism not pursued here (Peak Correlation Shifting, PCS); see Wildi (2026c).

In applications, complete decoupling is typically too extreme: pronounced left-shifts often arise through undesirable sign inversion. Here, $\Gamma_b(0) < 0$ for the complete decoupling DFP, inverting the mean or trend —a phenomenon eluded by time-shift DFP or Max-Tau.

	MSE	DFP weak decoupling	DFP complete decoupling
CCF at lag=0	0.86	0.43	0.00
CCF at h=5	0.51	0.42	0.26
Relative lead over process	0.00	2.51	5.03

Table 1: CCFs of the predictors at $\delta = 0$ and $\delta = h = 5$ for the MSE benchmark, DFP with weak decoupling, and DFP with complete decoupling. Forecast loss at the target horizon (second row) is minimized subject to the decoupling constraint at the present (first row). Empirical leads of the DFP predictors (third row) quantify the left shift (earlier timing) relative to the process, defined as the lead at which the sample correlation between the DFP predictors and the nowcast is maximized. The reduction in CCF at the forecast horizon in the second row alongside the increase in lead in the last row highlights the accuracy–timeliness trade-off.

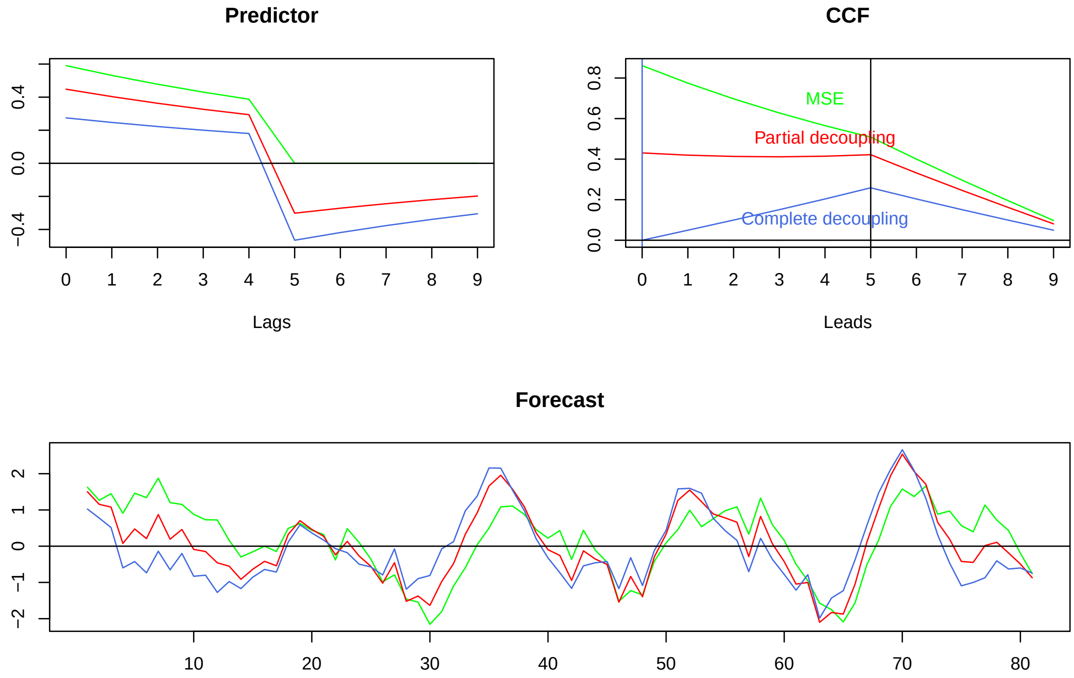

Figure 4: MSE (green) and DFP predictors under weak decoupling (red) and complete decoupling (blue). Top-left: forecast weights. Top-right: CCF across leads (the CCF under complete decoupling is zero at $\delta = 0$). Bottom: standardized forecasts. Increasing decoupling produces a leftward shift (advancement) of the DFP forecasts relative to the MSE benchmark.

5.2 Application to Real Time Business-Cycle Analysis

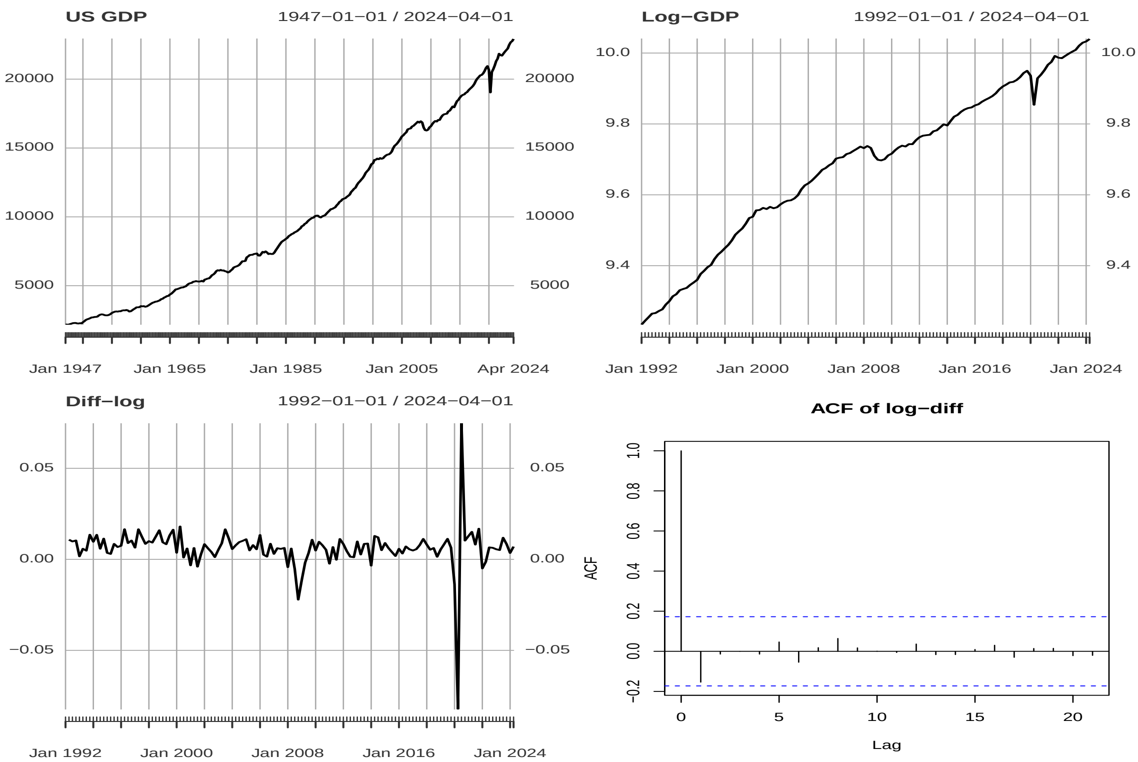

Figure 5: US GDP long time span (top left), log-GDP from 1992 to 2024 (top right), log-differences (bottom left) and ACF of log-differences (bottom right).

Business cycle analysis characterizes macroeconomic fluctuations either as deviations from a smooth potential-growth path (classical cycle) or via movements in growth rates (growth cycle), both requiring a reliable trend estimate. We extract the trend of quarterly U.S. real GDP using the Hodrick–Prescott (HP) filter (Hodrick and Prescott, 1997) with smoothing parameter $\lambda_{HP} = 1600$. The data, retrieved from FRED (https://fred.stlouisfed.org/), are displayed in Fig. 5, where log-differences (bottom left) cover the last three recession episodes from 1992-01-01 to 2024-04-01. HP is applied to log-differences to track trend-growth, whose sign indicates below/above-average growth (Wildi, 2024). While this application deviates from business-cycle orthodoxy—where the HP bandpass is typically applied to the original non-stationary GDP—we select this framework because it emphasizes lead/lag issues more clearly and because tracking growth through the lowpass addresses spurious cycles inherent to the classic approach (Wildi, 2024).

The HP filter is acausal and two-sided, making real-time application infeasible. It admits a one-sided approximation for nowcasting that tracks the two-sided design optimally under suitable assumptions (McElroy, 2008; Cornea-Madeira, 2017). Wildi (2024) develops a concurrent HP filter trading off accuracy and smoothness. We extend this by applying the DFP to induce a predictive lead relative to the MSE benchmark, comparing the MSE DFP $\mathbf{b} = \boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0$ with $\lambda = \lambda(\alpha_0)$ (classic decoupling, Eq.(7)) and the time-shift DFP with $\lambda = \lambda(\tau)$ derived directly from the lead τ (Eq. (18)).

We analyse a two-quarters ahead forecast framework, $h = 2$, at which classic GDP forecasts remain potentially informative (Heinisch et al., 2026). The nowcast and MSE(h) predictor are obtained from the causal part of the two-sided HP trend filter, truncated at leads of future data[10].

[10]The filter is also truncated at lags $k \geq L$ (nowcast) or $k \geq L - h$ (forecast), where weights become negligible

Truncation is MSE optimal under white noise, an assumption not rejected by the empirical ACF in Fig. 5. The resulting truncated nowcast substantially outperforms the classic concurrent HP in MSE, see Wildi (2024).

We derive DFP-MSE predictors with λ based on a range of α_0 and τ values, including the boundary cases of full decoupling ($\alpha_0 = 0$) and the infinity bound $\tau = \infty$. All predictors comply with strict positivity (Section 4.4) and the time-shift DFP based on $\lambda(\tau)$ preserves trend orientation and mean sign. Some MSE DFP near full decoupling allow $\lambda(\alpha_0) < \lim_{\tau\to\infty} \lambda(\tau)$: these designs push look-ahead further but at the cost of smaller target correlation $\mathrm{CCF}(h = 2)$ and trend/mean inversion, reducing interpretability.

Table 2 compares the MSE benchmark and MSE-DFP for various α_0, determined by experimentation. The top panels of Fig. 6 display the corresponding predictors and CCFs, and the top panels of Fig. 7 compare forecasts over the dotcom and financial crises[11]. As α_0 approaches zero, CCF(0) follows, confirming strong decoupling, but CCF(h) also decreases, indicating loose target tracking despite the DFP efficient frontier. This is due to trend autocorrelation linking CCF(0) to CCF(h) for small h. A negative $\Gamma(0)$ (third row) signals trend/mean inversion from excessive decoupling, leaving τ_b undefined (empty entries), though CCF(h) remains positive throughout. Small α_0 values produce strong crisis left-shifts but at the cost of evolving inversion, starting with trend/mean, reducing interpretability.

Similar results for the time-shift DFP appear in Table 3 for various leads (Lead= $\tau_b - \tau_h$), with filters, time-shifts (in $[0, \pi/5]$, covering business-cycle frequencies) and predictors shown in the bottom panels of Figs. 6 and 7. Time-shifts at frequency zero match the imposed constraints, with part of the lead spilling over to business-cycle frequencies, yielding controlled advancements during protracted crises. The infinite lead in the last column marks the asymptotic limit of trend/mean inversion (Proposition 5). Beyond this limit, the original DFP (Table 2) can further increase look-ahead while preserving a positive target correlation (Section 4.4), but at the cost of trend/mean inversion[12], reducing interpretability despite a positive CCF(h). CCF(0) (first row) remains above 0.8 in Table 3, contrasting with the more extreme designs in Table 2. The more controlled crisis left-shifts in Fig. 7 (bottom panels) suggest improved interpretability of the time-shift DFP, where $\alpha_0 = \alpha_0(\tau_b)$ is derived directly from τ_b.

While statistical evidence for classic GDP forecasts at horizons $h > 2$ may remain weak (Heinisch et al., 2026), the authors argue that forecasts may remain significant even beyond one year after lowpass filtering. Our example aligns with these findings: stronger look-ahead advances the predictor such that zero-crossings—transitions between growth and contraction—and recession dips can be anticipated relative to the MSE benchmark. The time-shift DFP enables interpretable fine-tuning by linking decoupling to lead while preserving signs and trend orientation, avoiding distortions during prolonged up- or down-cycle phases and remaining anchored at the target horizon. While stronger MSE-DFP decoupling can further improve look-ahead, fully decoupled designs ($\mathrm{CCF}(0) = 0$) are extreme and serve mainly for illustration.

for sufficiently large L.

[11] The COVID pandemic lockdown is excluded due to extreme outliers.

[12] With increasing decoupling, sign inversion starting at frequency zero ($\Gamma(0) < 0$) progressively spreads to adjacent and eventually all frequencies. While compatible with periodic processes, this may conflict with economic data, rendering extreme look-ahead designs difficult to interpret.

	MSE	DFP $\alpha_0 = 0.1$	0.05	0.02	0.017	0.005	0
CCF(0)	0.9940	0.9923	0.9701	0.8477	0.8053	0.3709	0.0000
CCF(h=2)	0.7333	0.7333	0.7266	0.6605	0.6346	0.3450	0.0804
$\Gamma(0)$	0.6179	0.5315	0.2119	0.0201	0.0009	-0.0758	-0.1078
τ	2.5858	2.9669	7.0802	72.4240	1618.0390		

Table 2: Original MSE-DFP: MSE (first column) vs. DFP. The constraint parameter α_0 controls decoupling strength: smaller α_0 yields smaller CCF(0) while minimizing spillover losses in CCF(h) (efficient frontier), with full decoupling at $\alpha_0 = 0$. Unlike unit-length DFP, α_0 is not directly interpretable due to scaling effects. $\Gamma(0) < 0$ indicates trend inversion, rendering τ undefined (empty entries). Small α_0 may produce arbitrarily large τ.

	MSE	DFP Lead = 1	2	6	∞
CCF(0)	0.994	0.989	0.984	0.962	0.803
CCF(h=2)	1.000	0.733	0.732	0.723	0.633
τ	2.586	3.586	4.586	8.586	Inf
$\alpha_0(\tau)$	0.114	0.085	0.069	0.044	0.017

Table 3: Time-shift DFP: MSE (first column) vs. DFP using the time-shift decoupling constraint. Lead= $\tau_b - \tau_h$ is the lead of DFP over MSE at frequency zero, controlling decoupling strength: larger Lead yields smaller CCF(0) and smaller $\alpha_0(\tau)$ as well as smaller TC or CCF(2) (efficient frontier).

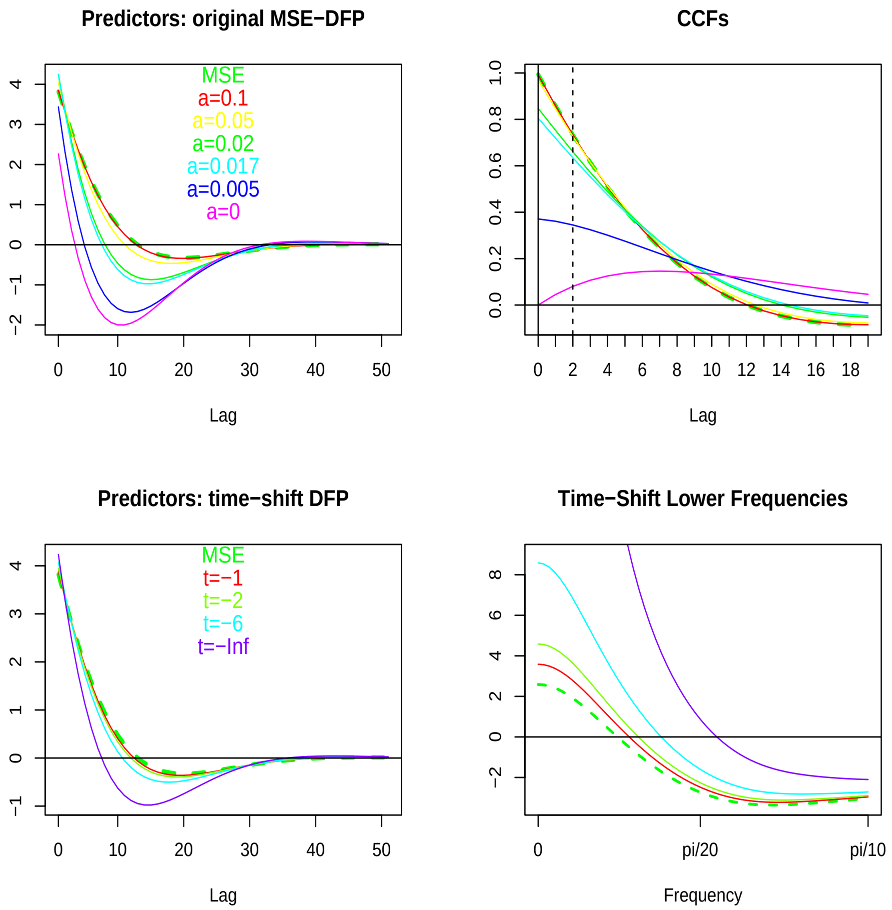

Figure 6: Top: filter coefficients (left) and CCFs (right) for the original MSE-DFP. Bottom: filter coefficients (left) and time-shifts in the interval $[0, \pi/5]$ (right) for the 'time-shift DFP'. The MSE benchmark is shown as a green bold dashed line. Predictors are normalized for visual clarity. Time-shifts at frequency zero (bottom right) match the imposed leads; CCF at lag zero (top right) reflects the decoupling imposed by α_0, with full decoupling at $\alpha_0 = 0$ (violet line).

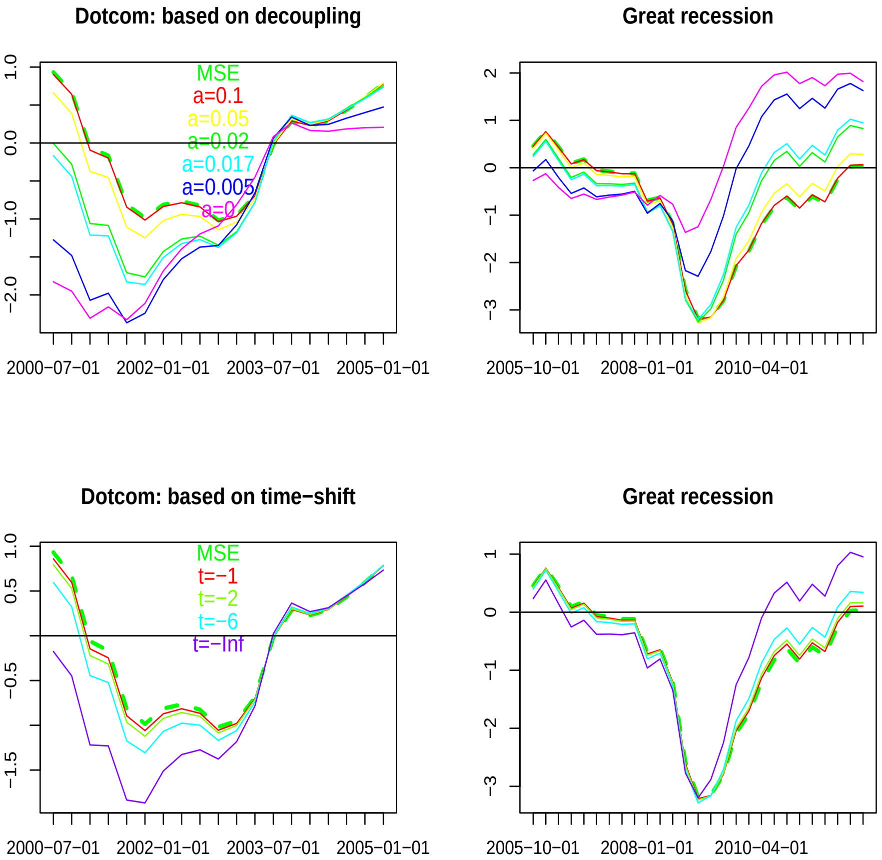

Figure 7: Predictors towards recessions. Top: original DFP; bottom: ‘time-shift DFP’; left: Dotcom crisis; right: Great Financial Crisis. The time-shift DFP avoids excessive leads by preserving trend orientation $\Gamma_b(0) > 0$, facilitating interpretability. The original DFP permits more extreme leads at the cost of trend orientation, making interpretation harder, though all designs comply with strict positivity (positive target correlation).

5.3 Time-Shift DFP vs. Max-Tau

We complete the previous business-cycle application by computing and comparing Max-Tau with time-shift DFP predictors. Max-Tau maximizes lead at a specified reference frequency ω_0; we select $\omega_0 = 0$ (trend) and $\omega_0 = \pi/20$ (business-cycle). No other linear predictor of the same length can outperform Max-Tau in lead *and* TC simultaneously at their respective frequency (efficient frontier).

Tables 4 and 5 report time-shifts of Max-Tau(0) and Max-Tau($\pi/20$), respectively, for target correlations TC (first column) matching those of the DFP-MSE from the previous section.[13] While Max-Tau maximizes shift at its specific frequency, DFP's shift consistently falls in between, suggesting averaged time-shift performance.

The predictors are compared in Fig. 8 (left), with frequency-domain time-shifts (right); forecasts over the dotcom and financial crises are compared in Fig. 9. For business-cycle applications

[13] The decoupling constraint in DFP normalizes by $1/\|\boldsymbol{\gamma}_0\|$, while the TC constraint in Max-Tau normalizes by $1/\|\boldsymbol{\gamma}_h\|$. For consistency with DFP, we normalize with the nowcast here, so TCs are directly comparable with Table 3 (second row: CCF($h = 2$)).

(duration under ten years), Max-Tau($\pi/20$) is faster at identical TC; for trend and long components (duration over twenty years), Max-Tau(0) is best.

Note that $L = 50$ is relatively large in this example, allowing for non-negligible overfitting of Max-Tau at the reference frequencies. This partly explains the worse performance of the Max-Tau(0) design, whose lead does not sufficiently spill over to business-cycle frequencies. Performance can be improved by controlling overfitting through an additional curvature constraint (Appendix C); results are shown in the accompanying R package.

TC	DFP	Max-Tau(0)	Max-Tau($\pi/20$)
0.733	3.586	6.941	2.463
0.732	4.586	11.800	2.465
0.723	8.586	39.309	3.157
0.633	Inf	Inf	21.689

Table 4: Time-shifts at $\omega_0 = 0$. Max-Tau optimized for frequency zero (middle) is largest for given target correlation (TC).

DFP	Max-Tau(0)	Max-Tau($\pi/20$)
-2.596	-2.845	-2.454
-2.388	-2.845	-2.117
-1.763	-2.779	-1.066
0.550	-5.512	4.264

Table 5: Time-shifts at $\omega_0 = \pi/20$. Max-Tau optimized for $\pi/20$ (rightmost) is largest.

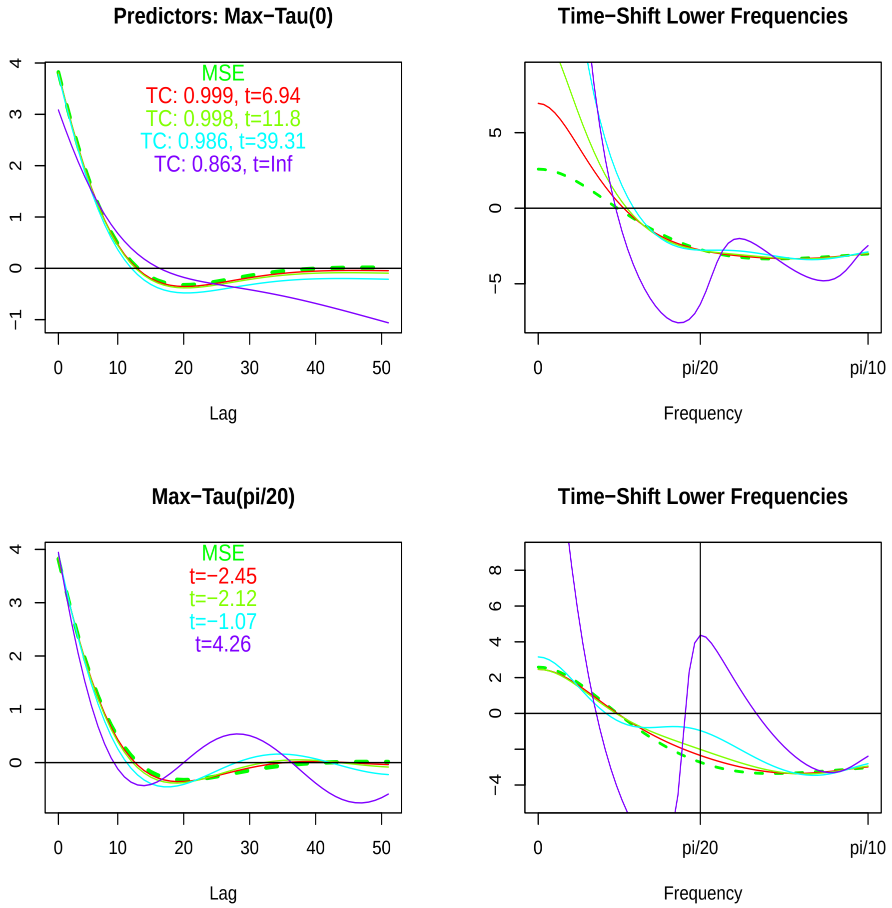

Figure 8: Filter coefficients (left) and time-shifts (right) for Max-Tau(0) (top) and Max-Tau($\pi/20$) (bottom), maximizing τ_b at $\omega_0 = 0$ and $\omega_0 = \pi/20$, respectively, for given target correlations TC matching the DFP-MSE. No other length-L linear predictor can increase τ_b at these frequencies for identical TC. As $\tau \to \infty$ in Max-Tau(0), the decoupling vectors $\mathbf{1}$ and $\mathbf{k} = (0, ..., L-1)'$ gain weight, making the profile more linear (violet line, top left). Similarly, for Max-Tau($\pi/20$), the decoupling vectors $\mathbf{cos}_{\omega_0} = (\cos(k\pi/20))_{k=0,...,L-1}$ and $\mathbf{sin}_{\omega_0} = (\sin(k\pi/20))_{k=0,...,L-1}$ gain importance, making the profile more cyclical (violet line, bottom left). The maximal lead of Max-Tau(0) at frequency zero is confined to a narrow interval and does not spill over to business-cycle frequencies (top right). In contrast, Max-Tau($\pi/20$) increasingly outpaces the MSE predictor at business-cycle frequencies $\omega > \pi/20$ (bottom right).

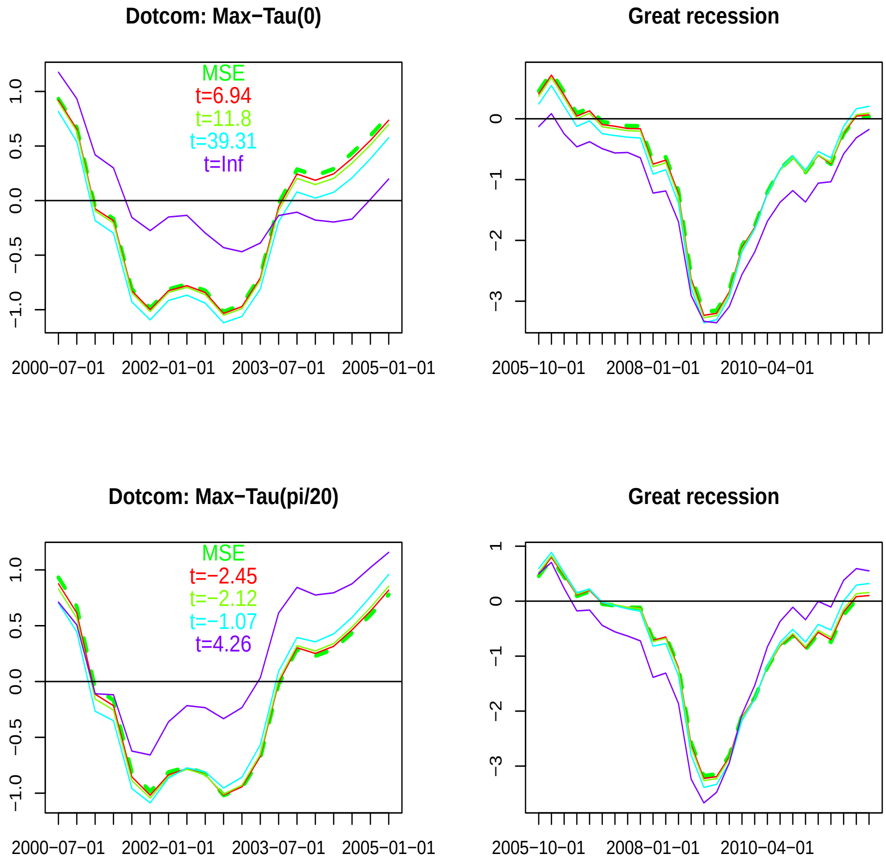

Figure 9: Predictors towards recessions. Top: Max-Tau(0); bottom: Max-Tau($\pi/20$); left: Dotcom crisis; right: Great Financial Crisis. The positive lead of Max-Tau(0) at the trend frequency does not spill over to business cycles, making it unsuitable as a leading business-cycle indicator. Max-Tau($\pi/20$), in contrast, shows leading behavior, maximized at the ten-year periodicity. Larger leads can be obtained by sacrificing TC, and alternative leading behaviour can be achieved by maximizing at other (higher) business-cycle frequencies or by controlling Max-Tau's curvature (see R-package).

5.4 Generalized DFP II and the Singular AR(1) Case

We demonstrate the generalized DFP II (Section 3.2) using an AR(1) process with $a_1 = 0.9$. The self-similarity of the AR(1) implies $\boldsymbol{\gamma}_h \propto \boldsymbol{\gamma}_0$ for $h > 0$, meaning target correlation is insensitive to the forecast horizon, violating the linear independence assumption (singular rank-one case). To re-establish independence and full-rank, we rely on the generalized Criterion (10), setting $h_1 = h = 12$. Fig. 10 displays predictor weights (top left), CCFs (top right) and forecasts (bottom) for a range of decoupling parameters α_0. Smaller α_0 (stronger decoupling) increasingly right-skews the CCF and decreases the time-shift, resulting in a left-shift of the forecasts. Fig. 11 provides two alternative time-shift measures: the classic frequency-domain measure (left) and a measure based on the $\tau(k)$-statistic (Wildi, 2024). The $\tau(k)$-statistic measures the distance between zero-crossings of DFP II and MSE when DFP is shifted by k time units: for each zero-crossing of MSE, $\tau(k)$ records the distance to the nearest DFP crossing, averaged over all crossings. Perfect synchronization gives $\tau(0) = 0$; a lag of exactly k units gives $\tau(k) = 0$. The dip—the lead or lag k

minimizing $\tau(k)$—measures the lead or lag of DFP II against MSE at zero-crossings specifically, which is useful when the peak-CCF concept, which aggregates over all (zero- *and* non-zero) levels, is uninformative due to self-similarity. Moreover, if the series corresponds to first differences of an economic time series, zero-crossings track turning points between growth and contraction, and a lead signifies earlier detection. The figure illustrates an increasing zero-crossing lead (a left-shift of the τ-dip) as decoupling strengthens, providing a complementary appraisal of the effective lead in this rank-deficient case.

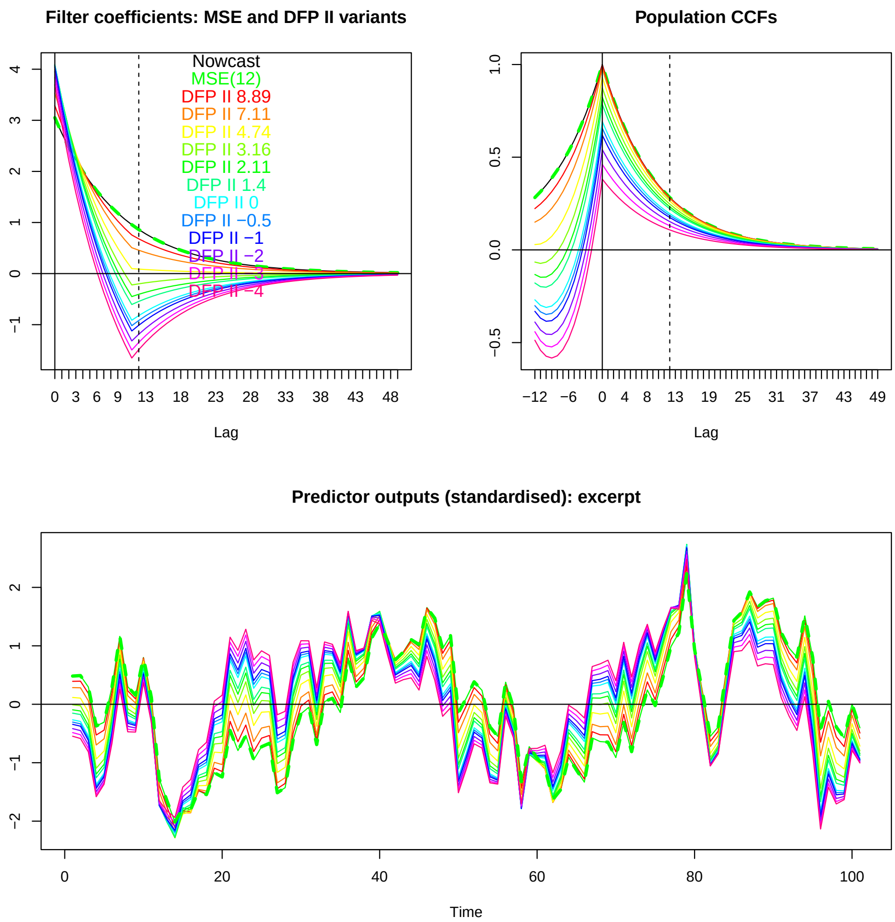

Figure 10: Generalized DFP II for the AR(1) process based on $h_1 = h = 12$. Top: predictor weights (left) and CCF (right); Bottom: predictors for a sample of the AR(1) process. Stronger decoupling enforces right-skeweness of the CCF by pulling down the CCF at negative lags. By removing positive mass from the past, this generates advancement of the predictors.

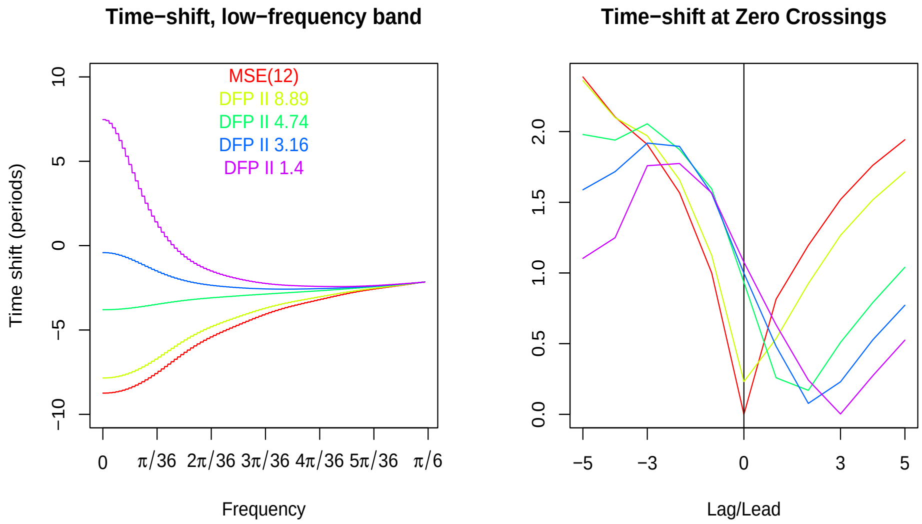

Figure 11: Generalized DFP II for the AR(1) process. Left: time-shift in $[0, \pi/6]$; right: τ-statistic. For each decoupling strength, the dip in the τ-statistic indicates the shift minimizing the distance between the DFP and the MSE benchmark at zero-crossings. The τ-statistic for the MSE (red) is centered and vanishing at zero. Dips at positive (leads) indicate a corresponding DFP lead. Increasing decoupling shifts the dip rightwards, signifying an increasing lead of DFP II over MSE at zero-crossings.

6 Conclusion

In applications, the classic h-step-ahead MSE predictor can be stuck at present, failing to lead the process irrespective of h. Decoupling the predictor from the present can unleash look-ahead behavior by instilling an effective lead. While counterintuitive—since x_T is typically highly informative—decoupling does not imply x_T is irrelevant; rather, it acknowledges that an effective lead requires a dynamic pattern distinct from the contemporaneous behavior of x_T.

Decoupling is achieved through MSE-DFP and unit-length DFP. The former yields a geometrically simpler, unique solution; the latter induces a richer geometry with two candidate solutions and greater interpretability of the constraint hyperparameter. However, MSE-DFP's more rigid sign convention is ill-suited to non-standard cases where the MSE benchmark lags the nowcast. Both approaches extend to integrated processes via I-DFP. Where the problem is infeasible due to linear dependence (rank one), the generalized DFP II re-establishes full rank via right-skewing of the CCF.

The DFP addresses the inherent accuracy–timeliness conflict as an objective–constraint optimization. A formal definition of time-shift and lead maps this dilemma to an explicit trade-off between MSE or TC and lead. Conditional on its geometric constraint—decoupling from the nowcast—the DFP traces an efficient frontier, extending the single-point MSE solution to the entire frontier; we derive an upper bound for the lead of a linear predictor in this space, under a strict positivity constraint, which the DFP approaches arbitrarily closely. The Max-Tau design resolves this geometric constraint, extending the frontier to all linear predictors by constructing optimal decoupling vectors that replace the original nowcast vector. These optimal vectors have surprisingly simple linear or sinusoidal profiles, determined by the reference frequency, with coefficients that do not decay to zero for increasing lag. In any case, decoupling can be interpreted as freeing MSE-optimal predictors from lag-inducing or lead-inhibiting components.

While Max-Tau leads DFP for given TC—sitting on the unconditional efficient frontier—DFP can nevertheless extend look-ahead behavior beyond Max-Tau's natural upper bound, at the cost of increasing sign change, beginning with trend and mean inversion. Also, in contrast to DFP, Max-Tau is subject to overfitting at the reference frequency, tightening the lead to an arbitrarily narrow interval for increasing filter length; this singularity can be addressed by extending the approach to multiple frequencies or by controlling curvature through an additional constraint, spreading lead to adjacent frequencies, and resulting in a formal trilemma between TC, lead and curvature at the optimum.

We derive the distribution of the DFP predictor, linking it to standard MSE-predictor theory through appropriate transformations, with a similar proceeding applying to Max-Tau, and illustrate applications to forecasting, signal extraction, and the singular rank-one case. The DFP package further covers leading indicator design, the non-standard case, the non-stationary I-DFP and the DFP's finite-length sampling distribution.

Our approach extends to multivariate problems, and the AT-dilemma can be combined with the AS-dilemma (Wildi, 2024–2026) to form a full AST prediction trilemma (in preparation).

A Technical Results and Proofs

A.1 Additional Propositions for Proofs of Main Results

The following proposition links the (frequency-domain) phase excess $\Phi_b(\omega) - \Phi_h(\omega)$ of DFP with the (frequency-domain) DFP triangle. This is used when linking the hyperparameter λ to the DFP lead $\tau = \tau_b - \tau_h$ in Proposition 3.

Proposition 4. *Let the baseline assumptions hold and define $\beta(\omega) := \Phi_b(\omega) - \Phi_h(\omega)$. Then, for ω in a neighborhood of zero,*

$$\beta(\omega) = \arctan\left(\frac{\sin(\gamma(\omega))}{a(\omega)/b(\omega) - \cos(\gamma(\omega))}\right) \approx \frac{\gamma(\omega)}{a(0)/b(0) - 1}, \tag{29}$$

where $\gamma(\omega) := \Phi_h(\omega) - \Phi_0(\omega)$, $b = |\lambda| A_0(\omega)$, and $a = A_h(\omega)$.

Proof: For $\omega > 0$ sufficiently small

$$\Phi_b(\omega) > \Phi_h(\omega) > \Phi_0(\omega). \tag{30}$$

The first inequality follows from the baseline assumptions $\Gamma_k(0) > 0$ (so $\Phi_k(0) = 0$) and $\tau = \tau_b - \tau_h > 0$ (lead), via a first-order Taylor expansion

$$\Phi_b(\omega) \approx \dot{\Phi}_b(0)\omega = \tau_b\omega > \tau_h\omega = \dot{\Phi}_h(0)\omega \approx \Phi_h(\omega)$$

and similarly for the second inequality, using $\tau_h > \tau_0$ (standard case). So $\Gamma_b(\omega) = \Gamma_h(\omega) + \lambda\Gamma_0(\omega)$ with $\lambda < 0$ corresponding to the time-domain DFP triangle 3 but now the fixed time-domain vectors $\mathbf{b}$, $\boldsymbol{\gamma}_0$, $\boldsymbol{\gamma}_h$ are replaced by their frequency dependent transfer-function counterparts: the triangle is frequency-dependent and the abscissa and ordinate correspond to the real and imaginary axes. We retain the naming convention from the figure with side lengths $a(\omega) = A_h(\omega)$, $b(\omega) = |\lambda| A_0(\omega)$, and $c(\omega) = A_b(\omega)$.

The strict inequality case, $\Phi_b(\omega) > \Phi_h(\omega) > \Phi_0(\omega)$, implies a non-degenerate triangle, and the angles opposite sides a, b, c are $\alpha(\omega)$ (irrelevant for our analysis), $\beta(\omega) = \Phi_b(\omega) - \Phi_h(\omega)$, and $\gamma(\omega) = \Phi_h(\omega) - \Phi_0(\omega)$. Given $a(\omega)$, $b(\omega)$, and included angle $\gamma(\omega)$,

$$\beta(\omega) = \arctan\left(\frac{\sin(\gamma(\omega))}{a(\omega)/b(\omega) - \cos(\gamma(\omega))}\right) \approx \frac{\gamma(\omega)}{a(0)/b(0) - 1}, \tag{31}$$

where the approximation uses first-order Taylor expansions and is well-defined because

$$a(0)/b(0) - 1 = \Gamma_h(0)/(|\lambda|\Gamma_0(0)) - 1 \neq 0; \tag{32}$$

otherwise $|\lambda| = \Gamma_h(0)/\Gamma_0(0)$ with $\lambda < 0$ would force $\Gamma_b(0) = 0$, contradicting the baseline assumptions. $\square$

Remarks: The frequency-domain triangle with vertices $\mathbf{0}$, $\Gamma_h(\omega)$, $\Gamma_b(\omega)$ deforms with ω, unlike the fixed time-domain triangle in Fig. 3. We therefore restrict attention to a sufficiently small neighborhood of $\omega = 0$, (avoiding $\omega = 0$ where the triangle collapses), so the ordering (30) is preserved and Γ_h lies between Γ_b and Γ_0, reflecting the geometry in Fig. 3. If $\Gamma(\omega)$ crosses a bound—either because $\omega > 0$ is too large or because baseline assumption (ii) or (iii) is violated—the triangle's orientation changes, and deriving $\beta(\omega)$ requires an alternative trigonometric identity (see Appendix E for the non-standard case $\tau_h < \tau_0$). Furthermore, for sufficiently large L, $\boldsymbol{\gamma}_h \approx \mathbf{F}^h\boldsymbol{\gamma}_0$, so $\Phi_h(\omega)$ and $A_h(\omega)$ are determined by $\boldsymbol{\gamma}_0$ up to negligible deviations, and $\beta(\omega)$ in (29) depends solely on λ and $\boldsymbol{\gamma}_0$.

The following propositions describe a limiting case when $|\tau_b|$, the time-shift of the DFP predictor, is unbounded, as relevant in Section 5.

Proposition 5. *Suppose there exists a finite-valued α such that $\lim_{\alpha_0\to\alpha} |\tau_{b(\alpha_0)}| = \infty$ for the MSE-DFP predictor $\mathbf{b}(\alpha_0)$ with constraint parameter α_0. Then $\lim_{\alpha_0\to\alpha} \Gamma_{b(\alpha_0)}(0) = 0$ and $\Gamma_{b(\alpha_0)}(0)$ changes sign exactly once, at $\alpha_0 = \alpha$.*

Proof: Finiteness of α and Proposition (2) imply that $\mathbf{b}(\alpha_0)$ remains finite. Then the first statement follows from Eq. (16), using boundedness of the numerator. The second follows from $\Gamma_b(0) = \Gamma_h(0) + \lambda\Gamma_0(0)$ and Eq. (9), which imply that $\Gamma_b(0)$ is linear in α_0. $\square$

Proposition 6. *Let $\boldsymbol{\gamma} = (\gamma_0, ..., \gamma_{L-1})'$ be such that $\Gamma(0) = \sum_{k=0}^{L-1} \gamma_k = 0$. Then either $|\tau| = \infty$ or $\boldsymbol{\gamma} = \mathbf{0}$.*

Proof: From (16), $\tau = -i\frac{\dot{\Gamma}(0)}{\Gamma(0)}$. If $\dot{\Gamma}(0) \neq 0$, then $|\tau| = \infty$ (as a limit). Otherwise, de l'Hôpital's rule yields $\tau = -i\frac{\ddot{\Gamma}(0)}{\dot{\Gamma}(0)}$. If $\ddot{\Gamma}(0) \neq 0$ then $|\tau| = \infty$. Otherwise, this is iterated at most $L-1$ times: if some higher-order derivative is nonzero, $|\tau| = \infty$; otherwise $\boldsymbol{\gamma} = \mathbf{0}$. The latter follows since the vectors $(1^j, 2^j, \ldots, (L-1)^j)$, $j = 1, \ldots, L-1$, are linearly independent, so vanishing derivatives of order 1 to $L-1$ imply $\gamma_1 = \cdots = \gamma_{L-1} = 0$, and then $\Gamma(0) = 0$ gives $\gamma_0 = 0$. $\square$

Proposition 7. *Let $\mathbf{v} = \mathbf{k} + \tau\mathbf{1}$ with $\mathbf{k} = (0, 1, ..., L-1)'$ and $\mathbf{1} = (1, ..., 1)'$ be a vector of length L of ones, with $L \geq 2$. Consider $e(\tau) := -\mathbf{v}'\mathbf{k}/\mathbf{v}'\mathbf{1}$. Then*

$$e(\tau) = -\frac{(L-1)((2L-1)+3\tau)}{3((L-1)+2\tau)}$$

The function has a pole at $\tau = -(L-1)/2$ and is monotonically increasing for $\tau > -(L-1)/2$ with $\lim_{\tau\to\infty} e(\tau) = -(L-1)/2$.

Proof: From

$$\begin{aligned}
\mathbf{v}'\mathbf{1} &= (\mathbf{k}+\tau\mathbf{1})'\mathbf{1} = \mathbf{k}'\mathbf{1} + \tau\mathbf{1}'\mathbf{1} = L\left(\frac{L-1}{2} + \tau\right) \\
\mathbf{v}'\mathbf{k} &= (\mathbf{k}+\tau\mathbf{1})'\mathbf{k} = \mathbf{k}'\mathbf{k} + \tau\mathbf{1}'\mathbf{k} = \frac{L(L-1)}{6}\Big[(2L-1) + 3\tau\Big]
\end{aligned}$$

we have

$$-\frac{\mathbf{v}'\mathbf{k}}{\mathbf{v}'\mathbf{1}} = -\frac{(L-1)\left[(2L-1)+3\tau\right]}{3\left[(L-1)+2\tau\right]}.$$

It follows that $\lim_{\tau\to\infty} e(\tau) = -(L-1)/2$. The ratio has a pole at $\tau = -(L-1)/2$. Note that

$$e(\tau) = -\frac{A\tau + B}{C\tau + D}$$

with $A = D = 3(L-1)$, $B = (L-1)(2L-1)$ and $C = 6$ is a Moebius function of τ with derivative

$$\dot{e}(\tau) = -\frac{AD - BC}{(C\tau + D)^2} = \frac{3(L^2-1)}{\big[6\tau + 3(L-1)\big]^2} > 0 \quad \text{for } \tau \neq -(L-1)/2,\ L \geq 2,$$

as claimed. $\square$

Proposition 8. *Let* $\mathbf{c} = (\cos(k\omega_0))_{k=0,\dots,L-1}$ *and* $\mathbf{s} = (\sin(k\omega_0))_{k=0,\dots,L-1}$ *be trigonometric vectors of lengh* L. *Then*

$$\frac{\mathbf{s}'\mathbf{c}}{\mathbf{c}'\mathbf{c}} = \frac{\sin(L\omega_0)\sin((L-1)\omega_0)}{L\sin(\omega_0) + \sin(L\omega_0)\cos((L-1)\omega_0)}$$

Proof:

Using the double-angle identity $\sin(2\theta) = 2\sin\theta\cos\theta$, the numerator becomes:

$$\mathbf{s}'\mathbf{c} = \frac{1}{2}\sum_{k=0}^{L-1}\sin(2k\omega_0)$$

This is a standard trigonometric series. The sum of sines of an arithmetic progression is given by the formula:

$$\sum_{k=0}^{L-1}\sin(kx) = \frac{\sin\left(\frac{Lx}{2}\right)\sin\left(\frac{(L-1)x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$$

Substituting $x = 2\omega_0$, we get:

$$\mathbf{s}^T\mathbf{c} = \frac{1}{2}\frac{\sin(L\omega_0)\sin((L-1)\omega_0)}{\sin(\omega_0)}.$$

Using the half-angle identity $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$, the denominator becomes:

$$\mathbf{c}'\mathbf{c} = \sum_{k=0}^{L-1}\frac{1+\cos(2k\omega_0)}{2} = \frac{L}{2} + \frac{1}{2}\sum_{k=0}^{L-1}\cos(2k\omega_0)$$

Using the standard formula for the sum of cosines of an arithmetic progression:

$$\sum_{k=0}^{L-1}\cos(kx) = \frac{\sin\left(\frac{Lx}{2}\right)\cos\left(\frac{(L-1)x}{2}\right)}{\sin\left(\frac{x}{2}\right)}$$

Substituting $x = 2\omega_0$, we get:

$$\mathbf{c}^T\mathbf{c} = \frac{L}{2} + \frac{1}{2}\frac{\sin(L\omega_0)\cos((L-1)\omega_0)}{\sin(\omega_0)}.$$

Combining both expressions gives

$$\frac{\mathbf{s}^T\mathbf{c}}{\mathbf{c}^T\mathbf{c}} = \frac{\frac{1}{2}\frac{\sin(L\omega_0)\sin((L-1)\omega_0)}{\sin(\omega_0)}}{\frac{L}{2} + \frac{1}{2}\frac{\sin(L\omega_0)\cos((L-1)\omega_0)}{\sin(\omega_0)}} = \frac{\sin(L\omega_0)\sin((L-1)\omega_0)}{L\sin(\omega_0) + \sin(L\omega_0)\cos((L-1)\omega_0)}$$

as claimed. $\square$

A.2 Proofs of Propositions, Corollaries and Theorems in Main Text

Proof of Proposition 1: We first assume $|\alpha_0| < 1$. Then, under criterion (2), the feasible set is the intersection of the unit sphere ($\|\mathbf{b}\| = 1$) and the circular cone defined by the decoupling constraint $\boldsymbol{\gamma}_0'\mathbf{b} = \alpha_0\|\boldsymbol{\gamma}_0\|\|\mathbf{b}\|$ (where $\|\mathbf{b}\| = 1$ is relaxed), i.e., the cone with axis $\boldsymbol{\gamma}_0$ and semi-angle $\theta_{0b} = \arccos(\alpha_0)$. Moreover, maximizing the projection $\boldsymbol{\gamma}_h'\mathbf{b}$ implies that the optimizer—i.e., the admissible unit vectors on the cone—lies in the two-dimensional subspace spanned by the axis $\boldsymbol{\gamma}_0$ of the cone and the objective direction $\boldsymbol{\gamma}_h$ (see Fig. 2). Consequently,

$$\mathbf{b} = \lambda_1\boldsymbol{\gamma}_h + \lambda_2\boldsymbol{\gamma}_0,$$

where λ_1 and λ_2 are determined by the decoupling and unit-norm constraints. Assuming $\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h \neq 0$, the decoupling constraint implies

$$\lambda_1 = \frac{\alpha_0\|\boldsymbol{\gamma}_0\| - \lambda_2\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0}{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h}. \tag{33}$$

From the unit-length constraint we deduce

$$1 = \mathbf{b}'\mathbf{b} = (\lambda_1\boldsymbol{\gamma}_h + \lambda_2\boldsymbol{\gamma}_0)'(\lambda_1\boldsymbol{\gamma}_h + \lambda_2\boldsymbol{\gamma}_0).$$

Substituting (33) into the unit-length condition and simplifying yields

$$\lambda_{2\pm} \quad = \quad \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

with $a = \boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0\left(\frac{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}{(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h)^2} - 1\right)$, $b = 2\alpha_0\|\boldsymbol{\gamma}_0\|\left(1 - \frac{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}{(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h)^2}\right)$, and $c = \frac{\alpha_0^2\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}{(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h)^2} - 1$. The correct sign maximizes the objective: for a fixed length $\|\mathbf{b}\| = 1$, it is the sign that maximizes the weight λ_1 on $\boldsymbol{\gamma}_h$. Since $a > 0$ by Cauchy-Schwarz, the minus sign shrinks λ_2, so λ_1 is larger when $\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h > 0$; otherwise, the plus sign maximizes λ_1 when $\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h < 0$, so

$$\lambda_2 \quad = \quad \frac{-b - \text{sign}(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h)\sqrt{b^2 - 4ac}}{2a},$$

as claimed.
On the other hand, if $\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_h = 0$, then the decoupling constraint implies

$$\lambda_2 = \alpha_0\frac{\|\boldsymbol{\gamma}_0\|}{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0} = \frac{\alpha_0}{\|\boldsymbol{\gamma}_0\|} \tag{34}$$

Inserted into the length constraint, we obtain

$$1 = \mathbf{b}'\mathbf{b} = (\lambda_1\boldsymbol{\gamma}_h + \lambda_2\boldsymbol{\gamma}_0)'(\lambda_1\boldsymbol{\gamma}_h + \lambda_2\boldsymbol{\gamma}_0) = \lambda_1^2\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h + \lambda_2^2\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0.$$

Inserting (34) then implies

$$\lambda_1 = \pm\sqrt{\frac{1 - \alpha_0^2}{\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h}},$$

where the plus sign maximizes the objective. Finally, if $|\alpha_0| = 1$, then the cone degenerates to a ray along $\boldsymbol{\gamma}_0$ and the decoupling constraint together with the unit-length uniquely determine

$$\mathbf{b} = \text{sign}(\alpha_0)\boldsymbol{\gamma}_0/\|\boldsymbol{\gamma}_0\|,$$

as claimed. □

Proof of Theorem 1: Assuming $d = 1$, define $e_{DFP,t} := y_{t,h,MSE} - y_t$. By the law of iterated projections, replacing $E[(\tilde{x}_{t+h} - y_t)^2]$ with $E[e^2_{DFP,t}]$ in the I-DFP objective does not alter the solution. Thus, the I-DFP filter $\mathbf{b}_{\tilde{z}}$ minimizes $E[e_{DFP,t}^2]$ subject to the decoupling constraint and the single I(1) cointegration constraint

$$\sum_{k=0}^{L-1} b_{\tilde{z}k} = \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x},h,k} = \sum_{k=-\infty}^{\infty} \gamma_k,$$

which ensures stationarity of $e_{DFP,t}$ (without the decoupling constraint, $y_t = y_{t,h,MSE}$ and $e_{DFP,t} = 0$ vanishes). Specifically:

$$e_{DFP,t} = (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\tilde{\mathbf{z}}_t = (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}'\mathbf{\Delta}'\tilde{\mathbf{z}}_t = (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}'\mathbf{z}_t, \tag{35}$$

where the final equality holds because the last entry $\sum_{k=0}^{L-1}(\hat{\gamma}_{\tilde{x},h,k} - b_{\tilde{z}k})$ of $(\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}'$ vanishes by the cointegration constraint, so the last entry $\tilde{z}_{t-(L-1)}$ of $\mathbf{\Delta}'\tilde{\mathbf{z}}_t$, which is non-stationary, may be replaced by the stationary $z_{t-(L-1)}$ without affecting the result. Hence $e_{DFP,t}$ is stationary with truncated Wold decomposition

$$(\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}'\mathbf{\Xi}'\boldsymbol{\epsilon}_t \approx (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}'\mathbf{z}_t$$

that approximates $e_{DFP,t}$ arbitrarily closely (in L^2-norm and for L sufficiently large). We then obtain:

$$\begin{aligned} e_{DFP,t} &= y_{t,h,MSE} - y_t = (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}'\mathbf{z}_t \approx (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}'\mathbf{\Xi}'\boldsymbol{\epsilon}_t \\ &= \hat{\boldsymbol{\gamma}}_{h,MSE}'\mathbf{\Sigma}'\mathbf{\Xi}'\boldsymbol{\epsilon}_t - \mathbf{b}_\epsilon'\mathbf{\Sigma}'\boldsymbol{\epsilon}_t. \end{aligned} \tag{36}$$

The last equality follows from commutativity of the convolution $\mathbf{\Sigma}\mathbf{\Xi} = \mathbf{\Xi}\mathbf{\Sigma}$. In Equation (36), $y_{t,h,MSE}$ and y_t are non-stationary cointegrated predictors of $\tilde{x}_{t+h}$, whereas the synthetic processes on the right, $\hat{\boldsymbol{\gamma}}_{h,MSE}'\mathbf{\Sigma}'\mathbf{\Xi}'\boldsymbol{\epsilon}_t$ and $\mathbf{b}_\epsilon'\mathbf{\Sigma}'\boldsymbol{\epsilon}_t$, are stationary. Crucially, their cross-sectional differences remain identical or virtually indistinguishable, if L is sufficiently large, which is key to optimization. Moreover, the finite (stationary) MA representations on the right-hand side of (36) yield valid covariance expressions, which we refer to as pseudo covariances between the original non-stationary predictors, for the objective and decoupling constraint, giving the equivalent I-DFP criteria:

$$\left.\begin{array}{cc} \begin{array}{c} \min_{\mathbf{b}_\epsilon} ||\boldsymbol{\gamma}_{\tilde{\Xi}h} - \mathbf{\Sigma}\mathbf{b}_\epsilon||^2 \\ \mathbf{b}_\epsilon'\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0} = \alpha_0 \end{array}, & \begin{array}{c} \min_{\mathbf{b}_\epsilon} ||\boldsymbol{\gamma}_{\tilde{\Xi}h} - \mathbf{\Sigma}\mathbf{b}_\epsilon||^2 \\ ||\boldsymbol{\gamma}_{\tilde{\Xi}0} - \mathbf{\Sigma}\mathbf{b}_\epsilon||^2 = \beta_0 \end{array} \\ & \\ \begin{array}{c} \max_{\mathbf{b}_\epsilon} \mathbf{b}_\epsilon'\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}h} \\ \mathbf{b}_\epsilon'\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0} = \nu_0 \\ ||\mathbf{\Sigma}\mathbf{b}_\epsilon|| = 1 \end{array}, & \begin{array}{c} \max_{\mathbf{b}_\epsilon} \mathbf{b}_\epsilon'\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}h} \\ ||\boldsymbol{\gamma}_{\tilde{\Xi}0} - \mathbf{\Sigma}\mathbf{b}_\epsilon||^2 = \mu_0 \\ ||\mathbf{\Sigma}\mathbf{b}_\epsilon|| = 1 \end{array} \end{array}\right\} \tag{37}$$

where $\tilde{\mathbf{\Xi}} := \mathbf{\Xi}\mathbf{\Sigma} = \mathbf{\Sigma}\mathbf{\Xi}$, $\boldsymbol{\gamma}_{\tilde{\Xi}\delta} := \tilde{\mathbf{\Xi}}\hat{\gamma}_{\delta,MSE}$, and $\boldsymbol{\gamma}_{\Xi\delta} := \mathbf{\Xi}\hat{\gamma}_{\delta,MSE}$, with $\delta \in \{0, h\}$. Here, $\mathbf{b}_\epsilon'\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}\delta}$ is the covariance between the finite (stationary) MA forms of the MSE nowcast ($\delta = 0$) or forecast ($\delta = h$) and the I-DFP (which we refer to as pseudo covariance between the non-stationary predictors). The additional unit-length constraint in the bottom criteria ensures proportionality of the objective and pseudo target correlation, thus generalizing the original unit-length DFP (2). Note that α_0 and β_0 represent the MSE and pseudo covariance relative to the nowcast, respectively, with generally differing numerical values (and similarly for ν_0, μ_0 under the additional unit-length constraint). The cointegration constraint is not yet explicitly enforced (see below), so the criteria remain incomplete at this stage. The Lagrangians of the four criteria yield the normal equations

$$\begin{aligned} \mathbf{\Sigma}'\mathbf{\Sigma}\mathbf{b}_\epsilon - \mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}h} - \frac{\lambda_1}{2}\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0} &= \mathbf{0} \\ (1 - \lambda_1)\mathbf{\Sigma}'\mathbf{\Sigma}\mathbf{b}_\epsilon - \mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}h} + \lambda_1\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0} &= \mathbf{0} \end{aligned}$$

for the top two criteria and

$$\begin{aligned} \frac{1}{2}\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}h} - \frac{\lambda_1}{2}\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0} - \lambda_2\mathbf{\Sigma}'\mathbf{\Sigma}\mathbf{b}_\epsilon &= \mathbf{0} \\ \frac{1}{2}\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}h} - \lambda_1\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0} - (\lambda_1 + \lambda_2)\mathbf{\Sigma}'\mathbf{\Sigma}\mathbf{b}_\epsilon &= \mathbf{0} \end{aligned}$$

for the bottom two, where we used the equivalent squared unit-length specification $||\mathbf{\Sigma b}_\epsilon||^2 = 1$. These expressions are solvable for $\mathbf{b}_\epsilon$ since $\mathbf{\Sigma}'\mathbf{\Sigma}$ has full rank. After suitable re-scaling, all four solutions take the generic form

$$\mathbf{b}_\epsilon = (\mathbf{\Sigma}'\mathbf{\Sigma})^{-1}(\tilde{\lambda}_1 \boldsymbol{\gamma}_{\tilde{\Xi}h} + \tilde{\lambda}_2 \boldsymbol{\gamma}_{\tilde{\Xi}0}).$$

The Lagrange multipliers account for differences in the decoupling constraints and the presence or absence of a (unit-) length constraint. However, all four predictors lie in $\text{span}\{\boldsymbol{\gamma}_{\tilde{\Xi}0}, \boldsymbol{\gamma}_{\tilde{\Xi}h}\}$, so for identical decoupling, i.e., angle θ_{0b} between I-DFP and nowcast, they must be identical up to rescaling. This generalizes the stationary DFP, where $\mathbf{\Sigma}$ is replaced by $\mathbf{I}$.

It remains to enforce the cointegration constraint explicitly. Since the approaches are equivalent, we consider the first (top-left) criterion in (37) and derive its closed-form solution directly, without a Lagrangian multiplier. Let $\Gamma(0) := \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x},h,k}$ (where $\Gamma(0)$ denotes the filter's transfer function at frequency zero, see McElroy and Wildi, 2016), so the I(1)-DFP cointegration constraint reads $\sum_{k=0}^{L-1} b_{\tilde{z}k} = \Gamma(0)$. In vector notation:

$$\mathbf{b}_{\tilde{z}} = \Gamma(0)\mathbf{e}_1^L + \mathbf{B}\tilde{\mathbf{b}}, \tag{38}$$

where $\mathbf{B}$ is an $L \times (L-1)$ matrix defined as $\mathbf{B} = \begin{pmatrix} -1 & -1 & -1 & \cdots & -1 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$ with the first row of -1s stacked on the $(L-1) \times (L-1)$ identity matrix, $\mathbf{e}_1^L = (1, 0, ..., 0)'$ of length L, and $\tilde{\mathbf{b}} = (\tilde{b}_1, ..., \tilde{b}_{L-1})'$ of length $L-1$. Using $\mathbf{b}_\epsilon = \mathbf{\Xi b}_{\tilde{z}}$ and inserting into the covariance decoupling constraint (top-left criterion) gives

$$(\Gamma(0)\mathbf{\Xi e}_1^L + \mathbf{\Xi B}\tilde{\mathbf{b}})'\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0} = \alpha_0. \tag{39}$$

Define $\mathbf{c} := \mathbf{B}'\mathbf{\Xi}'\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0}$ of length $L-1$ and $\tilde{\alpha}_0 := \alpha_0 - \Gamma(0)\mathbf{e}_1^L{}'\mathbf{\Xi}'\mathbf{\Sigma}'\boldsymbol{\gamma}_{\tilde{\Xi}0}$, so (39) becomes

$$\tilde{\mathbf{b}}'\mathbf{c} = \tilde{\alpha}_0. \tag{40}$$

Since $\hat{\boldsymbol{\gamma}}_{0,MSE} \neq \mathbf{0}$ (full rank assumption) we deduce $\boldsymbol{\gamma}_{\tilde{\Xi}0} \neq \mathbf{0}$ and hence $\mathbf{c} \neq \mathbf{0}$. Assuming $c_1 \neq 0$ (otherwise select k such that $c_k \neq 0$), (40) can be rewritten as

$$\tilde{\mathbf{b}} = \tilde{\mathbf{B}}\tilde{\tilde{\mathbf{b}}} + \frac{\tilde{\alpha}_0}{c_1}\mathbf{e}_1^{L-1} \tag{41}$$

where $\tilde{\mathbf{B}} = \begin{pmatrix} -c_2/c_1 & -c_3/c_1 & -c_4/c_1 & \cdots & -c_{L-1}/c_1 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$ is $(L-1) \times (L-2)$, $\tilde{\tilde{\mathbf{b}}}$ has length $L-2$, and $\mathbf{e}_1^{L-1}$ has length $L-1$. Note that $L > d+1 = 2$, by assumption, so expressions are well defined. Combining both constraints yields

$$\mathbf{b}_\epsilon = \Gamma(0)\mathbf{\Xi e}_1^L + \frac{\tilde{\alpha}_0}{c_1}\mathbf{\Xi B e}_1^{L-1} + \mathbf{\Xi B}\tilde{\mathbf{B}}\tilde{\tilde{\mathbf{b}}}. \tag{42}$$

Inserting (42) into $||\boldsymbol{\gamma}_{\tilde{\Xi}h} - \mathbf{\Sigma b}_\epsilon||^2$, the objective of the top-left criterion in (37), and setting the derivative to zero leads to

$$\tilde{\tilde{\mathbf{b}}} = \left(\tilde{\mathbf{B}}'\mathbf{B}'\mathbf{\Xi}'\mathbf{\Sigma}'\mathbf{\Sigma\Xi B}\tilde{\mathbf{B}}\right)^{-1}\tilde{\mathbf{B}}'\mathbf{B}'\mathbf{\Xi}'\mathbf{\Sigma}'\mathbf{\Sigma\Xi}\left(\hat{\gamma}_{h,MSE} - \Gamma(0)\mathbf{e}_1^L - \frac{\tilde{\alpha}_0}{c_1}\mathbf{B e}_1^{L-1}\right). \tag{43}$$

The I-DFP solution $\mathbf{b}_{\tilde{z}}$ in the case $d = 1$ is then obtained by inserting (43) into (41), and the latter into (38).
Extensions to higher orders of integration $d > 1$ are obtained similarly, replacing $\mathbf{\Sigma}$ with $\mathbf{\Sigma}^d$ and imposing additional cointegration constraints

$$\sum_{k=0}^{L-1} b_{\tilde{z}k}k^j = \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x},h,k}k^j = \sum_{k=-\infty}^{\infty} \gamma_k k^j, \quad j = 0, ..., d-1,$$

to eliminate the higher-order unit roots of $\tilde{z}_t$ in the filter error $e_{DFP,t} = y_{t,h,MSE} - y_t$. Specifically,

$$\begin{aligned} y_{t,h,MSE} - y_t &= (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\tilde{\mathbf{z}}_t = (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}^{d\ \prime}\mathbf{\Delta}^{d\ \prime}\tilde{\mathbf{z}}_t \\ &= (\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}^{d\ \prime}\mathbf{z}_t. \end{aligned} \tag{44}$$

The first $L - d$ entries of $\mathbf{\Delta}^{d\ \prime}\tilde{\mathbf{z}}_t$ are the stationary d-th order differences $z_t, z_{t-1}, ..., z_{t-(L-1-d)}$. The last d non-stationary entries are replaced by $z_{t-(L-d)}, ..., z_{t-(L-1)}$ due to the structure of $\mathbf{\Sigma}^{d\ \prime}$, whose last d columns are polynomials of order $d-1$ in the lag, so that the last d entries of $(\hat{\boldsymbol{\gamma}}_{h,MSE} - \mathbf{b}_{\tilde{z}})'\mathbf{\Sigma}^{d\ \prime}$ vanish by the cointegration constraints. The problem structure mirrors the I(1) case, with $\mathbf{\Sigma}$ replaced by $\mathbf{\Sigma}^d$ in (44). Since the processes on the right-hand side of (44) are stationary, meaningful (pseudo) covariances can be formed in the objective and decoupling constraint. The I(d)-DFP criteria follow from (37) with $\mathbf{\Sigma}^d$ replacing $\mathbf{\Sigma}$, and the richer cointegration constraints extend (38).

For illustration, consider $d = 2$, with constraints $\sum_{k=0}^{L-1} b_{\tilde{z}k} = \sum_{k=0}^{L-1} \hat{\gamma}_{\tilde{x},h,k} =: \Gamma(0)$ and $\sum_{k=1}^{L-1} kb_{\tilde{z}k} = \sum_{k=1}^{L-1} k\hat{\gamma}_{\tilde{x},h,k} =: \dot{\Gamma}(0)$, which annihilate a first-order trend polynomial in the cross-sectional difference (filter error). Solving for the first two entries $b_{\tilde{z}0}$ and $b_{\tilde{z}1}$ gives $\mathbf{b}_{\tilde{z}} = (\Gamma(0) - \dot{\Gamma}(0))\mathbf{e}_1 + \dot{\Gamma}(0)\mathbf{e}_2 + \mathbf{B}\tilde{\mathbf{b}}$, where $\tilde{\mathbf{b}}$ has length $L-2$ and $\mathbf{B}$ is:

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 & ... & L-2 \\ -2 & -3 & -4 & ... & -(L-1) \\ 1 & 0 & 0 & ... & 0 \\ 0 & 1 & 0 & ... & 0 \\ ... & & & & \\ 0 & 0 & 0 & ... & 1 \end{pmatrix}.$$

This $L \times (L-2)$ matrix, with $\mathbf{e}_1$, $\mathbf{e}_2$ the first two canonical unit vectors of length L, allows a straightforward extension of the cointegration constraint (38), the ensuing decoupling constraint (39) and the final steps leading to $\tilde{\tilde{\mathbf{b}}}$, which has now length $L - d - 1 > 0$.

Remarks: While I-DFP optimization relies on finite stationary MA inversions, the solution $\mathbf{b}_{\tilde{z}}$ is applied to the original non-stationary levels, with a seamless transition ensured by the cointegration constraints. The hyperparameter α_0 in the decoupling constraint of the top left I-DFP criterion represents the pseudo covariance between the finite-length MA-inversions of the I-DFP and the nowcast. Decoupling is obtained by setting $\alpha_0 < \boldsymbol{\gamma}'_{\tilde{\Xi}h}\boldsymbol{\gamma}_{\tilde{\Xi}0}$, the pseudo covariance of the MSE predictor and nowcast (analogously, $\beta_0 > ||\boldsymbol{\gamma}_{\tilde{\Xi}0} - \boldsymbol{\gamma}_{\tilde{\Xi}h}||^2$ generates decoupling via the mean-square decoupling constraint in (37)). The finite-length MA inversion of an I(d) process comprises a non-deterministic part, based on past innovations, and a deterministic polynomial of order $d-1$ in the (assumed to be fixed) start values $\tilde{z}_0, ..., \tilde{z}_{d-1}$, see Bell (1984).[14] The cointegration constraints cancel this polynomial in the filter error $e_{DFP,t}$, see (35) (I(1)) and (44) (I(2)), ensuring stationarity and leaving the optimization unaffected.

[14] A random-walk $\tilde{z}_t = \sum_{k=0}^{t-1} \epsilon_{t-k} + \tilde{z}_0$ has constant term $\tilde{z}_0$, and an integrated random-walk $\tilde{z}_t = \sum_{k=0}^{t-1}(k+1)\epsilon_{t-k} + \tilde{z}_0 + t(\tilde{z}_0 - \tilde{z}_{-1})$ has linear polynomial $\tilde{z}_0 + t(\tilde{z}_0 - \tilde{z}_{-1})$ in start values $\tilde{z}_0, \tilde{z}_{-1}$.

Proof of Proposition 3: Observe that

$$\begin{aligned}\tau &= \tau_b - \tau_h = \lim_{\omega\to 0}(\tau_b(\omega) - \tau_h(\omega)) = \lim_{\omega\to 0}(\Phi_b(\omega)/\omega - \Phi_h(\omega)/\omega)\\ &= \lim_{\omega\to 0}\frac{\beta(\omega)}{\omega} = \lim_{\omega\to 0}\frac{\gamma(\omega)/\omega}{A_h(0)/(|\lambda|A_0(0))-1} = \frac{\tau_h-\tau_0}{\Gamma_h(0)/(|\lambda|\Gamma_0(0))-1},\end{aligned}$$

where $\beta(\omega),\gamma(\omega)$ are as in Proposition 4. In the present setting, with $\omega\to 0$, the Taylor approximation on the right hand side of (29) is exact. The quotient is well-defined when $\Gamma_b(0) > 0$; see (32). Solving for λ gives

$$\lambda = -\frac{\tau\Gamma_h(0)}{(\tau+\tau_h-\tau_0)\Gamma_0(0)},$$

where the negative sign is implied by the lead (phase excess) baseline assumption. This expression is well defined because of the baseline assumptions and $\tau > 0$, so the denominator cannot vanish. □

Proof of Corollary 2, second claim:
At $\tau = 0$, $\lambda = 0$ and $\theta_{hb} = 0$ (no phase excess, $\mathbf{b} = \boldsymbol{\gamma}_h$). Define

$$f(\lambda) := \frac{\boldsymbol{\gamma}_h'(\boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0)}{\|\boldsymbol{\gamma}_h\|\|\boldsymbol{\gamma}_h+\lambda\boldsymbol{\gamma}_0\|}.$$

Then $f(0) = 1$ and $f(-\infty) = -\cos(\theta_{0h})$. We now show that f is strictly monotonically increasing in $\lambda < 0$. Define scalars:

$$a = \boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h, \quad b = \boldsymbol{\gamma}_h'\boldsymbol{\gamma}_0, \quad c = \boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0$$

so that:

$$f(\lambda) = \frac{a+\lambda b}{\sqrt{a}\sqrt{a+2\lambda b+\lambda^2 c}}$$

Setting $D(\lambda) = a + 2\lambda b + \lambda^2 c > 0$—the length cannot vanish by the full rank assumption— and differentiating:

$$f'(\lambda) = \frac{b\cdot D(\lambda) - (a+\lambda b)(b+\lambda c)}{\sqrt{a}\cdot D(\lambda)^{3/2}} = \frac{\lambda\,(b^2-ac)}{\sqrt{a}\cdot D(\lambda)^{3/2}}$$

By the Cauchy–Schwarz inequality and $\boldsymbol{\gamma}_h \not\propto \boldsymbol{\gamma}_0$ (full rank):

$$b^2 = (\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_0)^2 < (\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h)(\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0) = ac$$

Hence $f'(\lambda) > 0$ for $\lambda < 0$, so f is strictly increasing. Since

$$\lambda = -\frac{\tau\Gamma_h(0)}{(\tau+\tau_h-\tau_0)\Gamma_0(0)}$$

in (18) is strictly decreasing in $\tau > 0$ under the baseline assumptions, $f(\tau) := f(\lambda(\tau))$ is also strictly decreasing. Strict positivity ensures $|\theta_{hb}| < \pi/2$, and $\lambda < 0$ implies $\theta_{hb} > 0$ so $\theta_{hb}(\tau) = \arccos(f(\lambda(\tau))) \in]0, \pi/2[$ is the correct branch, strictly increasing in τ. □

Proof of Theorem 2: We split the proof into several parts.

I) Primal Problem: Max-Rho Predictor

By Proposition 7, the assumption $\tau_h > -(L-1)/2$ implies $\boldsymbol{\gamma}_h$ is linearly independent of $\mathbf{1} = (1, ..., 1)'$ and $\mathbf{k} := (0, 1, ..., L-1)'$ (full rank). We then consider the primal optimization problem:

$$\max_{\mathbf{b}} \mathbf{b}'\boldsymbol{\gamma}_h \tag{45}$$

$$\mathbf{b}'(\mathbf{k} + \tau_b\mathbf{1}) = 0 \tag{46}$$

$$\mathbf{b}'\mathbf{b} = 1,$$

where $\mathbf{k} = (0, 1, ..., L-1)'$ and $\mathbf{1} = (1, ..., 1)'$ are length-L vectors, and $\tau_b = -\mathbf{b}'\mathbf{k}/\mathbf{b}'\mathbf{1}$ is the time-shift at frequency zero. The unit-length constraint enforces proportionality $\mathbf{b}'\boldsymbol{\gamma}_h \propto \rho(x_{t+h}, y_t)$ between the linearized objective and the target correlation TC, denoted ρ. Solving via Lagrange multipliers, the stationarity condition gives

$$\boldsymbol{\gamma}_h = \lambda\mathbf{v} + \mu\mathbf{b} \tag{47}$$

for multipliers λ, μ. Ignoring the length constraint (we are mainly interested in direction) and setting $\mu = 1$, inserting (47) into (46) gives $\lambda = \frac{\boldsymbol{\gamma}_h'\mathbf{v}}{\mathbf{v}'\mathbf{v}}$, so that

$$\mathbf{b} = \boldsymbol{\gamma}_h - \frac{\boldsymbol{\gamma}_h'\mathbf{v}}{\mathbf{v}'\mathbf{v}}\mathbf{v} = \left(\mathbf{I} - \frac{\mathbf{v}\mathbf{v}'}{\mathbf{v}'\mathbf{v}}\right)\boldsymbol{\gamma}_h = \mathbf{P}_{v^\perp}\boldsymbol{\gamma}_h, \tag{48}$$

where $\mathbf{P}_{v^\perp} = \mathbf{I} - \frac{\mathbf{v}\mathbf{v}'}{\mathbf{v}'\mathbf{v}}$ is the orthogonal projection along $\mathbf{v}_\perp$. The multiplier μ allows unit-norm scaling, giving the Max-Rho predictor solving the primal problem:

$$\mathbf{b}^*(\tau) = \frac{\mathbf{p}_\perp}{\|\mathbf{p}_\perp\|}, \tag{49}$$

with $\mathbf{p}_\perp := \mathbf{P}_{v^\perp}\boldsymbol{\gamma}_h$. Note that $\mathbf{b}^{*\,\prime}\boldsymbol{\gamma}_h = \boldsymbol{\gamma}_h'\mathbf{P}_{v^\perp}\boldsymbol{\gamma}_h > 0$, since $\boldsymbol{\gamma}_h \not\propto \mathbf{v}$ by assumption: strict positivity is preserved, as in the time-shift DFP. A sign change—a negative μ, implying a negative sign on $\boldsymbol{\gamma}_h$ as in the non-standard case of the original DFP—is excluded, since by assumption $\tau_h > -(L-1)/2$, and by Proposition 7 the shift τ_v of the decoupling vector $\mathbf{v}$ satisfies $-(L-1)/2 \geq \tau_v$; hence $\tau_h > \tau_v$, establishing the standard case of the baseline assumptions. Geometrically, as illustrated in Fig.2, rotating $\mathbf{b}$ in the plane spanned by $\{\boldsymbol{\gamma}_h, \mathbf{v}\}$ away from $\boldsymbol{\gamma}_h$, in the same sense as $\boldsymbol{\gamma}_h$ from $\mathbf{v}$, increases the lead $\tau_b > \tau_h$, confirming the positive sign on $\boldsymbol{\gamma}_h$ in $\mathbf{b}^*$. In summary, unlike the classic DFP, the Max-Rho predictor $\mathbf{b}^*$ ensures strict positivity and avoids sign inversion in the non-standard case.

II) Uniqueness

From the above

$$\mathbf{b}^{*\,\prime}\boldsymbol{\gamma}_h = \mathbf{b}^{*\,\prime}\mathbf{p}_\perp$$

so that

$$\|\mathbf{p}_\perp\| = \mathbf{b}^{*\,\prime}\mathbf{p}_\perp \leq \|\mathbf{b}^*\|\,\|\mathbf{p}_\perp\| = \|\mathbf{p}_\perp\|,$$

by Cauchy–Schwarz (CS). Uniqueness at the CS bound implies uniqueness of $\mathbf{b}^*$, provided $\mathbf{p}_\perp \neq 0$, i.e., $\boldsymbol{\gamma}_h \not\propto \mathbf{v}$ (full rank condition). If instead $\mathbf{p}_\perp = \mathbf{P}_{v^\perp}\boldsymbol{\gamma}_h = 0$, then $\boldsymbol{\gamma}_h$ lies entirely in the span of $\mathbf{v}$, violating full rank. The objective $\mathbf{b}'\boldsymbol{\gamma}_h$ then vanishes for every feasible $\mathbf{b}$ (since $\mathbf{b}'\mathbf{v} = 0$ by the constraint), so every feasible $\mathbf{b}$ is optimal — uniqueness fails.

III) Max-TC $\rho(\tau)$ and Critical Points

We analyze the target correlation $\rho(\tau)$ for $\mathbf{b}^* = \mathbf{b}^*(\tau)$ as a function of the time-shift $\tau = \tau_b$. The maximized objective is

$$\rho(\tau)\|\boldsymbol{\gamma}_h\| = f(\tau) = \max_{\mathbf{b}} \mathbf{b}'\boldsymbol{\gamma}_h = \left\|\boldsymbol{\gamma}_h - \frac{(\boldsymbol{\gamma}_h'\mathbf{v})}{\mathbf{v}'\mathbf{v}}\mathbf{v}\right\| = \sqrt{\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h - \frac{(\boldsymbol{\gamma}_h'\mathbf{v})^2}{\mathbf{v}'\mathbf{v}}}.$$

The term under the root is the squared length of $\mathbf{p}_\perp$, hence positive, so the root is well-defined. Since $f(\tau) > 0$, its critical points coincide with those of $f(\tau)^2$:

$$f(\tau)^2 = |\boldsymbol{\gamma}_h|^2 - \frac{(a + \tau c)^2}{S_1 + 2\tau S_2 + \tau^2 S_0},$$

where $a = \boldsymbol{\gamma}_h'\mathbf{k}$, $c = \boldsymbol{\gamma}_h'\mathbf{1}$, $S_1 = \mathbf{k}'\mathbf{k} = (L-1)L(2L-1)/6$, $S_2 = \mathbf{k}'\mathbf{1} = (L-1)L/2$, $S_0 = \mathbf{1}'\mathbf{1} = L$. Thus,

$$f(\tau)^2 = |\boldsymbol{\gamma}_h|^2 - g(\tau), \qquad g(\tau) = \frac{(a+\tau c)^2}{Q(\tau)}, \qquad Q(\tau) = S_0\tau^2 + 2S_2\tau + S_1.$$

Now $g(\tau) = (\boldsymbol{\gamma}_h'\mathbf{v})^2/\mathbf{v}'\mathbf{v} \geq 0$ and $\lim_{\tau\to\pm\infty} g(\tau) = c^2/S_0$ (same on both sides). Setting $\dot{g}(\tau) = 0$ gives

$$2c(a+\tau c)(S_1 + 2\tau S_2 + \tau^2 S_0) - (a+\tau c)^2(2S_2 + 2\tau S_0) = 0,$$

which factors as

$$(a+\tau c)\Big[c(S_1 + 2\tau S_2 + \tau^2 S_0) - (a+\tau c)(S_2 + \tau S_0)\Big] = 0.$$

Hence the critical points are

$$\tau_1 = \frac{aS_2 - cS_1}{cS_2 - aS_0} \qquad \text{and} \qquad \tau_2 = -a/c,$$

where τ_1 arises from the bracket term (the squared term cancels, leaving $\dot{g}$ quadratic). Note $\tau_2 = -a/c$ is well-defined since $c = \boldsymbol{\gamma}_h'\mathbf{1} = \Gamma_h(0) > 0$ (baseline assumption). Since $g(\tau_2) = 0$, τ_2 minimizes $g \geq 0$ and hence maximizes f and ρ.

Given $g \to c^2/S_0$ at both infinities, two cases arise: i) $\tau_1 < \tau_2$ and ii) $\tau_1 > \tau_2$ (the degenerate case $\tau_1 = \tau_2$ is addressed below). In case i), $\rho(\tau)$ decreases strictly monotonically (s-m) from $\rho(\pm\infty) = f(\pm\infty)/\|\boldsymbol{\gamma}_h\| = \sqrt{1 - \Gamma_h(0)^2/(\|\boldsymbol{\gamma}_h\|^2 L)}$ to a minimum on $(-\infty, \tau_1)$, rises s-m to its maximum $\rho(\tau_2) = 1$ (where $\mathbf{b}^*(\tau_2) \propto \boldsymbol{\gamma}_h$), then decreases s-m again toward $\rho(\pm\infty)$ as $\tau \to +\infty$ (Fig.12, left panel, $\boldsymbol{\gamma}_h$ an AR(1) with $a_1 = 0.8$). In case ii), $\rho(\tau)$ increases s-m from $\rho(\pm\infty)$ to its maximum on $(-\infty, \tau_2)$, decreases s-m to a minimum at τ_1, then increases s-m again toward $\rho(\pm\infty)$ as $\tau \to +\infty$ (Fig.12, right panel, $\boldsymbol{\gamma}_h = \mathbf{k} = (0, ..., L-1)'$). These two cases, with their reverted graph orientations, are formalized below. Notably, the shape of $\rho(\tau)$ closely resembles the optimized target correlation in the smoothness-accuracy dilemma (efficient frontier) of the Smooth Sign Accuracy (SSA) approach in Wildi (2026a,b).

We briefly discuss the degenerate cases. i) If $c = \boldsymbol{\gamma}_h'\mathbf{1} = 0$ (violating $\Gamma_h(0) > 0$), then $\tau^* = -a/c$ is undefined (unless $a = 0$ too), and $g(\tau) = a^2/Q(\tau)$ has a single critical point at the vertex $\tau_v = -S_2/S_0$ of $Q(\tau)$ (ρ is strictly monotonic on $(-\infty, \tau_v]$ and $[\tau_v, \infty)$). ii) If $a = c = 0$ simultaneously ($\boldsymbol{\gamma}_h$ orthogonal to both $\mathbf{k}$ and $\mathbf{1}$), then $g \equiv 0$, $f(\tau) = \|\boldsymbol{\gamma}_h\|$, $\rho = 1$, and $\mathbf{b}^* = \boldsymbol{\gamma}_h$ are all independent of τ. This also covers $\tau_1 = \tau_2$, since that equality forces $a = c = 0$ (a degenerate, undefined case rather than a genuine repeated root). To see this, set

$$\tau_1 = \frac{aS_2 - cS_1}{cS_2 - aS_0} = -\frac{a}{c} = \tau_2$$

and cross-multiply:

$$c(aS_2 - cS_1) = -a(cS_2 - aS_0) \implies a^2 S_0 - 2ac\, S_2 + c^2 S_1 = 0.$$

The left side is a quadratic form $Q(a, c)$ with matrix

$$M = \begin{pmatrix} S_0 & -S_2 \\ -S_2 & S_1 \end{pmatrix}, \qquad \det M = S_0 S_1 - S_2^2 = (\mathbf{1}'\mathbf{1})(\mathbf{k}'\mathbf{k}) - (\mathbf{k}'\mathbf{1})^2.$$

By Cauchy–Schwarz, $\det M \geq 0$, with equality only if $\mathbf{k}$ and $\mathbf{1}$ are linearly dependent. Since $\mathbf{k} = (0, 1, \ldots, L-1)'$ is not proportional to $\mathbf{1}$ for $L \geq 2$, $\det M > 0$; also $S_0 = L > 0$. Hence $M \succ 0$, so

$$Q(a, c) = 0 \iff a = c = 0.$$

Thus for any $\boldsymbol{\gamma}_h$ with $(a, c) \neq (0, 0)$ (full rank assumption), $Q(a, c) > 0$ strictly, so $\tau_1 \neq \tau_2$ always.

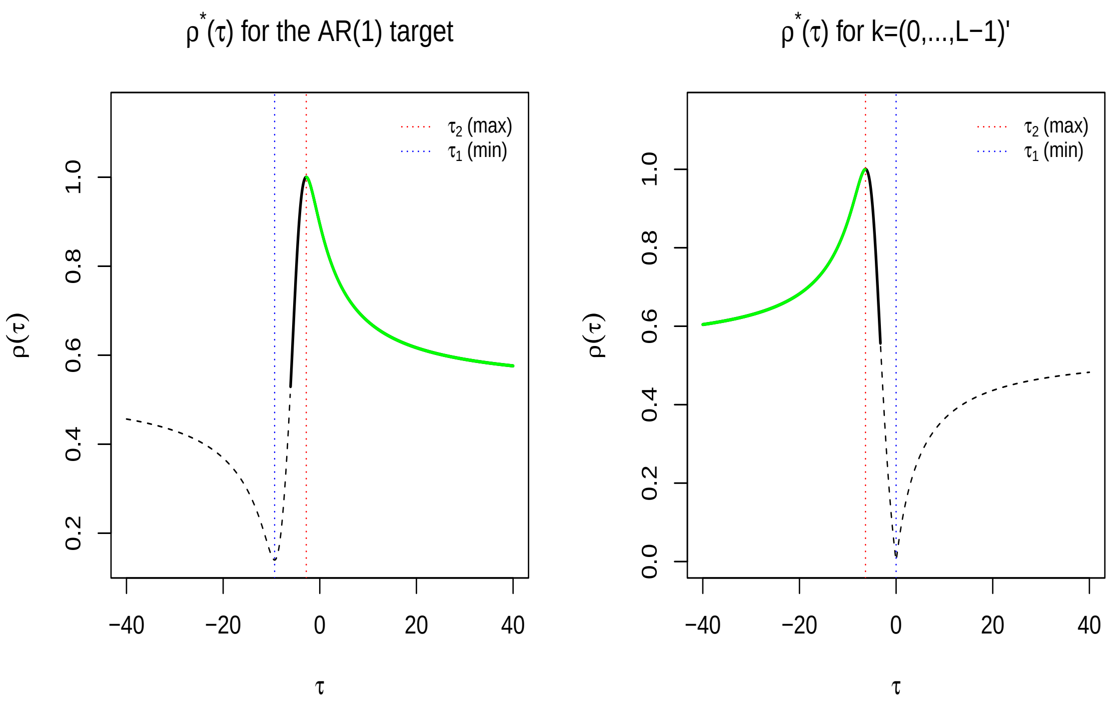

Figure 12: The maximized target correlation $\rho(\tau)$ for $\boldsymbol{\gamma}_h$ is shown for two cases (two different data generating processes): $\boldsymbol{\gamma}_h$ as an AR(1) (left; $\tau_2 > \tau_1$) and $\boldsymbol{\gamma}_h = \mathbf{k} = (0, \ldots, L-1)'$ (right; $\tau_2 < \tau_1$). The solid green and black portions indicate $\rho(\tau) > \rho(\infty)$. The green segments correspond to the efficient frontiers without time inversion (left panel) and with time inversion (right panel).

IV) Time orientation: $\tau_1 < \tau_2$ vs. $\tau_2 < \tau_1$

Fig. 12 shows an inverted orientation of $\rho(\tau)$ depending on the sign of $\tau_1 - \tau_2$. We relate this sign to the decay pattern of the coefficients of $\boldsymbol{\gamma}_h$—i.e., whether more weight falls on the remote past than on current observations, interpretable as time inversion. Starting from

$$\begin{aligned}\tau_1 - \tau_2 &= \frac{aS_2 - cS_1}{cS_2 - aS_0} + \frac{a}{c} = \frac{c(aS_2 - cS_1) + a(cS_2 - aS_0)}{c(cS_2 - aS_0)}\\ &= \frac{2acS_2 - c^2 S_1 - a^2 S_0}{c(cS_2 - aS_0)},\end{aligned}$$

we rewrite the numerator by factoring out $-c^2 S_0$:

$$\begin{aligned}2acS_2 - c^2 S_1 - a^2 S_0 &= -c^2 S_0 \left[\frac{S_1}{S_0} - 2\frac{a}{c}\frac{S_2}{S_0} + \frac{a^2}{c^2}\right]\\ &= -c^2 S_0 \left[\frac{S_1}{S_0} - \left(\frac{S_2}{S_0}\right)^2 + \left(\frac{S_2}{S_0} - \frac{a^2}{c^2}\right)^2\right]\\ &= -c^2 S_0 \left[(L^2 - 1)/12 + \left(\frac{S_2}{S_0} - \frac{a^2}{c^2}\right)^2\right],\end{aligned}$$

where the bracketed term is strictly positive for $L \geq 2$. Since $c = \boldsymbol{\gamma}_h'\mathbf{1} = \Gamma_h(0) > 0$ under the baseline assumptions, the numerator is always strictly negative, hence

$$\operatorname{sign}(\tau_1 - \tau_2) = -\operatorname{sign}\big(c(cS_2 - aS_0)\big).$$

Using $\tau_2 = -a/c$,

$$c(cS_2 - aS_0) = c(cS_2 + c\tau_2 S_0) = c^2 S_0 \left(\tau_2 + \frac{S_2}{S_0}\right) = c^2 S_0 \left(\tau_2 + \frac{L-1}{2}\right), \tag{50}$$

so

$$\operatorname{sign}(\tau_1 - \tau_2) = -\operatorname{sign}\left(\tau_2 + \frac{L-1}{2}\right) = -\operatorname{sign}\left(\tau_h + \frac{L-1}{2}\right), \tag{51}$$

since $\tau_2 = \tau_h$ because $\mathbf{b}^*(\tau_2) \propto \boldsymbol{\gamma}_h$ at the maximum τ_2. Thus $\tau_2 > \tau_1$ (left panel) versus $\tau_2 < \tau_1$ (right panel) is determined by the shift τ_h of $\boldsymbol{\gamma}_h$: by assumption $\tau_h > -(L-1)/2$, so $\tau_2 > \tau_1$ as in the left panel. This means the lag/retardation cannot exceed $(L-1)/2$, so the filter assigns more mass to earlier lags—typically the case when target coefficients decay (not necessarily monotonically) with increasing lag. Assigning more mass to the past, i.e., $\tau_h < -(L-1)/2$, would reverse the time direction, matching the right panel's orientation.

The excluded boundary case $\tau_h = -(L-1)/2$ (equivalently $\tau_2 = -(L-1)/2$) is degenerate: τ_1 diverges to $\pm\infty$ (since $cS_2 - aS_0 = 0$), and the "second-branch" structure collapses, leaving only the finite critical point τ_2 (when $\mathbf{b}^* \propto \boldsymbol{\gamma}_h$).

V) Primal Inversion on the Efficient Frontier: Max-Tau Predictor

The primal problem maximizes ρ for given τ. One may instead exchange objective and constraint, maximizing τ for given ρ, with the unit-length constraint identifying $\mathbf{b}^*$ (noting τ, ρ are both scale invariant). However, $\tau = -\mathbf{b}'\mathbf{k}/\mathbf{b}'\mathbf{1}$ is non-linear, making direct optimization tedious (see below for a formal treatment). We here instead solve by 'inverting' the primal.

For inversion consider

$$\rho(\tau)^2 \|\boldsymbol{\gamma}_h\|^2 = f(\tau)^2 = \|\boldsymbol{\gamma}_h\|^2 - \frac{(a+\tau c)^2}{S_0\tau^2 + 2S_2\tau + S_1}. \tag{52}$$

Set $f(\tau)^2 = f_0^2 = \rho_0^2 \|\boldsymbol{\gamma}_h\|^2$ and solve for $\tau = \tau(\rho_0)$ from

$$(a+\tau c)^2 = \|\boldsymbol{\gamma}_h\|^2 (1-\rho_0^2)(S_0\tau^2 + 2S_2\tau + S_1).$$

Rewriting,

$$\begin{aligned}
&A\tau^2 + B\tau + C = 0\\
&A = c^2 - \|\boldsymbol{\gamma}_h\|^2(1-\rho_0^2)S_0, \quad B = 2ac - \|\boldsymbol{\gamma}_h\|^2(1-\rho_0^2)2S_2, \quad C = a^2 - \|\boldsymbol{\gamma}_h\|^2(1-\rho_0^2)S_1\\
&\text{with roots: } \tau_i^* = \frac{-B + (-1)^i\sqrt{B^2 - 4AC}}{2A}, \quad i = 1, 2,
\end{aligned} \tag{53}$$

noting that squaring in (52) introduces no spurious roots since $\rho > 0$ under the primal specification.

A tighter bound follows from (52):

$$\lim_{\tau\to\infty} \rho_0(\tau) = \sqrt{1 - \frac{(\boldsymbol{\gamma}_h'\mathbf{1})^2}{\|\boldsymbol{\gamma}_h\|^2 \mathbf{1}'\mathbf{1}}},$$

so using $c = \boldsymbol{\gamma}_h'\mathbf{1} = \Gamma_h(0)$ and $S_0 = L$,

$$\rho_0 \in \left]U, 1\right], \qquad U = \sqrt{1 - \frac{\Gamma_h(0)^2}{\|\boldsymbol{\gamma}_h\|^2 L}} < 1, \tag{54}$$

where the upper bound follows from Cauchy–Schwarz and the full-rank assumption $\boldsymbol{\gamma}_h \not\propto \mathbf{1}$. The boundary $\rho_0 = U$ corresponds to $|\tau| = \infty$, i.e., $\Gamma_{b^*}(0) = 0$ (trend elimination), contradicting the baseline assumptions and thus excluded. The case $\rho_0 < U$ has no practical relevance: for any $\tau_0 < \infty$ a predictor exists with $\tau > \tau_0$ and $\rho_0 > U$[15], so $\rho_0 < U$ never lies on the efficient frontier. We nonetheless give an explicit solution for arbitrary ρ_0 below, including $\rho_0 < U$.

Under restriction (54), the two solutions $\tau_i^*(\rho_0)$ in (53) invert the solid (black and green) parts of the graphs in Fig. 12, with the two roots on opposite sides of the peak at τ_2. The theorem's assumptions imply $\tau_2 > \tau_1$ (left panel); we focus on $\tau_b > \tau_h = \tau_2$, i.e., a *lead* of $\mathbf{b}^* = \mathbf{b}^*(\rho_0)$ over $\boldsymbol{\gamma}_h$. Hence we restrict to the decreasing green half $\tau \geq \tau_2$, so the correct root in (53) lies on the green line, right of the peak. Since $A > 0$ for $\rho_0 \in]U, 1]$, this is the plus sign:

$$\tau^*(\rho_0) = \max(\tau_1^*(\rho_0), \tau_2^*(\rho_0)) = \frac{-B + \sqrt{B^2 - 4AC}}{2A}. \tag{55}$$

The green segment in the left panel is the efficient frontier: strict monotonicity ensures no linear predictor can increase both ρ and τ, and the resulting Max-Tau predictor lies on this frontier right of τ_2, as formalized by (55). Increasing $\rho = \rho_0$ decreases $\tau = \tau(\rho_0)$ and vice versa. Viewing $\mathbf{b}^*(\rho_0)$ as a function of ρ_0, inversion on the frontier yields the decoupling vector $\mathbf{v}^* := \mathbf{k} + \tau^*(\rho_0)\mathbf{1}$, and then the resulting Max-Tau $\mathbf{b}^*(\rho_0)$ from (48). If needed, the MSE-optimal design follows by rescaling:

$$\mathbf{b}^*_{MSE}(\rho_0) = (\mathbf{b}^*(\rho_0)'\boldsymbol{\gamma}_h)\, \mathbf{b}^*(\rho_0),$$

using $\|\mathbf{b}^*(\rho_0)\| = 1$.

Finally, an anti-symmetric efficient frontier exists in the right panel for $\tau < \tau_2$, with the green curve now monotonically increasing. There, no linear predictor achieves a larger *lag* (rather than lead) for given ρ; this corresponds to time reversion when $\tau_h \leq -(L-1)/2$, i.e., $\tau_2 < \tau_1$, so $\boldsymbol{\gamma}_h$ places more weight on the remote past. This case is irrelevant in forecast applications.

Uniqueness of the Max-Tau predictor is inherited from strict monotonicity on the frontier and uniqueness of the primal. □

Remark: When time-shift DFP reaches a lead of $\tau = \infty$, so does Max-Tau. The two generally differ, however, since they lie in different spaces (span($\boldsymbol{\gamma}_0, \boldsymbol{\gamma}_h$) vs. span($\boldsymbol{\gamma}_h, \mathbf{1}, \mathbf{k}$)), yielding two distinct predictors with the same infinite lead and target correlation. This does not contradict uniqueness of Max-Tau since we assume $\Gamma_b(0) > 0$ throughout (baseline assumption: trend preservation), giving $\tau < \infty$: for identical α_h, Max-Tau, lying on the efficient frontier, generally yields a larger (i.e., different) but finite lead $\tau(\alpha_h)$. The case $\Gamma_b(0) = 0$ is ill-defined, as the signal vanishes; thus $\tau = \infty$ is only a limiting result, and uniqueness holds under the theorem's assumptions.

Proof of Corollary 3: We consider the case with $\omega_0 > 0$, where

$$\tau_b(\omega_0) = \Phi_b(\omega_0)/\omega_0 = \arctan(\Im(\Gamma_b(\omega_0))/\Re(\Gamma_b(\omega_0)))/\omega_0.$$

The primal optimization criterion is

$$\max_{\mathbf{b}} \mathbf{b}'\boldsymbol{\gamma}_h \tag{56}$$

$$\mathbf{b}'(\mathbf{s} + d_b\mathbf{c}) = 0 \tag{57}$$

$$\mathbf{b}'\mathbf{b} = 1,$$

where $\mathbf{s} = (\sin(k\omega_0))_{k=0,\ldots,L-1}$, with $\mathbf{b}'\mathbf{s} = -\Im(\Gamma_b(\omega_0))$, and $\mathbf{c} = (\cos(k\omega_0))_{k=0,\ldots,L-1}$, with $\mathbf{b}'\mathbf{c} = \Re(\Gamma_b(\omega_0))$. Here $d_b = \Im(\Gamma_b(\omega_0))/\Re(\Gamma_b(\omega_0)) = \tan(\omega_0\tau_b(\omega_0))$ links the time-shift constraint (57) to $\tau_b(\omega_0)$.

This problem shares the formal structure of the zero-frequency case, with two modifications: i) $\mathbf{1}$ and $\mathbf{k}$ in the decoupling vector $\mathbf{v} = \mathbf{k} + \tau\mathbf{1}$ are replaced by $\mathbf{c}$ and $\mathbf{s}$ in $\tilde{\mathbf{v}} := \mathbf{s} + d_b\mathbf{c}$; and ii)

[15] For given τ_0 select $\epsilon > 0$ such that $\tau(U+\epsilon) > \tau_0$ (always possible since $\tau(U) = \infty$). Then $\rho(\tau(U+\epsilon)) = U + \epsilon > \rho_0$, as claimed.

τ_b is replaced by $d_b = d_b(\tau_b, \omega_0) = \tan(\omega_0 \tau_b(\omega_0))$. We rely on the proof of Theorem 2, noting that it applies formally to d_b, yielding a so-called Max-d predictor; we thus maximize d_b instead of τ_b. Since $\mathbf{b} = \boldsymbol{\gamma}_h$ when $d_b = d_h$ (the d-value of $\boldsymbol{\gamma}_h$), and $-P/4 < \tau_h < P/4$, by assumption, varying $d_b \in]-\infty, \infty[$ implies $\tau \in]-P/4, P/4[$ by continuity, with tan/arctan strictly monotonic on this single branch. Hence Max-d and Max-Tau predictors coincide: maximizing d_b also maximizes τ_b. In particular, $\tau_b \in]-P/4, P/4[$ ensures the predictor retains positive correlation with the identity filter when applied to a sinusoid of frequency ω_0.

The proof of Theorem 2 gives the efficient frontier between d_b and ρ for $\rho \in]U, 1]$, with $U = \sqrt{1 - \frac{\Re\left(\Gamma_h(0)\right)^2}{\|\boldsymbol{\gamma}_h\|^2 \mathbf{c}'\mathbf{c}}}$, obtained by replacing $\mathbf{1}, \mathbf{k}$ with $\mathbf{s}, \mathbf{c}$. In principle the proof is complete, but for clarity we distinguish the two cases $d_2 > d_1$ and $d_1 > d_2$, where d_1, d_2 (the minimum and maximum of the objective) correspond to τ_1, τ_2 in (51).

Consider first $d_2 > d_1$, corresponding to the left-panel orientation in Fig.12 (now as a function of d). Using (50), we obtain

$$\mathrm{sign}(d_2 - d_1) = \mathrm{sign}(c(cS_2 - aS_0)) = \mathrm{sign}(\boldsymbol{\gamma}_h'\mathbf{c}(\boldsymbol{\gamma}_h'\mathbf{c} \cdot \mathbf{s}'\mathbf{c} - \boldsymbol{\gamma}_h'\mathbf{s} \cdot \mathbf{c}'\mathbf{c})) = \mathrm{sign}(\boldsymbol{\gamma}_h'\mathbf{c} \cdot \mathbf{s}'\mathbf{c} - \boldsymbol{\gamma}_h'\mathbf{s} \cdot \mathbf{c}'\mathbf{c})$$

using $\boldsymbol{\gamma}_h'\mathbf{c} > 0$ (baseline assumption). Dividing by $\mathbf{c}'\mathbf{c} > 0$,

$$\mathrm{sign}(d_2 - d_1) = -\,\mathrm{sign}\left(\boldsymbol{\gamma}_h'\left(\mathbf{s} - \frac{\mathbf{s}'\mathbf{c}}{\mathbf{c}'\mathbf{c}}\mathbf{c}\right)\right) = -\,\mathrm{sign}\left(\boldsymbol{\gamma}_h'\mathbf{s}_{\perp\mathbf{c}}\right) \to_{L\to\infty} -\,\mathrm{sign}\left(\boldsymbol{\gamma}_h'\mathbf{s}\right),$$

where $\mathbf{s}_{\perp\mathbf{c}}$ is the projection of $\mathbf{s}$ orthogonal to $\mathbf{c}$. The limit as L increases follows from Proposition 8, which gives $\mathbf{s}'\mathbf{c} = O(1/L)$ (with strict equality at Fourier frequencies $\omega_0 = 2k\pi/L$, where $\mathbf{s}'\mathbf{c} = 0$). Thus $d_2 > d_1$ (left panel) is determined by

$$-\Im(\Gamma_h(\omega_0)) = \boldsymbol{\gamma}_h'\mathbf{s} \approx \boldsymbol{\gamma}_h'\mathbf{s}_{\perp\mathbf{c}} < 0.$$

Since $\boldsymbol{\gamma}_h'\mathbf{c} = \Re(\Gamma_h(\omega_0)) > 0$ by assumption, we conclude that $\Gamma_h(\omega_0)$ lies in the first quadrant, $-\Phi_h(\omega_0) \in]0, \pi/2[$ and $-\tau_h(\omega_0) \in]0, P/4[$, i.e., $\boldsymbol{\gamma}_h$ lags at ω_0 (the typical fore-/nowcasting case). The efficient frontier between d_b (equivalently τ_b) and ρ_b then corresponds to the green line in the left panel of Fig.12.

If instead $d_2 < d_1$, then by analogy $\Gamma_h(\omega_0)$ lies in the fourth quadrant, $-\Phi_h(\omega_0) \in]-\pi/2, 0[$ and $\tau_h(\omega_0) \in]0, P/4[$, i.e., $\boldsymbol{\gamma}_h$ leads at ω_0. The efficient frontier here corresponds to the solid black portion (between the solid green and the shaded black regions) in the right panel of Fig.12, i.e., $\tau_b \in [\tau_h, P/4[$. Values of d_b beyond this portion (shaded area) cross the singularity at $\tau_b = P/4$ and are discarded.

As with the case $\omega_0 = 0$ (where $\tau_1 = \tau_2$ is excluded), $d_1 = d_2$ is excluded when $\omega_0 > 0$: $\boldsymbol{\gamma}_h'\mathbf{s}_{\perp\mathbf{c}} = 0$ would violate the full rank assumption.

These findings show that for $\omega_0 > 0$ (finite wavelength $P < \infty$), with τ linked indirectly and monotonically to the hyperparameter d_b in the constraint, both panels of the figure become relevant to the efficient frontier, depending on whether $\boldsymbol{\gamma}_h$ lags (left panel) or leads (right panel) at ω_0.

Finally, as $\omega_0 \to 0$, $\mathbf{c} = (\cos(k\omega_0))_{k=0,\ldots,L-1} \to \mathbf{1}$ and $\mathbf{s}/\omega_0 = (\sin(k\omega_0)/\omega_0)_{k=0,\ldots,L-1} \to \mathbf{k}$, recovering the original formulation with decoupling vector $\mathbf{v} = \mathbf{k} + \tau\mathbf{1}$ as the limiting case $\omega_0 = 0$. □

Proof of Corollary 4: This follows directly from Corollary 3, noting that the shift of Γ_b/Γ_h addressed by the constraint is the excess time-shift $\tau(\omega_0) = \tau_b(\omega_0) - \tau_h(\omega_0)$. In particular, the maximum $\rho(\tau) = 1$ of ρ is obtained at $\tau = \tau_2 = 0$ when $\mathbf{b} = \boldsymbol{\gamma}_h$, so $\tau \in]0, P/4[$ on the frontier and therefore $\tau_b = \tau + \tau_h \in]\tau_h, \tau_h + P/4[$, as claimed. □

As $L \to \infty$, $U \to 1$, so the efficient frontier 'shrinks' with increasing L: larger L provides more degrees of freedom, letting the maximization of the time-shift confine to an arbitrarily narrow band around ω_0 (overfitting). A possible fix is to limit adaptivity by controlling the curvature at the reference frequency, see Appendix C. Instead, we here discuss an extension of Max-Tau to multiple frequencies. Let $\omega_i \in \Omega, i = 1, ..., |\Omega|$ be a set of reference frequencies and $\mathbf{s}_i = (\sin(k\omega_i))_{k=0,...,L-1}$, $\mathbf{c}_i = (\cos(k\omega_i))_{k=0,...,L-1}$ the corresponding trigonometric support vectors of length L. Let

$$\mathbf{S} := \sum_{i=1}^{|\Omega|} w_i \mathbf{s}_i \quad , \quad \mathbf{C} := \sum_{i=1}^{|\Omega|} w_i \mathbf{c}_i,$$

where $w_i \geq 0$ are weights. We can now insert $\mathbf{S}, \mathbf{C}$ instead of $\mathbf{s}, \mathbf{c}$ in the time-shift constraint (57), to address a weighted lead of the Max-Tau predictor over $\omega_i \in \Omega$, without affecting the derivation or proof of Max-Tau. Specifically, each individual summand $w_i\mathbf{b}'(\mathbf{s}_i+d_b\mathbf{c}_i)$ of $\mathbf{b}'(\mathbf{S}+d_b\mathbf{C})$ addresses the lead τ_i at ω_i through $d_b = \tan(\tau_i\omega_i)$, and the weights w_i assign more or less importance to ω_i when maximizing the common d_b through inversion on the efficient frontier. While a common d_b may seem unnecessarily rigid, this approach extends the previous results straightforwardly, and its simplicity also guards against overfitting—the main motivation for this extension (a version with frequency-dependent d_{bi} lacks a closed-form solution, mirroring a similar shortcoming in the DFA trilemma of McElroy and Wildi, 2019). In applications we typically set $w_i \equiv 1$, though allowing arbitrary weights alleviates some of the aforementioned rigidity. An alternative to equal-weighting is proposed in McElroy and Wildi (2019), where w_i account for the amplitudes of target and predictor, as well as for the spectrum of the process at ω_i, though their frequency-domain approach does not allow for a simple, numerically tractable, closed-form solution; nor do the authors explicitly aim at maximizing the lead, as here for Max-Tau.

B Max-Tau 'Dual' Problem

Having explored inversion of the primal problem, we analyze an alternative route to the Max-Tau predictor based on the *dual*, where the objective and the time-shift constraint are swapped. We first assume $\omega_0 = 0$ (relaxed below):

$$\begin{aligned}
&\max \tau_b \\
&\mathbf{b}'\boldsymbol{\gamma}_h = \alpha_h \\
&\mathbf{b}'\mathbf{b} = 1.
\end{aligned}$$

Directly maximizing τ_b, rather than inverting the primal as before, offers further insight into the 'Max-Tau' prediction problem. Unfortunately, the objective $\tau_b = -\mathbf{b}'\mathbf{k}/\mathbf{b}'\mathbf{1}$ is non-linear in $\mathbf{b}$. We 'linearize' the criterion by introducing an additional constraint:

$$\begin{aligned}
&\min \mathbf{b}'\mathbf{k}/f \\
&\mathbf{b}'\boldsymbol{\gamma}_h = \alpha_h \\
&\mathbf{b}'\mathbf{1} = f \\
&\mathbf{b}'\mathbf{b} = 1,
\end{aligned}$$

where the minimization accounts for the sign change, and $f > 0$ is fixed for now (its value is optimized later, so the scaling $1/f$ must be retained in the objective). For brevity, the two linear constraints are grouped as $\mathbf{C}'\mathbf{b} = \mathbf{d}$, with $\mathbf{C} = (\boldsymbol{\gamma}_h, \mathbf{1})$ and $\mathbf{d} = (\alpha_h, f)'$. The Lagrangian is

$$\mathcal{L}(\mathbf{b}, \boldsymbol{\lambda}, \mu) = \frac{1}{f}\mathbf{b}'\mathbf{k} + \boldsymbol{\lambda}'(\mathbf{C}'\mathbf{b} - \mathbf{d}) + \frac{\mu}{2}(\mathbf{b}'\mathbf{b} - 1),$$

where $\boldsymbol{\lambda} = (\lambda_1, \lambda_2)'$ and μ are the multipliers for the linear and quadratic constraints, respectively. The normal equations are:

$$\nabla_{\mathbf{b}}\mathcal{L} = \frac{1}{f}\mathbf{k} + \mathbf{C}\boldsymbol{\lambda} + \mu\mathbf{b} = \mathbf{0},$$

giving

$$\mathbf{b} \;\; = \;\; -\frac{1}{\mu}\left(\frac{1}{f}\mathbf{k} + \mathbf{C}\boldsymbol{\lambda}\right) \quad . \tag{58}$$

Substituting (58) into $\mathbf{C}'\mathbf{b} = \mathbf{d}$:

$$\mathbf{C}'\left[-\frac{1}{\mu}\left(\frac{1}{f}\mathbf{k} + \mathbf{C}\boldsymbol{\lambda}\right)\right] = \mathbf{d}$$

$$-\frac{1}{\mu f}\mathbf{C}'\mathbf{k} - \frac{1}{\mu}\mathbf{C}'\mathbf{C}\boldsymbol{\lambda} = \mathbf{d}.$$

Multiplying by $-\mu$ and solving for $\boldsymbol{\lambda}$:

$$\mathbf{C}'\mathbf{C}\boldsymbol{\lambda} = -\mu\mathbf{d} - \frac{1}{f}\mathbf{C}'\mathbf{k}$$

$$\boldsymbol{\lambda} = -(\mathbf{C}'\mathbf{C})^{-1}\left(\mu\mathbf{d} + \frac{1}{f}\mathbf{C}'\mathbf{k}\right). \tag{59}$$

Note that $\mathbf{C}'\mathbf{C} = \begin{pmatrix} \boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h & \boldsymbol{\gamma}_h'\mathbf{1} \\ \boldsymbol{\gamma}_h'\mathbf{1} & \mathbf{1}'\mathbf{1} \end{pmatrix}$ has full rank, since

$$\det(\mathbf{C}'\mathbf{C}) = (\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h)(\mathbf{1}'\mathbf{1}) - (\boldsymbol{\gamma}_h'\mathbf{1})^2 > 0$$

whenever $\boldsymbol{\gamma}_h \not\propto \mathbf{1}$, as assumed. Substituting back into (58):

$$\mathbf{b} = -\frac{1}{\mu f}\mathbf{k} + \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{d} + \frac{1}{\mu f}\mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}'\mathbf{k}.$$

Introducing the least-squares solution $\mathbf{b}_{LS}$ and the null-space projection matrix $\mathbf{P}_C^{\perp}$: $\mathbf{b}_{LS} = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{d}$ (depending on f through $\mathbf{d}$), $\mathbf{P}_C^{\perp} = \mathbf{I} - \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}'$, we group the $\mathbf{k}$-terms:

$$\mathbf{b} = \mathbf{b}_{LS} - \frac{1}{\mu f}\mathbf{P}_C^{\perp}\mathbf{k}. \tag{60}$$

Enforcing the quadratic constraint: since $\mathbf{b}_{LS}$ and $\mathbf{P}_C^{\perp}\mathbf{k}$ are orthogonal, the squared norm of $\mathbf{b}$ gives

$$\mathbf{b}'\mathbf{b} = |\mathbf{b}_{LS}|^2 + \left(\frac{1}{\mu f}\right)^2 |\mathbf{P}_C^{\perp}\mathbf{k}|^2 = 1,$$

so

$$\frac{1}{\mu f} = \pm\sqrt{\frac{1 - |\mathbf{b}_{LS}|^2}{|\mathbf{P}_C^{\perp}\mathbf{k}|^2}}.$$

A real solution requires $|\mathbf{b}_{LS}|^2 < 1$ (otherwise the constraints do not intersect), setting a lower bound on f for given α_h. Decomposing $\mathbf{b}_{LS} = \mathbf{v}f + \mathbf{u}$ with $\mathbf{u} = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}(\alpha_h, 0)'$ and $\mathbf{v} = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{e}_2$, with $\mathbf{e}_2 = [0, 1]'$, the lower bound for admissible f satisfies

$$|\mathbf{b}_{LS}|^2 - 1 = \mathbf{v}'\mathbf{v}f^2 + 2\mathbf{v}'\mathbf{u}f + \mathbf{u}'\mathbf{u} - 1 = 0.$$

Note that $\mathbf{v} \neq 0$, since otherwise

$$\mathbf{v} = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{e}_2 = \mathbf{0} \implies \mathbf{0} = \mathbf{C}'\mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{e}_2 = \mathbf{e}_2$$

contradicts $\mathbf{e}_2 = (0, 1)'$, using the invertibility of $\mathbf{C}'\mathbf{C}$.
Assuming a solution exists, i.e., f exceeds the lower bound, the linearized dual criterion yields

$$\mathbf{b}(f) = \mathbf{b}_{LS} - \nu P_C^{\perp}\mathbf{k}.$$

We now determine the optimal f maximizing the time-shift. For given f, the minimum objective achieved by $\mathbf{b}(f)$ is:

$$V(f) = \frac{\mathbf{k}'\mathbf{u}}{f} + \mathbf{k}'\mathbf{v} - \frac{\sqrt{K_\perp}}{f}\sqrt{1 - |\mathbf{u}|^2 - 2f(\mathbf{u}'\mathbf{v}) - f^2|\mathbf{v}|^2}.$$

Setting $x = 1/f$ $(x > 0)$ and moving $1/f$ inside the square root as x^2, the function simplifies to

$$W(x) = (\mathbf{k}'\mathbf{u})x + \mathbf{k}'\mathbf{v} - \sqrt{K_\perp}\sqrt{(1 - |\mathbf{u}|^2)x^2 - 2(\mathbf{u}'\mathbf{v})x - |\mathbf{v}|^2}.$$

Defining scalar constants $D = \mathbf{k}'\mathbf{u}$, $A = 1 - |\mathbf{u}|^2$, $B = -2\mathbf{u}'\mathbf{v}$, $E = -|\mathbf{v}|^2$, the function to minimize becomes

$$W(x) = Dx + \mathbf{k}'\mathbf{v} - \sqrt{K_\perp}\sqrt{Ax^2 + Bx + E}.$$

The feasible region is $F := \{x|Ax^2 + Bx + E \geq 0\}$, representing the non-empty intersection of the constraints. We now assume $A \neq 0$. Setting the derivative to zero:

$$\frac{dW}{dx} = D - \sqrt{K_\perp}\frac{2Ax + B}{2\sqrt{Ax^2 + Bx + E}} = 0. \tag{61}$$

The second derivative is

$$\frac{d^2W}{d^2x} = -\sqrt{K_\perp}\frac{4A(Ax^2 + Bx + E) + (2Ax + B)^2}{4(Ax^2 + Bx + E)^{3/2}} = \sqrt{K_\perp}\frac{B^2 - 4AE}{4(Ax^2 + Bx + E)^{3/2}}.$$

We now show $B^2 - 4AE \geq 0$. If instead $B^2 - 4AE < 0$, then $M(x) := Ax^2 + Bx + E$ has no zero, so $M(x)$ is either negative everywhere or positive everywhere. Since $\mathbf{v} \neq \mathbf{0}$ and $E = -\|\mathbf{v}\|^2$, we have $M(0) < 0$, hence $M(x) < 0$ everywhere—contradicting the existence of a feasible region. Note that $B^2 - 4AE > 0$ implies the feasible region is a proper interval, versus a single point when $B^2 - 4AE = 0$. In the latter case, $x_{min} := -B/(2A)$ solves the minimization problem. Otherwise, if $B^2 - 4AC > 0$, $W(x)$ is convex (strictly positive second derivative), and the feasible region F is the interval $[z_1, z_2]$, where z_1, z_2 are the roots of $M(x)$. We now seek the minimum of $W(x)$ for $x \in [z_1, z_2]$, first deriving the roots of (61) by rearranging terms:

$$2D\sqrt{Ax^2 + Bx + E} = \sqrt{K_\perp}(2Ax + B).$$

Squaring both sides:

$$4D^2(Ax^2 + Bx + E) = K_\perp(4A^2x^2 + 4ABx + B^2).$$

Since squaring discards the sign, we must select the root satisfying sign(D) =sign$(2Ax + B)$. Grouping by powers of x:

$$4[A(D^2 - K_\perp A)]x^2 + [4B(D^2 - K_\perp A)]x + [4D^2E - K_\perp B^2] = 0,$$

with coefficients $a = 4[A(D^2 - K_\perp A)]$, $b = [4B(D^2 - K_\perp A)]$, $c = [4D^2E - K_\perp B^2]$, giving

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

provided $b^2 - 4ac \geq 0$. We select the root x_1 satisfying sign(D) =sign$(2Ax_1 + B)$. If $x_1 \in [z_1, z_2]$, $W(x_1)$ is a minimum by convexity, so $x_{min} := x_1$. Otherwise (i.e., $b^2 - 4ac < 0$ or $x_1 \notin [z_1, z_2]$), the minimum lies at a boundary of F. Specifically, if $\frac{dW}{dx} < 0$ at an interior point $x \in]z_1, z_2[$, the derivative is negative throughout F, so $x_{min} = \max(z_1, z_2)$; otherwise $x_{min} = \min(z_1, z_2)$. In conclusion, linearizing the dual through an additional constraint yields a stacked optimization problem: the first level specifies feasibility, i.e., F, and the second determines the solution's location within F.

We now analyze the case $A = 0$, so $M(x) = Bx + E$. We first show $B > 0$ under the posited assumptions. From above,

$$(\mathbf{C}'\mathbf{C})^{-1} = d\begin{pmatrix} \mathbf{1}'\mathbf{1} & -\boldsymbol{\gamma}_h'\mathbf{1} \\ -\boldsymbol{\gamma}_h'\mathbf{1} & \boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h \end{pmatrix},$$

where the inverse of the determinant $d = \frac{1}{\mathbf{1}'\mathbf{1}\cdot\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h - (\boldsymbol{\gamma}_h'\mathbf{1})^2} > 0$ by Cauchy-Schwarz and the full-rank assumption. It follows that

$$\begin{aligned} B &= -2\mathbf{u}'\mathbf{v} = -2(\alpha_h, 0)(\mathbf{C}'\mathbf{C})^{-1}\mathbf{e}_2 \\ &= 2d\alpha_h\boldsymbol{\gamma}_h'\mathbf{1} > 0 \end{aligned}$$

since $\alpha_h > 0$ under strict positivity. So $M(x)$ degenerates to the affine function $M(x) = Bx + E$ with $B > 0$, $E < 0$, and $F = [-E/B, \infty[$, the feasible region. The critical points of $W(x)$ now follow from

$$\frac{dW}{dx} = D - \sqrt{K_\perp}\frac{B}{2\sqrt{Bx + E}} = 0, \tag{62}$$

with second derivative

$$\frac{d^2W}{d^2x} = \sqrt{K_\perp}\frac{B^2}{4(Bx + E)^{3/2}} > 0.$$

We proceed as above to find the critical point x_1: existence requires sign(D) =sign(B), i.e., positive. If x_1 exists and $x_1 \in F$, it is a minimum by convexity, so $x_{min} = x_1$. Otherwise, if (62) has no root or the root lies outside F, then $x_{min} = -E/B$ or $x_{min} = \infty$, depending on the sign of dW/dx in F.
In either case, we set $f^* := 1/x_{min} \geq 0$ (noting $x_{min} > 0$ within the feasible region, ensuring finite unit-length of the predictor), and the time-shift-maximizing Max-Tau predictor is $\mathbf{b}(f^*)$ in (60).
As shown in the primal inversion, Theorem 2, the efficient frontier corresponds to $\rho \in]U, 1]$ with $U = \sqrt{1 - \frac{\Gamma_h(0)^2}{\|\boldsymbol{\gamma}_h\|^2 L}}$, so that $\tau_b > \tau_h$ (a lead). On the frontier, the dual also solves the inverted primal specification (identical length and target correlation through the constraints, and maximized time-shift); uniqueness therefore follows from uniqueness of the inverted primal. We can also derive $\tau = \tau(\rho)$ on the frontier, following the inverted primal formulation (55)—see Fig.13.

The above findings can be extended to $\omega_0 > 0$, noting that the max-shift is bounded from above by $\tau \leq P/4$, where P is the full-wave length at ω_0 (not shown).

Geometric interpretation of f: the dual Max-Tau lies on the intersection of the unit sphere and two cones with axes $\boldsymbol{\gamma}_h$ and $\mathbf{1}$, and angles given by the correlations $f/\sqrt{L} = \cos(\theta_f)$, where $\sqrt{L} = \sqrt{\mathbf{1}'\mathbf{1}}$, and $\alpha_h/\|\boldsymbol{\gamma}_h\| = \cos(\theta_h)$. Feasibility requires $|\alpha_h|/\|\boldsymbol{\gamma}_h\| \leq 1$ and $|f|/\sqrt{L} \leq 1$ (giving correlations); the latter is generally not an issue, since we minimize $f \geq 0$. The constraints are feasible if the cones intersect at the unit sphere, i.e., $\theta_f + \theta_h \geq \theta_{1h}$, where θ_{1h} is the angle between the two fixed axes. For optimization, we want $f > 0$ minimal, so $\theta_f + \theta_h = \theta_{1h}$, giving $f = \sqrt{L}\cos(\theta_f) = \sqrt{L}\cos(\theta_{1h} - \theta_h)$. If $\theta_h > \theta_{1h}$ any $f \geq 0$ is admissible, so we set $f = 0$.

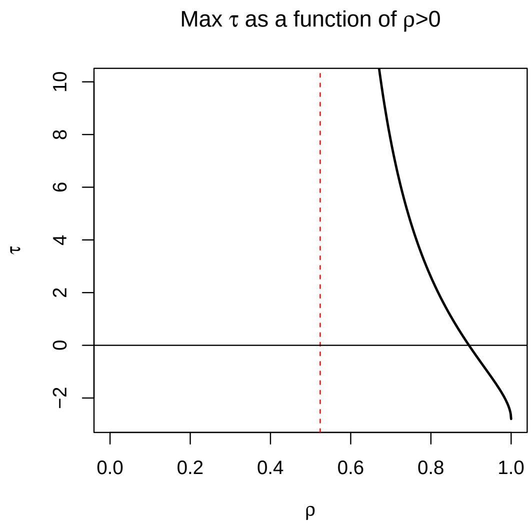

Figure 13: Max-Tau predictor: the maximized time-shift $\tau(\rho)$ at the reference frequency $\omega_0 = 0$ for $\boldsymbol{\gamma}_h$ is shown for a truncated AR(1) of length $L = 10$ and $a_1 = 0.8$. The curve is truncated at $\tau = 10$; the asymptote $\tau = \infty$ is indicated by the red line, corresponding to $\rho = \sqrt{1 - (\boldsymbol{\gamma}_h'\mathbf{1})^2/(L\boldsymbol{\gamma}_h'\boldsymbol{\gamma}_h)} \approx 0.52$. At $\rho = 1$, $\tau = -\boldsymbol{\gamma}_h\mathbf{k}/\boldsymbol{\gamma}_h\mathbf{1} \approx -2.8$, the shift (lag) of $\boldsymbol{\gamma}_h$. For $\rho < 0.894$, $\tau(\rho) > 0$, indicating a lead at the reference frequency. The shown portion of the curve corresponds to the inversion of the efficient frontier in the inverted primal solution. To the left of the asymptote, i.e., for smaller ρ, $\tau = \infty$—unlike the inverted primal, whose $\tau < \infty$ remains finite away from the asymptote. Dual and inverted primal match only on the frontier, for $\rho \in [0.52, 1]$ in this example.

We briefly discuss the dual Max-Tau predictor. For $\rho < U$ (away from the efficient frontier), the inverted primal and dual Max-Tau differ, since the former has a finite shift—the black shaded portion to the right of the minimum in Fig.12—while the latter has an infinite shift. This would infringe the trend preservation rule (baseline assumption) $\Gamma_b(0) > 0$[16], so an additional constraint $\Gamma_b(0) = \nu > 0$ must be imposed in the dual formulation when $\rho < U$, yielding $\tau < \infty$ and trend preservation. In any case, dual and inverted primal generally differ away from the frontier, $\rho < U$.

As discussed in the inverted primal approach, Max-Tau overfits when $L \to \infty$. One remedy, already used in the primal, is to extend the optimization to multiple reference frequencies $\omega_i \in \Omega$, with $1 < |\Omega| < \infty$. Specifically, we add $|\Omega|$ time-shift constraints $\mathbf{b}'\mathbf{c}_i = f$, with $\mathbf{c}_i = (\cos(k\omega_i))_{k=0,\ldots,L-1}$, and append the aggregate $\sum_{i=1}^{|\Omega|} w_i \mathbf{b}'\mathbf{s}_i/f$ to the objective, where $w_i > 0$ are weights (typically $w_i \equiv 1$ in applications) and $\mathbf{s}_i = (\sin(k\omega_i))_{k=0,\ldots,L-1}$. The purely periodic $\mathbf{s}, \mathbf{c}$ decoupling vectors are relevant when $\omega_i > 0$, see Corollary 3.

A similar stacked (two-step) optimization applies here, with obvious modifications (not shown). Note that overfitting may still occur if $|\Omega| < \infty$ as $L \to \infty$; thus either the reference frequency set must grow unboundedly too, exhausting degrees of freedom, or Max-Tau's flexibility must be restricted differently, see Appendix C.

The common f across constraints, while partially guarding against overfitting, may seem overly restrictive; the problem can be generalized to frequency-dependent f_i via $\mathbf{b}'\mathbf{c}_i = f_i$ and $\sum_{i=1}^{|\Omega|} w_i \mathbf{b}'\mathbf{s}_i/f_i$, though it then generally lacks a simple closed form.

Alternatively, we present a closed-form solution based on controlling curvature at the (single) reference frequency ω_0 (extensible to multiple frequencies).

[16] As $\tau \to \infty$, $\mathbf{v} \propto \mathbf{1}$, hence $\mathbf{v}'\mathbf{b} = 0$ means $\Gamma_b(0) = 0$.

C Dual Max-Tau Curvature and a Trilemma

For ease of exposition we assume $\omega_0 = 0$ (the discussion extends to $\omega_0 > 0$, not shown). As $L \to \infty$, the filter's degrees of freedom grow, $U \to 1$ so that $\rho \to 1$, and $\tau(\rho, \omega_0 = 0) \to \infty$; yet even as $\rho \to 1$, $\mathbf{b} \not\to \boldsymbol{\gamma}_h$. This is because $\Gamma_b(\omega) \to \begin{cases} 0 & \omega = \omega_0 = 0 \\ \Gamma_h(\omega) & \omega > 0 \end{cases}$ asymptotically, and since $\Gamma_h(0) > 0$ (baseline assumption), the steepness of $\Gamma_b(\omega)$ toward $\omega = 0$ diverges asymptotically (discontinuity), with $\tau(\omega) = \begin{cases} \infty & \omega = 0 \\ \tau_h(\omega) & \omega > 0 \end{cases}$ affected only at $\omega = 0$. However, eliminating the signal (here the trend) and narrowing the lead maximization to a single frequency are generally undesirable. In fact, imposing the TC constraint $\mathbf{b}'\boldsymbol{\gamma}_h = \alpha_h$ alone, does not warrant sufficient control of the filter's shape, allowing for undesirable overfitting at ω_0 for large L. We here extend the dual Max-Tau by providing better control of the bandwidth of the timeliness effect at the reference frequency, addressed via an additional curvature constraint on the second-order derivative $d^2/d\omega^2\Gamma_b(\omega)\big|_{\omega=0} = \mathbf{b}'\mathbf{k}^2 = e > 0$, with $\mathbf{k}^2 = (0^2, 1^2, ..., (L-1)^2)'$: a smaller curvature e regulates adaptivity, widening the lead to adjacent frequencies $\omega > 0$. In general, this also cares about a (desirable) non-vanishing $\Gamma_b(0) = \mathbf{b}'\mathbf{1} = f > 0$, with $f = f(e, \alpha_h)$ depending on α_h and e. For fixed α_h and $L < \infty$, the cost of increasing e signifies a smaller τ, so the efficient frontier extendeds to a trilemma, involving TC ρ, through α_h, time-shift τ, trough $f = f(\alpha_h, e)$, and curvature, via e.

Specifically, we consider the following *smooth dual Max-Tau* optimization problem:

$$\min_{\mathbf{b}, f} \quad \frac{\mathbf{b}'\mathbf{k}}{f}$$

subject to the constraints:

$$\begin{aligned} \mathbf{b}'\mathbf{1} &= f \\ \mathbf{b}'\boldsymbol{\gamma}_h &= \alpha_h \\ \mathbf{b}'\mathbf{k}^2 &= e \\ \mathbf{b}'\mathbf{b} &= 1 \end{aligned}$$

The proof follows the lines of the original dual with two minor modifications: $\mathbf{C} = (\mathbf{1}, \boldsymbol{\gamma}_h, \mathbf{k}^2)$ and $\mathbf{d} = (f, \alpha_h, e)'$. All intermediary steps remain unchanged by this extension, giving a stacked two-step optimization for the smooth Max-Tau predictor. In particular, $\mathbf{1}, \mathbf{k}, \mathbf{k}^2$ are linear independent for $L \geq 3$ and $\boldsymbol{\gamma}_h \not\propto \mathbf{k}^2$ in typical prediction, so full rank is ensured and $\mathbf{C}'\mathbf{C}$ remains invertible. However, for completeness we here provide an extensive formal derivation, bypassing some of the earlier, technically more cumbersome steps.

As for the dual, we consider an orthogonal decomposition of the *linear* constraints: let $\mathbf{C} = [\mathbf{1}, \boldsymbol{\gamma}_h, \mathbf{k}^2]$, with full column rank (see above), so that the linear constraints can be rewritten as:

$$\mathbf{C}'\mathbf{b} = \mathbf{d}(f)$$

where $\mathbf{d}(f) = [f, \alpha_h, e]'$. Decompose

$$\mathbf{d}(f) = \mathbf{d}_0 + f\mathbf{d}_1, \quad \text{with} \quad \mathbf{d}_0 = [0, \alpha_h, e] \quad \text{and} \quad \mathbf{d}_1 = [1, 0, 0]'.$$

Any vector $\mathbf{b}$ satisfying the linear constraints can be uniquely decomposed into a component residing in the column space of $\mathbf{C}$ and an orthogonal component $\mathbf{z}$ residing in the null space of $\mathbf{C}'$:

$$\mathbf{b} = \mathbf{v}(f) + \mathbf{z},$$

where $\mathbf{v}(f) = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{d}(f)$ and $\mathbf{C}'\mathbf{z} = \mathbf{0}$. Substituting this decomposition into the unit-length constraint yields:

$$\mathbf{b}'\mathbf{b} = \|\mathbf{v}(f)\|^2 + \|\mathbf{z}\|^2 = 1 \implies \|\mathbf{z}\|^2 = 1 - \|\mathbf{v}(f)\|^2.$$

The squared norm of the projected vector is a quadratic function of f:

$$\|\mathbf{v}(f)\|^2 = \mathbf{d}(f)'(\mathbf{C}'\mathbf{C})^{-1}\mathbf{d}(f) = C + Bf + Af^2,$$

with $C = \mathbf{d}_0'(\mathbf{C}'\mathbf{C})^{-1}\mathbf{d}_0$, $B = 2\mathbf{d}_1'(\mathbf{C}'\mathbf{C})^{-1}\mathbf{d}_0$, and $A = \mathbf{d}_1'(\mathbf{C}'\mathbf{C})^{-1}\mathbf{d}_1$. Because $\|\mathbf{z}\|^2 \geq 0$, the feasible domain for f is restricted by the inequality:

$$Af^2 + Bf + C \leq 1. \tag{63}$$

We can project $\mathbf{k}$ onto the column space of $\mathbf{C}$ and its orthogonal complement: $\mathbf{k} = \mathbf{k_C} + \mathbf{k}_\perp$, where $\mathbf{k_C} = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}'\mathbf{k}$ and $\mathbf{k}_\perp = (I - \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}')\mathbf{k}$. The numerator of the objective function becomes:

$$\mathbf{b}'\mathbf{k} = (\mathbf{v}(f) + \mathbf{z})'(\mathbf{k_C} + \mathbf{k}_\perp) = \mathbf{v}(f)'\mathbf{k_C} + \mathbf{z}'\mathbf{k}_\perp$$

Let $\mathbf{w} = (\mathbf{C}'\mathbf{C})^{-1}\mathbf{C}'\mathbf{k}$. The first term is purely a function of f:

$$\mathbf{v}(f)'\mathbf{k_C} = (\mathbf{d}_0 + f\mathbf{d}_1)'\mathbf{w} = w_0 + w_1 f,$$

where $w_0 = \mathbf{d}_0'\mathbf{w}$ and $w_1 = \mathbf{d}_1'\mathbf{w}$. To minimize the overall objective for a given f, the first step of the stacked optimization, we minimize the inner product $\mathbf{z}'\mathbf{k}_\perp$. By the Cauchy-Schwarz inequality, subject to the fixed norm $\|\mathbf{z}\| = \sqrt{1 - C - Bf - Af^2}$, this is minimized when $\mathbf{z}$ is anti-parallel to $\mathbf{k}_\perp$:

$$\mathbf{z}^* = -\|\mathbf{z}\| \frac{\mathbf{k}_\perp}{\|\mathbf{k}_\perp\|}.$$

Let $K = \|\mathbf{k}_\perp\|$, so that the objective function becomes:

$$g(f) = \frac{w_0 + w_1 f - K\sqrt{1 - C - Bf - Af^2}}{f}.$$

The global minimum of $g(f)$ may occur either at an interior critical point (where $\dot{g}(f) = 0$) or on the boundary of the feasible domain (where $\|\mathbf{z}\| = 0$). Interior critical points are derived from

$$\dot{g}(f) = \frac{f\left(w_1 - K\frac{-B-2Af}{2\sqrt{1-C-Bf-Af^2}}\right) - \left(w_0 + w_1 f - K\sqrt{1 - C - Bf - Af^2}\right)}{f^2} = 0.$$

Setting the numerator to zero:

$$\begin{aligned} w_0\sqrt{1 - C - Bf - Af^2} &= \frac{K}{2}(Bf + 2Af^2) + K(1 - C - Bf - Af^2) \\ &= K\left(1 - C - \frac{1}{2}Bf\right). \end{aligned} \tag{64}$$

Squaring both sides yields:

$$\begin{aligned} K^2\left(1 - C - \frac{1}{2}Bf\right)^2 &= w_0^2\left(1 - C - Bf - Af^2\right) \\ &= Q_2 f^2 + Q_1 f + Q_0 = 0, \quad \text{with } Q_2 = K^2\frac{B^2}{4} + w_0^2 A, \\ Q_1 &= w_0^2 B - K^2 B(1 - C) \quad \text{and} \quad Q_0 = K^2(1 - C)^2 - w_0^2(1 - C). \end{aligned}$$

Roots of this quadratic —the second step of the stacked optimization— are valid (interior) candidates if they satisfy the pre-squared condition in Equation (64): if one of the two roots, say z_1 is valid and a minimum, then $f^* = z_1$. If it is a maximum or no root is valid (infeasible), the minimum of the objective will lie on the boundary defined by the equality condition of Equation (63), where the orthogonal component vanishes ($\|\mathbf{z}\| = 0$):

$$\eta_{1,2} = \frac{-B \pm \sqrt{B^2 - 4A(C - 1)}}{2A}. \tag{65}$$

In this boundary case, select the root that minimizes the objective, say η_1, and set $f^* = \eta_1$. At this point, the vector $\mathbf{b}$ lies entirely within the column space of $\mathbf{C}$, i.e., $\mathbf{b} = \mathbf{v}(f^*)$, and the objective function simplifies to $g(f^*) = (w_0 + w_1 f^*)/f^*$. Compactly, the smooth dual Max-Tau is

$$\mathbf{b}^* = \mathbf{C}(\mathbf{C}'\mathbf{C})^{-1}\mathbf{d}(f^*) - \frac{\sqrt{\max(0, 1 - C - Bf^* - A(f^*)^2)}}{K}\mathbf{k}_\perp. \tag{66}$$

D DFP Distribution

The distribution of the MSE-DFP predictor follows from Corollary 1 with $\lambda_1 = 1$ and $\lambda_2 = \lambda$ fixed. We extend this to incorporate the randomness of λ when defined as a function of the empirical nowcast $\hat{\boldsymbol{\gamma}}_0$.

Corollary 5. *Suppose $\boldsymbol{\gamma}_0$ and $\boldsymbol{\gamma}_h$ $(h > 0)$ are linearly independent. Let $\hat{\boldsymbol{\gamma}}_0$ be an estimator of $\boldsymbol{\gamma}_0$ with mean $\boldsymbol{\mu}_{\gamma_0} \to \boldsymbol{\gamma}_0 \neq \mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}_{\gamma_0}$ with $\boldsymbol{\Sigma}_{\gamma_0} \to \mathbf{0}$ for $L \to \infty$. Then, for L and T sufficiently large, the mean and covariance of the DFP-MSE* (8)*, based on stochastic $\lambda = \lambda(\hat{\boldsymbol{\gamma}}_0)$ in* (9)*, are approximated by*

$$\boldsymbol{\mu}_b \approx (\mathbf{F}^h + \tilde{\lambda}\mathbf{I})\boldsymbol{\gamma}_0 \tag{67}$$

$$\boldsymbol{\Sigma}_b \approx \mathbf{J}\boldsymbol{\Sigma}_{\gamma_0}\mathbf{J}' \tag{68}$$

where

$$\tilde{\lambda} = \frac{\alpha_0 - (\mathbf{F}^h\boldsymbol{\gamma}_0)'\boldsymbol{\gamma}_0}{\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0} \tag{69}$$

$$\mathbf{J} = \mathbf{F}^h + \boldsymbol{\gamma}_0(\nabla\tilde{\lambda})' + \tilde{\lambda}\mathbf{I} \tag{70}$$

where $\nabla\tilde{\lambda}$ is the gradient with entries

$$\partial\tilde{\lambda}/\partial\gamma_k = -\frac{I_{\{k \geq h\}}\gamma_{k-h} + I_{\{k \leq L-1-h\}}\gamma_{k+h}}{\|\boldsymbol{\gamma}_0\|^2} - \frac{\alpha_0 - (\mathbf{F}^h\boldsymbol{\gamma}_0)'\boldsymbol{\gamma}_0}{\|\boldsymbol{\gamma}_0\|^4}2\gamma_k, \tag{71}$$

$k = 0, ..., L-1$ and $I_{\{\}}$ the indicator function. If the lag-zero entry of the nowcast is unity, $\gamma_0 = 1$, then the first row and column of $\boldsymbol{\Sigma}_{\gamma_0}$ are zero, and in (70) *the first column of $\boldsymbol{\gamma}_0(\nabla\tilde{\lambda})' + \tilde{\lambda}\mathbf{I}$ is set to zero.*

Proof: When $\lambda = \lambda(\hat{\boldsymbol{\gamma}}_0)$ is not treated as fixed, the approximation in (67) follows by continuity (the linear independence assumption implies $\boldsymbol{\gamma}_0'\boldsymbol{\gamma}_0 > 0$ in (69)) and by using $\boldsymbol{\gamma}_h \approx \mathbf{F}^h\boldsymbol{\gamma}_0$ when L is sufficiently large. To obtain the covariance matrix in (68), we apply the delta method (see Cox, 1990) and use the first-order Taylor expansion

$$\mathbf{b}(\hat{\boldsymbol{\gamma}}_0) \approx \mathbf{b}(\boldsymbol{\gamma}_0) + \mathbf{J}(\hat{\boldsymbol{\gamma}}_0 - \boldsymbol{\gamma}_0),$$

assuming T is large. The Jacobian $\mathbf{J}$ has entries

$$\mathbf{J}_{ij} = \partial b_{i-1}/\partial\gamma_{j-1} = \mathbf{F}^h_{ij} + \frac{\partial\tilde{\lambda}}{\partial\gamma_{j-1}}\gamma_{i-1} + \tilde{\lambda}\mathbf{I}_{ij},\ 1 \leq i, j \leq L, \tag{72}$$

with $\tilde{\lambda} = \tilde{\lambda}(\boldsymbol{\gamma}_0)$ as specified in (69), $\mathbf{I}_{ij} = \delta_{ij}$ the Kronecker symbol, and $\partial\tilde{\lambda}/\partial\gamma_{j-1}$ as in (71). It follows that

$$\boldsymbol{\Sigma}_b \approx \mathbf{J}\boldsymbol{\Sigma}_{\gamma_0}\mathbf{J}',$$

which establishes the claim. If the lag-zero entry of the nowcast is unity, $\gamma_0 = 1$, then the first row and the first column of $\boldsymbol{\Sigma}_{\gamma_0}$ vanish. In addition, the partial derivatives with respect to γ_0 in (72) disappear; equivalently, the first column of $\boldsymbol{\gamma}_0(\nabla\tilde{\lambda})' + \tilde{\lambda}\mathbf{I}$ is set to zero (the first row does generally not vanish because $\tilde{\lambda}$ depends on γ_k, $L > k > 0$). □

Remark: An extension of Corollary 1 to random λ_1, λ_2 can be obtained analogously, though $\lambda_i(\boldsymbol{\gamma}_0)$ and $\mathbf{b} = \lambda_1\boldsymbol{\gamma}_h + \lambda_2\boldsymbol{\gamma}_0$ become more involved functions of $\boldsymbol{\gamma}_0$; see Proposition 1. A finite-sample simulation study is provided in Wildi (2026c) as well as in the DFP R-package.

E Lead in the Non-Standard Case $\tau_h < \tau_0$

We argue that the final baseline condition $\tau_h > \tau_0$, signifying that the MSE predictor leads the nowcast, is not strictly necessary to derive an expression for the phase differential $\Phi_b(\omega) - \Phi_h(\omega)$ in Proposition (4), though it simplifies the analysis. Suppose instead that $\tau_h < \tau_0$ (non-standard situation). A simple example is the ARMA(1,1) process

$$x_t = a_1 x_{t-1} + \epsilon_t + b_1 \epsilon_{t-1}$$

with Wold decomposition

$$\gamma_k = \begin{cases} 1 & k = 0 \\ (a_1 + b_1) a_1^{k-1} & k > 0 \end{cases}$$

For $a_1 = 0.9$ and $b_1 = -0.5$ we obtain

$$\tau_k = \begin{cases} -7.826 & k = 0 \\ -9 & k > 0 \end{cases}$$

The non-standard case arises because the weights γ_{k+h}, $k = 0, ..., L-1$ in the MSE(h) predictor $\boldsymbol{\gamma}_h$, $h > 0$, decay exponentially, whereas those in the nowcast $\boldsymbol{\gamma}_0$ do not, since the first weight $\gamma_0 = 1$ assigns disproportionate mass to lag 0 when $b_1 < 0$, so that the nowcast effectively leads the MSE predictor—a rather counterintuitive result. More generally, using (16), we require

$$-\frac{\sum_{k=1}^{L-1} k\gamma_k}{\sum_{k=0}^{L-1} \gamma_k} = \tau_0 > \tau_h = -\frac{\sum_{k=1}^{L-1} k\gamma_{k+h}}{\sum_{k=0}^{L-1} \gamma_{k+h}}.$$

As an example, for $L = 2$ and $h = 1$, with $\gamma_0 = 1$, this reduces to $\gamma_1^2 < \gamma_2$. In that case the one-step predictor $\boldsymbol{\gamma}_1 = (\gamma_1, \gamma_2)'$ lags the nowcast $\boldsymbol{\gamma}_0 = (1, \gamma_1)'$ at frequency zero.

Assume therefore that $\tau_h < \tau_0$. In this non-standard case

$$\Phi_h(\omega) \approx \dot{\Phi}_h(0)\omega = \tau_h \omega < \tau_0 \omega = \dot{\Phi}_0(0)\omega \approx \Phi_0(\omega),$$

for small $\omega > 0$. For $\lambda < 0$ (phase excess) in $\Gamma_b = \Gamma_h + \lambda \Gamma_0$ this suggests $\Phi_b(\omega) < \Phi_h(\omega)$ and hence

$$\tau_b \approx \Phi_b(\omega)/\omega < \Phi_h(\omega)/\omega \approx \tau_h,$$

so the DFP would *lag* the MSE predictor at the reference (zero) frequency. Nevertheless, a lead is still attainable when flipping signs. Specifically, one can distinguish two scenarios: either i)

$$\Phi_0(\omega) > \Phi_b(\omega) \geq \Phi_h(\omega), \tag{73}$$

or ii)

$$\Phi_b(\omega) \geq \Phi_0(\omega) \geq \Phi_h(\omega), \tag{74}$$

both of which differ from the standard ordering (30).
In the first scenario, $\lambda > 0$ implies $\Phi_b(\omega) > \Phi_h(\omega)$[17] and

$$\tau_b \approx \Phi_b(\omega)/\omega > \Phi_h(\omega)/\omega \approx \tau_h$$

inducing a lead, as desired. However, $\Phi_b(\omega) < \Phi_0(\omega)$ so that

$$\tau_b \approx \Phi_b(\omega)/\omega < \Phi_0(\omega)/\omega \approx \tau_0, \tag{75}$$

[17]If $\lambda < 0$, the DFP rotates in the same direction as the MSE predictor relative to the nowcast: in that case $\theta_{0h} < 0$ and the 'phase excess' $\theta_{0b} = \theta_{0h} + \theta_{hb} < \theta_{0h} < 0$ increases the *lag*. Rotating in the opposite direction, i.e., $\theta_{hb} > 0$, to increase the *lead* therefore requires $\lambda > 0$ in the non-standard case.

that is, the DFP time shift does not exceed the nowcast. We now focus on the practically more relevant case ii). Going the other way in Fig. 2 means selecting the lower branch of the cone, i.e., $\theta_{0b} < 0$. In this non-standard setting, recovering the lower-branch solution from the unit-length criterion (2) requires switching the maximization to a *minimization*, with a negative weight $\lambda_1 < 0$ on $\boldsymbol{\gamma}_h$ in

$$\mathbf{b} = \lambda_1 \boldsymbol{\gamma}_h + \lambda_2 \boldsymbol{\gamma}_0$$

ensuring a lead. For the MSE-DFP, swapping sign by maximizing (7) fails since $\mathbf{b}$ is unconstrained in length. Instead, we swap the roles of $\boldsymbol{\gamma}_0$ and $\boldsymbol{\gamma}_h$:

$$\begin{aligned} & \min_{\mathbf{b}} (\mathbf{b} - \boldsymbol{\gamma}_0)'(\mathbf{b} - \boldsymbol{\gamma}_0) \\ \text{s.t.} \quad & \boldsymbol{\gamma}_h' \mathbf{b} = \alpha_h. \end{aligned} \tag{76}$$

Proposition 2 then yields

$$\mathbf{b}(\lambda) = \boldsymbol{\gamma}_0 + \lambda \boldsymbol{\gamma}_h. \tag{77}$$

For $\lambda < 0$, $\tau_b > \tau_0$, as desired. Fig. 14 illustrates the swap of nowcast and MSE predictor.

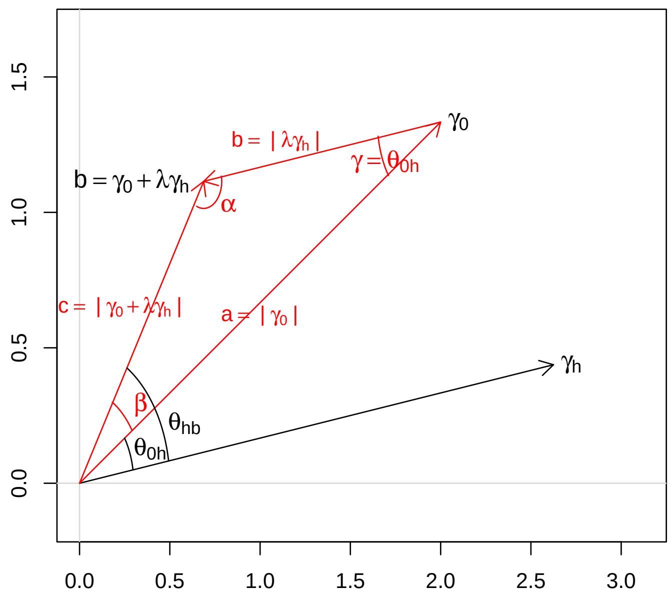

Figure 14: Non-standard case ii): Side lengths a, b and c (red) as well as angles α, β and γ of the DFP-triangle spanned by the vectors $\boldsymbol{\gamma}_0$ and $\mathbf{b} = \boldsymbol{\gamma}_0 + \lambda \boldsymbol{\gamma}_0$ (black), with $\lambda < 0$.

We extend Proposition 4 and Proposition 3, linking $\lambda = \lambda(\tau)$ to the DFP lead $\tau := \tau_b - \tau_0$, now referenced against the nowcast (instead of the MSE predictor). While a proof follows directly from swapping the roles of $\boldsymbol{\gamma}_0$, $\boldsymbol{\gamma}_h$ in the respective proofs, a separate derivation is instructive due to the modified geometry. Applying the law of cosines to Fig. 14 and the law of sines gives[18]

$$\beta = \arcsin\left(\frac{b\sin\gamma}{\sqrt{a^2+b^2-2ab\cos\gamma}}\right)$$

Transposing to the frequency-domain for small $\omega > 0$:

$$\Phi_b - \Phi_0 \quad = \quad \arcsin\left(\frac{|\lambda|A_h\sin(\Phi_0-\Phi_h)}{\sqrt{A_0^2+\lambda^2A_h^2-2|\lambda|A_hA_0\cos(\Phi_0-\Phi_h)}}\right) \approx \frac{|\lambda|A_h(\Phi_0-\Phi_h)}{|A_0-|\lambda|A_h|}$$

where $\Phi_b - \Phi_0$ is now referenced against the nowcast. Dividing by small $\omega > 0$:

$$\tau := \tau_b - \tau_0 \approx \frac{|\lambda|A_h(\tau_0-\tau_h)}{|A_0-|\lambda|A_h|}.$$

Using $A_i = \Gamma_i > 0$ (baseline assumptions) we can now solve for $\lambda = -|\lambda|$, assuming $A_0 - |\lambda|A_h = \Gamma_0 + \lambda\Gamma_h > 0$:

$$\lambda = -\frac{\tau\Gamma_0}{\Gamma_h(\tau+\tau_0-\tau_h)}$$

This expressions mirrors (18) in the standard case, with the roles of Γ_h and Γ_0 swapped and the sign of $\tau_0 - \tau_h$ inverted.

In summary, constructing large leads over the MSE predictor under the non-standard ordering $\tau_h < \tau_0$ (case ii)) requires either flipping the sign in (2) or swapping the roles of the nowcast and MSE predictor in (76). The baseline condition $\tau_h > \tau_0$ is not essential for deriving $\beta(\omega)$ in Proposition (4); it mainly streamlines the exposition by preserving the DFP-triangle geometry near zero frequency. In the standard case, $\mathbf{b} = \boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0$, and $\lambda < 0$ produces a lead $\tau_b > \tau_h$, consistent with the phase excess $\theta_{0b} - \theta_{0h} > 0$. For $\tau_h < \tau_0$, the reorderings (73) and (74) apply, giving $\mathbf{b} = \boldsymbol{\gamma}_h + \lambda\boldsymbol{\gamma}_0$ with $\lambda > 0$ (case i)) or $\mathbf{b} = \boldsymbol{\gamma}_0 + \lambda\boldsymbol{\gamma}_h$ with $\lambda < 0$ (case ii)). In both cases, λ can be linked to τ and MSE-optimal scalings can be obtained via $\mathbf{b}_{MSE-DFP} := \frac{\mathbf{b}'\boldsymbol{\gamma}_h}{\|\mathbf{b}\|^2}\mathbf{b}$. The counter-intuitive non-standard cases are illustrated in the DFP R-package (Tutorial 9).

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[18] Since arcsin returns values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, if $b > c$ one must verify whether β or $\pi - \beta$ is the valid solution, choosing the one satisfying $\alpha + \beta + \gamma = \pi$ with $\alpha > 0$.

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